Flight to Safety: Monetary Policy in an Open Economy*

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Abstract

In a global crisis, investors flee to the assets of a few countries. This paper asks what such a flight to safety does to the economy that receives it, and what its central bank can do about it. In an open-economy New Keynesian model, domestic assets provide convenience services to residents and foreigners. A flight makes the asset scarce at home and raises its convenience yield. Saving pays more at an unchanged policy rate, so demand falls as if the central bank had tightened: a portfolio event in normal times, a recession in a crisis. The flight also brings income, which appreciates the currency, but most of the appreciation comes from the lower prices the tightening causes. Calibrated to the Swiss franc in the Global Financial Crisis, the flight pushes the economy into a liquidity trap of foreign origin. The balance sheet, neutral in standard models, is here a second instrument. Supplying the asset the world wants stabilizes the economy, with an operation of 7% of annual GDP, close to the 6% the Swiss National Bank bought. A flight that does not end defeats interest-rate policy alone, but not the balance sheet.

JEL classification: E44, E52, E58, F31, F41

Keywords: Flight to safety, safe haven, convenience yield, liquidity trap, safe assets, zero lower bound, balance-sheet policy, foreign exchange intervention.

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1 Introduction

What happens to an economy when the rest of the world wants its assets and is willing to pay a premium for them? The question is best known from the United States. Investors hold dollar bonds for their safety and liquidity and accept a lower return on them than on other securities (Krishnamurthy and Vissing-Jorgensen, 2012). This discount earns the country an income, the exorbitant privilege (Gourinchas and Rey, 2007). Government bonds of other G10 countries carry a similar discount (Engel and Wu, 2023). In a global crisis investors flee to the safety of a few currencies, and the Swiss franc is among them (Ranaldo and Söderlind, 2010). The Swiss experience after the Global Financial Crisis (GFC) shows that this privilege can also carry a cost, and one that is hard for the central bank to manage. Foreign demand for the assets rises, the currency appreciates, inflation falls, and the policy rate is driven to its effective lower bound. In every major risk-off episode of the past two decades the franc appreciated (Figure 1).[1] The policy rate fell to zero and then below, to -0.75%, where it stayed from 2015 to 2022 (Figure 2a). At the same time, the balance sheet of the Swiss National Bank (SNB) grew to one and a half times GDP, the bulk of it foreign currency (Figure 2b). Two SNB chairmen, in 2015 and 2024, describe this pattern as a safe-haven story (Jordan, 2015; Schlegel, 2024). Their account is a narrative, and this paper gives it a model: it studies what such a flight to safety does to the economy that receives it, and what its central bank can do about it. The flight tightens monetary conditions at home with the policy rate unchanged, and in a large flight the economy falls into a liquidity trap of foreign origin. The central bank can offset it by supplying the asset the world wants.

Standard models do not produce the pattern seen in Figures 1 and 2. If home and foreign assets are perfect substitutes, a shift in foreign demand changes only who holds the asset, and the central bank's balance sheet has no role (Backus and Kehoe, 1989; Wallace, 1981). With a financial friction the shift moves the exchange rate, and through it trade, but the shift itself does not move the return households earn on saving at home. The balance sheet then works through the friction. Two links are missing. One runs from the inflow to domestic demand. The other runs from the central bank's balance sheet to the economy where no friction binds.

One ingredient supplies both. Domestic assets provide convenience services to residents and to the rest of the world, and their holders accept a lower return on them than on an ordinary bond. That discount is the convenience yield. A flight to safety is a rise in

[1] Documented for the franc by Grisse and Nitschka (2015). Over the sample of Figure 1 the monthly correlation between NEER changes and VIX changes is 0.23, significant at the one percent level, measured despite a policy that often leaned against it, including the minimum exchange rate of 2011 to 2015, which suppressed it.

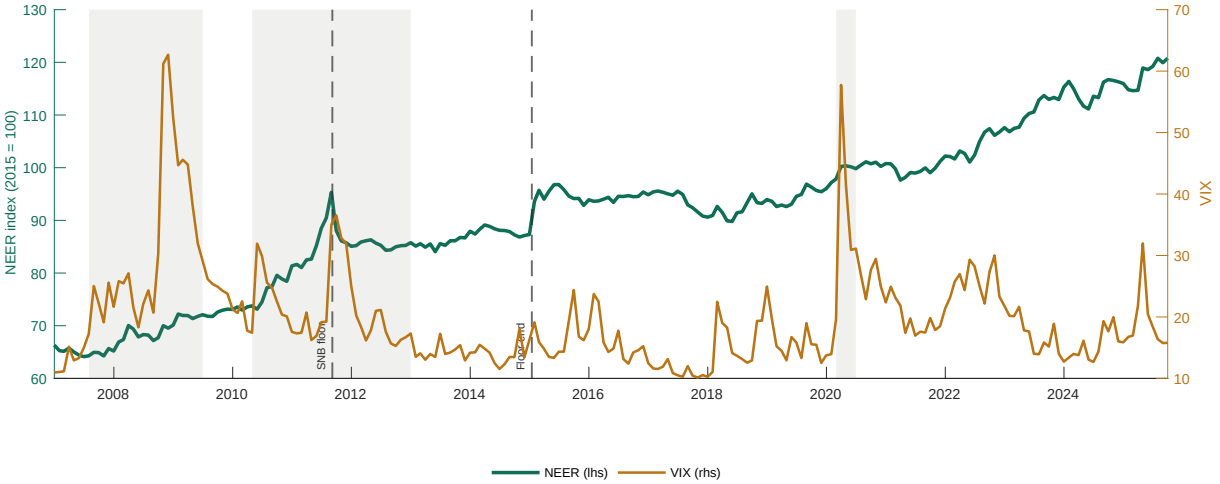
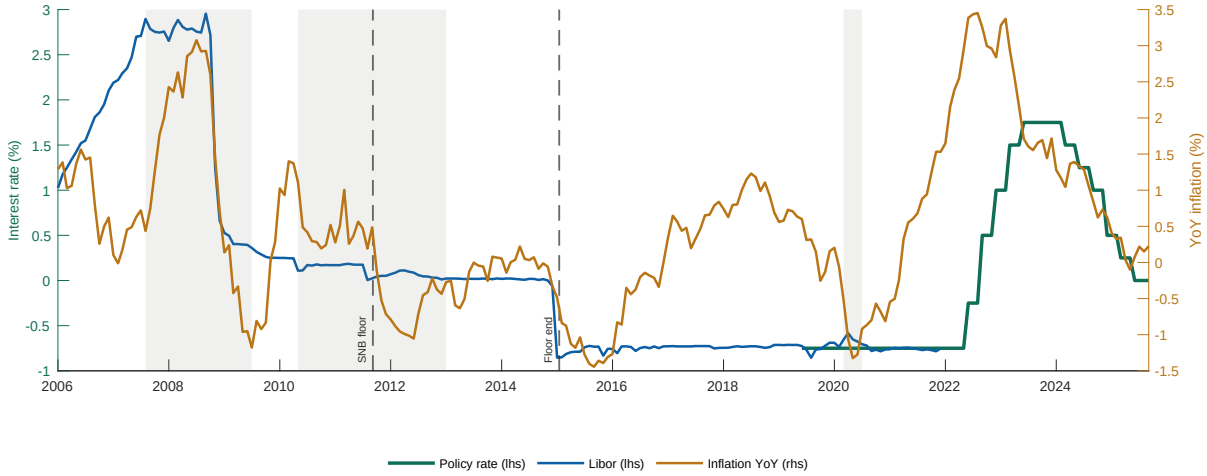


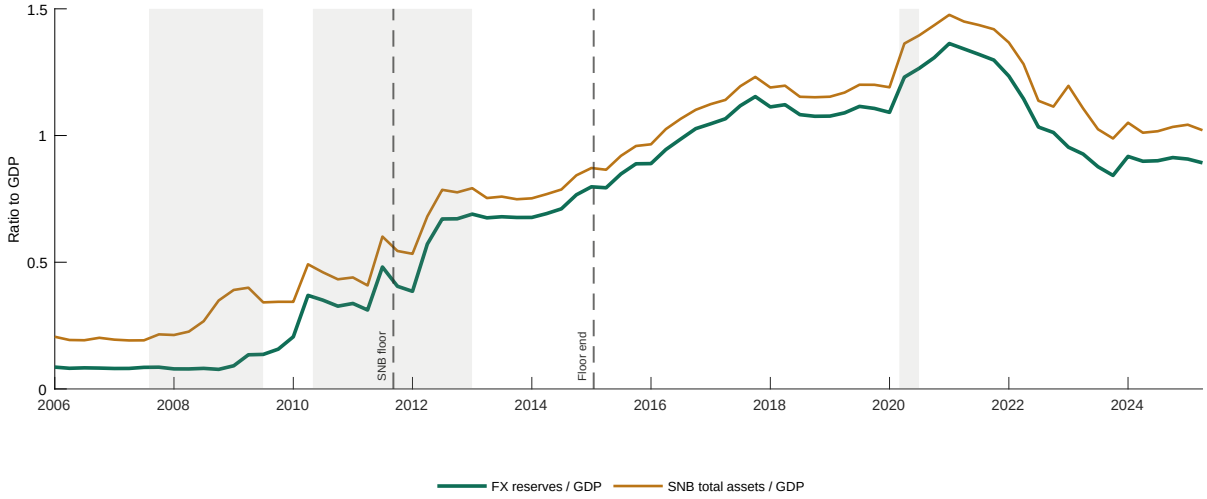
Figure 1: Safe-haven dynamics of the Swiss franc. The figure plots the nominal effective exchange rate (NEER) together with the VIX index, a measure of global financial risk. Shaded areas indicate major global risk-off episodes (GFC, euro area debt crisis, COVID-19). Sources: NEER, BIS (Bank for International Settlements, 2025); VIX, CBOE via FRED (series VIXCLS) (Chicago Board Options Exchange, 2025).

foreign demand for the services, and it works through two channels. The first is external. Foreigners pay for the convenience services by accepting a lower return, so the country earns an income, the model's counterpart of the exorbitant privilege. Households spend part of it on traded goods, which come in at the world price. Home goods cannot be imported, so with more traded goods per home good, traded goods become cheaper relative to home goods. Non-traded prices are sticky, so the adjustment falls on the franc price of traded goods: the currency appreciates. The second channel is domestic. The flight raises the convenience yield at home, because the units residents keep are worth more to them. Households count the services as part of the return on saving, so saving pays more while the policy rate is unchanged. They spend less, and output and inflation in the non-traded sector fall, as if the central bank had raised its rate. Both add to the appreciation. The lower output makes home goods scarcer relative to traded goods, which strengthens the franc on impact, and the lower non-traded prices pull the franc price of traded goods down with them for as long as inflation stays below target. On impact the two channels are of the same order. Over time the domestic one is most of the appreciation, and the income accounts for less than a tenth of the peak. The central bank can offset the tightening with its rate, until the rate reaches its lower bound, or by supplying the asset the world wants, which keeps the convenience yield from rising.

I develop an open-economy New Keynesian model with this ingredient, simple enough to prove its main results analytically. Four propositions show exactly how the flight reaches the domestic economy, what monetary policy can do about it with the interest rate and with the balance sheet, when the financing of the balance sheet matters, and when the interest-rate-only policy fails. The model prices the world's willingness to hold



(a) Policy rate and inflation



(b) SNB balance sheet

Figure 2: Monetary policy in Switzerland. Panel (a): CPI inflation (year-on-year) and the SNB policy rate (before June 2019, the three-month Libor target). Panel (b): SNB foreign exchange reserves and total assets, share of annual GDP. Shaded areas indicate major global risk-off episodes. Sources: SNB data portal (Swiss National Bank, 2025a,c,d); author's calculations.

the asset at a discount, whatever its source. It applies to any economy whose assets the rest of the world holds for their convenience services, from safe havens to the issuer of the reserve currency. I calibrate the model to Switzerland, where the predicament appears at full scale, and size the flight to the franc's appreciation in the acute phase of the GFC, from 2008Q3 to 2009Q2. Three main results follow.

The first result is that the flight reaches the domestic economy only through the convenience yield. The non-traded sector follows a New Keynesian model with one extra term, the change in the convenience yield. The flight enters through that term and nowhere else. The term lowers demand, so the flight lowers output, inflation and the policy rate, while the foreign demand inflow appreciates the currency. The effect of a unit of inflow is proportional to the steady-state convenience yield, so the same inflow is a portfolio event at normal-times levels and a recession at crisis levels (Proposition 1). The

policy rate cut prescribed by the Taylor rule lies below zero. With the zero lower bound the economy falls into a liquidity trap of foreign origin: output falls nine times as much as without the lower bound, non-traded inflation falls below zero, and the bound binds for thirteen quarters. The trap also costs the country its income from the asset. At the zero lower bound the sharper appreciation raises the real value of what the economy already owes abroad, and the net inflow turns negative. Compared to the acute phase of the GFC in Switzerland, the economy at the bound has the sign of every domestic variable right and the sizes to the order of magnitude, although only the appreciation is fitted.

The second result is that the balance sheet is an independent monetary policy instrument that can close the domestic channel. When the central bank sets the supply of the asset to stabilize the domestic economy, non-traded output and inflation stay at target and the policy rate never has to move, for any path of the flight (Proposition 2). The supply keeps the convenience yield at its steady-state value, so there is nothing for the rate to offset. For the flight of the acute phase of the GFC the operation is 7% of annual GDP. Over the same four quarters the SNB bought foreign currency worth 6% of annual GDP. The appreciation shrinks to less than a tenth of its peak without policy and reverses, and the country's income from the asset is kept and enlarged. Holding the exchange rate instead takes 13% of annual GDP and leaves a mild boom. A balance-sheet expansion is inflationary only when it supplies more of the convenient asset than what the flight takes from the domestic residents. What the central bank does with the proceeds matters little. It can rebate them to households or invest them in foreign assets, as the SNB did. Without portfolio costs the two are equivalent, because only the asset the central bank issues affects the economy (Proposition 3). With portfolio costs, households consume more of the country's income when the proceeds are rebated: tradable consumption rises three times as much as under reserve backing.

The third result is that a flight that does not end is the case the interest rate cannot handle alone. A permanent flight-to-safety shock permanently lowers the real rate the economy needs. Under the Taylor rule it comes with inflation permanently below target and a policy rate permanently below its steady state. Beyond a threshold, the same rule with the lower bound leaves the economy with no steady state and no bounded path (Proposition 4). In my calibration the threshold is two fifths of the fitted flight, and even flights of known length approach it fast: the recession is 0.9% for a flight of two years and 13% for one of five. The balance sheet still stabilizes, with a permanent operation of 7% of annual GDP for the flight of 2008.[2]

The paper builds on the open-economy New Keynesian framework (Benigno and Be-

[2]After 2009 the Swiss balance sheet grew with the flight, as the model prescribes for a flight that does not end. Its larger size reflects a different episode: later flights arrived on positions the earlier ones had enlarged, and calibrated to those positions the operation scales up.

nigno, 2003; Clarida et al., 2002; Galí and Monacelli, 2005) by embedding a convenience yield on domestic assets and a foreign demand for their services. It draws on the safe-asset literature that measures convenience yields and ties them to exchange rates (Engel and Wu, 2023; Jiang et al., 2021; Krishnamurthy and Vissing-Jorgensen, 2012; Valchev, 2020).

The flight to safety enters the foreign household's optimality condition as a deviation from uncovered interest parity, which connects the paper to models of exchange rates driven by financial frictions (Gabaix and Maggiori, 2015; Itskhoki and Mukhin, 2021), to the literature on foreign exchange intervention (Amador et al., 2020; Cavallino, 2019; Fanelli and Straub, 2021), to the instrument assignments of Itskhoki and Mukhin (2023) and Yakhin (2025), and to the integrated policy framework of Basu et al. (2020). In these papers the deviation from parity is a premium paid to intermediaries. It moves the exchange rate, and through it trade. Here it is a discount that holders accept for the services of the asset. It also moves the return households earn on saving at home, and it is attached to a quantity the central bank issues.

The liquidity trap connects the paper to work on safe-asset shortages (Caballero and Farhi, 2018; Caballero et al., 2021; Farhi and Maggiori, 2018). There the demand for safe assets comes from savers who want a store of value or are extremely risk averse, and a shortage pushes safe rates to the lower bound. Here the demand is for the convenience services of the asset, and a surge in foreign demand tightens monetary conditions in the issuing economy. The income from supplying the asset is the model counterpart of the exorbitant privilege (Gourinchas and Rey, 2007, 2022), and the lower bound is a reason why it weakens in crises.

The analysis is closest to Kekre and Lenel (2024) and to the note of Benigno (2022). Kekre and Lenel (2024) study the same kind of shock, a rise in the demand for the services of safe dollar bonds, with a focus on risk premia and international risk sharing. This paper asks what monetary policy can do about the flight: the interest rate and the balance sheet, the lower bound, and a flight that does not end. Its linear structure makes the channel explicit (Proposition 1). The paper builds on Benigno (2022), which Benigno (2025) develops to analyze the optimal long-run supply of a reserve currency. The present model studies the transmission of the flight and the interaction between interest-rate and balance-sheet policy.

The remainder of the paper proceeds as follows. Section 2 presents the model and Section 3 characterizes the equilibrium. Section 4 describes the calibration. Section 5 contains the results: the baseline transmission, the role of the convenience yield, the lower bound, balance-sheet policy, reserve-backed operations, the comparison with the acute phase in the data, and a flight that does not end. Section 6 concludes.

2 Model

2.1 General Framework

The model consists of a home economy and the rest of the world, and its convenience-asset structure builds directly on the note of Benigno (2022). The model is designed to isolate the mechanism through which foreign demand for home-currency assets becomes a macroeconomic shock. Country H issues a government asset that provides convenience benefits to both domestic and foreign households, in the tradition of Sidrauski (1967). Country F represents the rest of the world. There are three classes of assets:

1. **Domestic convenient asset Q_t:** Convenience-bearing government asset denominated in domestic currency, issued by the domestic government/central bank. It yields the policy-controlled gross return $(1 + i_t^R)$ and provides convenience benefits $V(q_t)$ to households.

2. **Foreign policy bond F_t:** Foreign government asset denominated in foreign currency, yielding the foreign policy rate $(1 + i_t^{R*})$. It does not generate convenience benefits,[3] and adjusting foreign bond holdings is costly for domestic households, reflecting financial frictions.

3. **Domestic ordinary bond B_t:** Private-sector issued bonds denominated in domestic currency. These bonds yield the gross return $(1 + i_t)$, but do not provide convenience benefits.

Households choose consumption, labour, and asset holdings $\{Q_t, B_t, F_t\}$ subject to their budget constraint. Firms operate under monopolistic competition and Calvo price stickiness, generating a New Keynesian Phillips curve. The domestic government sets the policy rate i_t^R and supplies convenient assets Q_t^s. The foreign authority sets the policy rate i_t^{R*} and supplies F_t^s. Domestic households have access to the foreign bond F_t, but they face portfolio adjustment costs. Following Schmitt-Grohé and Uribe (2003), the cost function in real terms is given in units of the traded good by

$$C\left(\frac{S_t F_t}{P_{T,t}}\right) = \frac{\psi}{2}\left(\frac{S_t F_t}{P_{T,t}} - \bar{f}\right)^2,$$

where $\psi > 0$ governs the size of the adjustment cost and $\bar{f}$ is the steady-state level of

[3] A normalization. Only the relative convenience of the two assets matters for the wedge, so endowing the foreign bond with services as well would only relabel $V(\cdot)$.

foreign bond holdings in real terms,

$$\bar{f} \equiv \frac{S\bar{F}}{P_T} = \frac{\bar{F}}{P_T^*}.$$

2.2 Country H

Households. The representative household living in country H has the following preferences:

$$E_t \sum_{t=0}^{\infty} \beta^t \left[\ln C_t - H_t + \theta_t V\left(\frac{Q_t}{P_{T,t}}\right) \right],$$

where β is the rate of time preference with $0 < \beta < 1$, C_t is a composite consumption bundle consisting of the traded and non-traded good according to

$$C_t = C_{N,t}^{\gamma} C_{T,t}^{1-\gamma}, \tag{1}$$

with $0 < \gamma < 1$ and where $C_{N,t}$ is consumption of the non-traded and $C_{T,t}$ of the traded good. H_t is labour supplied to produce the non-traded good. Households derive utility from real holdings of the convenient asset according to the convenience function $V(\cdot)$, which captures its convenience services: liquidity, safety, collateral value. The preference formulation is a reduced-form way of capturing the non-pecuniary value attached to holding the asset. Q_t is the nominal demand for this asset and $P_{T,t}$ denotes the price level of the traded good in country H. Define $q_t \equiv \frac{Q_t}{P_{T,t}}$.

Households are subject to the budget constraint

$$\begin{aligned} B_t + Q_t + S_t F_t + P_t C_t = (1 + i_{t-1}) B_{t-1} + (1 + i_{t-1}^R) Q_{t-1} + S_t (1 + i_{t-1}^{R*}) F_{t-1} \\ + W_t H_t + P_{T,t} Y_{T,t} + \Pi_t^N - T_t - P_{T,t} \frac{\psi}{2} \left(\frac{S_t F_t}{P_{T,t}} - \bar{f} \right)^2, \end{aligned} \tag{2}$$

where B_t are the risk-free holdings of the bonds issued by households and denominated in currency h, which carry an interest rate i_t, but do not provide convenience services. Q_t are risk-free bonds that are issued by the government in H and denominated in currency h, which provide convenience services and yield the policy rate i_t^R. F_t are risk-free bonds issued by the government in F and denominated in currency f. They do not provide convenience services and yield the foreign policy rate i_t^{R*}. It is costly for domestic households to invest in foreign bonds, and they need to pay a portfolio adjustment cost. P_t is the domestic price level of the aggregate consumption good C_t, W_t are nominal wages, $Y_{T,t}$ is the exogenous endowment of the traded good, Π_t^N are nominal profits of firms operating in the non-traded good sector and T_t denotes lump-sum taxes. Solving the household

expenditure minimization problem associated with the composite consumption good implies that the aggregate price level is a Cobb-Douglas index of tradable and non-tradable prices,

$$P_t = \omega P_{N,t}^{\gamma} P_{T,t}^{1-\gamma}, \tag{3}$$

and that nominal expenditures on each good satisfy constant expenditure shares,

$$P_{N,t}C_{N,t} = \gamma P_t C_t, \qquad P_{T,t}C_{T,t} = (1-\gamma) P_t C_t, \tag{4}$$

where $\omega \equiv \gamma^{-\gamma}(1-\gamma)^{-(1-\gamma)}$. Details of the derivation are reported in Appendix A.1.

The household's problem yields the following first-order conditions. The intratemporal optimality condition implies $W_t = P_t C_t$. The optimal asset pricing condition for B_t is given by

$$1 = E_t \left[M_{t+1}(1+i_t) \right], \tag{5}$$

where $M_{t+1} = \frac{\beta}{\Pi_{t+1}} \frac{C_t}{C_{t+1}}$ is the nominal stochastic discount factor (SDF) and $\Pi_{t+1} = \frac{P_{t+1}}{P_t}$ denotes gross inflation.

The optimal asset pricing condition for Q_t is given by

$$1 = L_t + E_t \left[M_{t+1}(1+i_t^R) \right], \tag{6}$$

where $L_t = \frac{\theta_t P_t C_t}{P_{T,t}} V_q(q_t)$ is the non-pecuniary value of convenience. It captures the marginal benefit of holding the convenient asset and drives a wedge between the returns on Q_t and B_t.

Furthermore, the optimal asset pricing condition for the bond issued by the government in F (F_t) is given by

$$E_t \left[M_{t+1} \frac{S_{t+1}}{S_t} (1+i_t^{R*}) \right] = 1 + \psi \left(\frac{F_t}{P_{T,t}^* \bar{f}} - \bar{f} \right). \tag{7}$$

The portfolio adjustment cost separates the expected exchange-rate-adjusted return on F_t from unity.

Firms. A representative non-traded final goods producer aggregates a continuum of intermediate goods $i \in [0,1]$ using a CES technology with substitution elasticity $\theta > 1$.

Cost minimization implies demand for each variety,

$$Y_{N,t}(i) = \left(\frac{P_{N,t}(i)}{P_{N,t}}\right)^{-\theta} Y_{N,t}, \tag{8}$$

and defines the CES aggregate price index

$$P_{N,t} = \left(\int_0^1 P_{N,t}(i)^{1-\theta}\, di\right)^{\frac{1}{1-\theta}}. \tag{9}$$

Intermediate non-traded goods are produced by a continuum of firms of measure one with linear technology $Y_{N,t}(i) = A_t H_t(i)$ for each $i \in [0,1]$, where $H_t(i)$ is labour and A_t productivity. Firms pay nominal wage W_t and set prices à la Calvo (1983): in each period a fraction $1-\alpha$ with $0 < \alpha < 1$ can reset to a common price $\tilde{P}_{N,t}$, while the remaining fraction α keeps its previous price indexed by steady-state inflation Π. A firm that is allowed to reset its price at time t chooses $\tilde{P}_{N,t}$ to maximize the expected present value of future profits while its price remains in effect:

$$\max_{\tilde{P}_{N,t}} E_t \sum_{k=0}^{\infty} \alpha^k M_{t,t+k} \Big[P_{N,t+k}(i) Y_{N,t+k}(i) - W_{t+k} H_{t+k}(i) \Big]. \tag{10}$$

The objective reflects that with probability α^k the same price remains effective k periods ahead. Profits at horizon $t+k$ are given by revenues $P_{N,t+k}(i)Y_{N,t+k}(i)$ minus wage costs $W_{t+k}H_{t+k}(i)$, and are discounted by the household's stochastic discount factor $M_{t,t+k}$ since firms are household-owned. At horizon $t+k$, if the firm is still constrained by the last reset, its nominal price is given by $P_{N,t+k}(i) = \Pi^k \tilde{P}_{N,t}$. The individual demand curve implied by CES aggregation is $Y_{N,t+k}(i) = \left(\frac{\Pi^k \tilde{P}_{N,t}}{P_{N,t+k}}\right)^{-\theta} Y_{N,t+k}$, and production is linear in labour so that $H_{t+k}(i) = Y_{N,t+k}(i)/A_{t+k}$.

The optimal reset price satisfies

$$\frac{\tilde{P}_{N,t}}{P_{N,t}} = \mu \frac{E_t \sum_{k=0}^{\infty} (\alpha\beta)^k \frac{1}{A_{t+k}} \frac{W_{t+k} Y_{N,t+k}}{P_{t+k} C_{t+k}} \left(\frac{P_{N,t}}{P_{N,t+k}} \Pi^k\right)^{-\theta}}{E_t \sum_{k=0}^{\infty} (\alpha\beta)^k \frac{P_{N,t+k} Y_{N,t+k}}{P_{t+k} C_{t+k}} \left(\frac{P_{N,t}}{P_{N,t+k}} \Pi^k\right)^{1-\theta}}, \tag{11}$$

with $\mu = \frac{\theta}{\theta-1}$ the steady-state markup.

Finally, the aggregate price index under Calvo pricing evolves according to

$$P_{N,t}^{1-\theta} = (1-\alpha)\, \tilde{P}_{N,t}^{1-\theta} + \alpha\, \Pi^{1-\theta}\, P_{N,t-1}^{1-\theta}. \tag{12}$$

Equations (11) and (12) are the key building blocks for deriving the New Keynesian Phillips curve. Detailed derivations can be found in Appendix A.2.

Government. Consider an integrated budget constraint of the central bank and treasury. Nothing hinges on which institution issues the convenient claims: any issuer whose liabilities carry the convenience, treasury, central bank, or a third agency, supports the same equilibrium, and the consolidated constraint is the standard device. The central-bank framing is kept because central banks are the institutions that conduct balance-sheet operations in practice. The government issues debt Q_t^s, buys assets from the domestic market, B_t^g, as well as from the foreign market, F_t^g, and taxes households in country H. The flow budget constraint is given by

$$B_t^g + S_t F_t^g - Q_t^s = (1 + i_{t-1})B_{t-1}^g + S_t(1 + i_{t-1}^{R*})F_{t-1}^g - (1 + i_{t-1}^R)Q_{t-1}^s + T_t. \tag{13}$$

2.3 Country F

Households. Further simplifying assumptions are used to model country F, representing the rest of the world. First, the model abstracts from non-traded goods. Second, utility is assumed linear with respect to the traded good. Preferences in country F are given by

$$E_t \sum_{s=0}^{\infty} \beta^s \left[C_{T,t+s}^* + \theta_{t+s}^* V\left(\frac{Q_{t+s}^*}{S_{t+s} P_{T,t+s}^*} \right) \right],$$

in which variables have the same definition as before but with the *-superscript indicating foreign-country variables. The price level of the traded good in units of foreign currency is denoted by $P_{T,t}^*$ and S_t is the nominal exchange rate defined as the price for one unit of foreign currency f in units of home currency h. Foreign households derive utility from convenience-bearing securities issued in the home country, where Q^* denotes households' holdings in units of currency h. The budget constraint in units of currency f is given by

$$F_t^* + \frac{Q_t^*}{S_t} + P_{T,t}^* C_{T,t}^* = (1 + i_{t-1}^{R^*})F_{t-1}^* + (1 + i_{t-1}^R)\frac{Q_{t-1}^*}{S_t} + P_{T,t}^* Y_{T,t}^* - T_t^*, \tag{14}$$

where F^* are risk-free securities denominated in units of currency f, which provide no convenience benefits and carry the risk-free policy rate i^{R^*}.

The optimal asset pricing condition for F_t^* is given by

$$1 = E_t \left[M_{t+1}^* (1 + i_t^{R^*}) \right], \tag{15}$$

where $M_{t+1}^* \equiv \frac{\beta}{\Pi_{T,t+1}^*}$ is the stochastic discount factor and $\Pi_{T,t+1}^* \equiv \frac{P_{T,t+1}^*}{P_{T,t}^*}$ denotes gross inflation in country F. The asset pricing condition for the convenient asset is

$$1 = L_t^* + E_t \left[M_{t+1}^* \frac{S_t}{S_{t+1}} (1 + i_t^R) \right], \tag{16}$$

where $L_t^* = \theta_t^* V_q(q_t^*)$ is the non-pecuniary value of convenience benefits and $q_t^* = \frac{Q_t^*}{P_{T,t}}$ denotes real convenient asset holdings.

Government. Conceptually, governments in both countries are alike. The only difference is that the foreign government issues ordinary debt F_t^s and thus has the following flow budget constraint:

$$F_t^s = (1 + i_{t-1}^{R^*}) F_{t-1}^s - T_t^*. \tag{17}$$

2.4 Asset Market Relationships

Combining the household optimality conditions yields a set of asset-market relationships that characterize the convenience yield and the departures from standard UIP. A household in country H is indifferent between investing in either B_t or Q_t. Thus equating (5) with (6) yields

$$L_t = (i_t - i_t^R) E_t \left[M_{t+1} \right]. \tag{18}$$

Equation (18) illustrates that L_t corresponds to the convenience yield: because both L_t and M_{t+1} are non-negative, the interest spread $i_t - i_t^R$ is non-negative.

The formulation does not microfound which service the holder values. Safety, liquidity, and collateral value enter the problem identically, and to first order any preference for the asset's services generates the same wedge: what is priced is the willingness to hold the asset at a pecuniary discount, not its source. This is the aggregation the empirical convenience-yield literature relies on from Krishnamurthy and Vissing-Jorgensen (2012) onward. Nothing below depends on why the world wants the asset, only on how much.

A household in H that is indifferent between investing in either the domestic (Q_t) or the foreign government (F_t) liability gives rise to an uncovered interest rate parity (UIP) relationship. Combining equations (6) and (7) gives

$$\frac{1 + i_t^R}{1 + i_t^{R^*}} \left[1 + \frac{L_t + \psi \left(\frac{F_t}{P_{T,t}^*} - \bar{f} \right)}{1 - L_t} \right] = \tilde{E}_t \left[\frac{S_t}{S_{t+1}} \right]. \tag{19}$$

Standard UIP does not hold even if $L_t = 0$ under the risk-neutral measure. A deviation that depends on $\frac{F_t}{P_{T,t}^*}$ remains even if convenience demand is fully satiated. Closing this UIP deviation would also require $\psi = 0$.

Furthermore, a household in H being indifferent between the domestic ordinary bond (B_t) and the foreign policy bond (F_t) yields another UIP relationship:

$$\frac{1+i_t}{1+i_t^{R*}}\left[1+\psi\left(\frac{F_t}{P_{T,t}^*}-\bar{f}\right)\right]=\tilde{E}_t\left[\frac{S_t}{S_{t+1}}\right]. \tag{20}$$

The adjustment cost prevents UIP from holding, and the deviation closes when $\psi = 0$. Neither B_t nor F_t provides convenience benefits, but B_t is privately issued at rate i_t, whereas F_t is a government bond returning the foreign policy rate i_t^{R*}.

A household in country F is indifferent between investing in either F_t^* or Q_t^*:

$$\frac{1+i_t^{R^*}}{1+i_t^R}\left(1-L_t^*\right)=\tilde{E}_t^*\left[\frac{S_t}{S_{t+1}}\right]. \tag{21}$$

There is also a connection between the convenience benefits in both countries. Equating equations (6) and (16) links the convenience benefits between countries:

$$L_t-L_t^*=(1+i_t^R)E_t\left[M_{t+1}^*\frac{S_t}{S_{t+1}}-M_{t+1}\right]. \tag{22}$$

Note that if international asset markets are complete, then the convenience benefits are equalized across countries ($L_t = L_t^*$).

A similar indifference equation between countries for asset F_t follows from linking (7) and (15):

$$E_t\left[M_{t+1}\frac{S_{t+1}}{S_t}\right]-\frac{\psi\left(\frac{F_t}{P_{T,t}^*}-\bar{f}\right)}{(1+i^{R^*})}=E_t[M_{t+1}^*]. \tag{23}$$

Note that the exchange-rate-adjusted SDFs now differ.

3 Equilibrium

This section characterizes the equilibrium conditions. An equilibrium is a sequence of prices and allocations such that households and firms optimize, governments satisfy their budget constraints, and all goods and asset markets clear.

The following market clearing conditions need to be satisfied:

$$Y_{T,t} + Y^*_{T,t} = C_{T,t} + C^*_{T,t}, \tag{24}$$

$$Y_{N,t} = C_{N,t}, \tag{25}$$

$$B_t + B^g_t = 0, \tag{26}$$

$$F^s_t = F^*_t + F_t + F^g_t, \tag{27}$$

$$Q^s_t = Q_t + Q^*_t. \tag{28}$$

The first two equations clear the traded and non-traded goods markets. The remaining conditions clear the markets for the domestic ordinary bond, the foreign bond, and the domestic convenient asset. The relative price of non-traded goods is pinned down by the expenditure shares

$$\frac{P_{N,t}Y_{N,t}}{P_{T,t}C_{T,t}} = \frac{\gamma}{1-\gamma}, \tag{29}$$

and for traded goods the law of one price $P_{T,t} = S_t P^*_{T,t}$ holds. Together, these two equations connect the non-traded sector to the exchange rate.

Using the household Euler equation for the domestic ordinary bond in country H (5) together with non-traded goods market clearing, domestic aggregate demand can be written as

$$\frac{1}{Y_{N,t}} = \beta E_t \left[\frac{(1+i_t)}{\Pi_{N,t+1}} \frac{1}{Y_{N,t+1}} \right]. \tag{30}$$

Equation (30) connects output in the non-tradable sector to its future value and the related real interest rate.[4] The convenient asset Euler equation (6) can be expressed as

$$\frac{C_{T,t}\theta_t}{1-\gamma} V_q \left(\frac{Q_t}{P_{T,t}} \right) = 1 - \frac{1+i^R_t}{1+i_t}, \tag{31}$$

showing that the demand for liquidity depends positively on tradable output and negatively on the spread between the ordinary bond and the convenient asset.[5] The Euler equation for the foreign bond (7) gives

[4] Note that i is not the policy rate.

[5] A higher spread between the ordinary bond rate and the policy rate raises the required marginal convenience yield and, given concavity of $V(\cdot)$, lowers desired convenient asset holdings.

$$\frac{1}{C_{T,t}}\left[1+\psi\left(\frac{F_t}{P^*_{T,t}}-\bar{f}\right)\right]=\beta E_t\left[\frac{(1+i_t^{R*})}{\Pi_{T,t+1}}\frac{1}{C_{T,t+1}}\frac{S_{t+1}}{S_t}\right]. \tag{32}$$

This condition governs the domestic household's demand for foreign bonds. The portfolio adjustment cost introduces a wedge in the return on foreign assets. As a result, foreign-asset positions, tradable consumption, and expected exchange-rate movements are jointly determined. This completes the aggregate demand side of the model for the country H.

Aggregate supply in the non-traded sector in country H is given by the following equations:

$$\left[\frac{1-\alpha\left(\frac{\Pi_{N,t}}{\Pi}\right)^{\theta-1}}{1-\alpha}\right]^{\frac{1}{\theta-1}}=\frac{D_t}{K_t}, \tag{33}$$

$$D_t=\gamma+\alpha\beta E_t\left[\left(\frac{\Pi_{N,t+1}}{\Pi}\right)^{\theta-1}D_{t+1}\right], \tag{34}$$

$$K_t=\mu\left(\frac{Y_{N,t}}{A_t}\right)+\alpha\beta E_t\left[\left(\frac{\Pi_{N,t+1}}{\Pi}\right)^{\theta}K_{t+1}\right]. \tag{35}$$

These equations determine non-traded inflation as a function of marginal costs and expected future pricing decisions.

On the foreign side, households have linear utility over traded good and the foreign Euler equation (15) implies

$$1=\beta E_t\left[\frac{(1+i_t^{R^*})}{\Pi^*_{T,t+1}}\right], \tag{36}$$

in which the relevant rate is the policy rate $i_t^{R^*}$. Using the asset-pricing condition (16) we obtain the demand for liquid securities in country F, given by

$$\theta^*_t V_q\left(\frac{Q^*_t}{S_t P^*_{T,t}}\right)=1-\beta E_t\left[\frac{S_t}{S_{t+1}}\frac{1}{\Pi^*_{T,t+1}}\right](1+i_t^R). \tag{37}$$

This is the equation through which the flight-to-safety shock enters the model. An increase in θ^*_t raises the foreign marginal value of domestic liquidity.

Finally, aggregating the home budget constraint and imposing asset-market clearings

yields the current account. In nominal terms, it is given by

$$\begin{aligned}
&S_t[F_t + F_t^g - (1 + i_{t-1}^{R*})(F_{t-1} + F_{t-1}^g)] \\
&- [Q_t^* - (1 + i_{t-1}^R)Q_{t-1}^*] + P_{T,t}\frac{\psi}{2}\left(\frac{F_t}{P_{T,t}^*} - \bar{f}\right)^2 \\
&= P_{T,t}[Y_{T,t} - C_{T,t}].
\end{aligned}$$

Dividing by $P_{T,t}$, using the law of one price and defining $q_t^* \equiv \frac{Q_t^*}{P_{T,t}}$ and $f_t \equiv \frac{F_t}{P_{T,t}^*}$ gives the current account in real terms

$$\begin{aligned}
f_t + f_t^g - q_t^* = &\frac{(1 + i_{t-1}^{R*})}{\Pi_{T,t}^*}(f_{t-1} + f_{t-1}^g) \\
&- \frac{(1 + i_{t-1}^R)}{\Pi_{T,t}}q_{t-1}^* - \frac{\psi}{2}\left(f_t - \bar{f}\right)^2 \\
&+ Y_{T,t} - C_{T,t}. \qquad (38)
\end{aligned}$$

The left-hand side is the change in the home economy's net foreign asset position. The right hand side records interest payments on inherited positions, portfolio adjustment costs, and the trade balance. This equation is central for the transmission mechanism: when foreign households demand more domestic liquidity, the resulting capital inflow must be absorbed through foreign-asset positions, tradable consumption, and the exchange rate.

3.1 Log-Linearized General Equilibrium

This section derives the log-linearized equilibrium of the model above. Hatted variables denote log deviations from their steady-state values, solved in Appendix A.3.[6] The log-linearized dynamics of the competitive equilibrium are summarized by the following equations:

$$\hat{\pi}_{N,t} = \kappa \hat{Y}_{N,t} + \beta E_t \hat{\pi}_{N,t+1} \tag{39}$$

$$\hat{Y}_{N,t} = z E_t \hat{Y}_{N,t+1} - z\left(\hat{\imath}_t^R - E_t \hat{\pi}_{N,t+1}\right) + \frac{z}{\phi_2}\hat{q}_t - (1-z)\left(\hat{p}_{N,t} + \hat{\theta}_t\right) \tag{40}$$

$$\hat{C}_{T,t} = E_t \hat{C}_{T,t+1} + \psi f \hat{f}_t - \left(\hat{\imath}_t^{R*} - E_t \hat{\pi}_{T,t+1} + E_t \Delta \hat{S}_{t+1}\right) \tag{41}$$

$$\hat{q}_t^* = \phi_1^* \hat{\theta}_t^* + \varphi_2^*\left(E_t \Delta \hat{S}_{t+1} + \hat{\imath}_t^{R^*} - \hat{\imath}_t^R\right) \tag{42}$$

$$\begin{aligned}\hat{Y}_{T,t} - s_c \hat{C}_{T,t} = {} & s_f\left(\hat{f}_t - \frac{1}{\beta}\left[\hat{f}_{t-1} + \hat{\imath}_{t-1}^{R*} - \hat{\pi}_{T,t}^*\right]\right) \\ & + s_f^g\left(\hat{f}_t^g - \frac{1}{\beta}\left[\hat{f}_{t-1}^g + \hat{\imath}_{t-1}^{R*} - \hat{\pi}_{T,t}^*\right]\right) \\ & - s_q\left(\hat{q}_t^* - \frac{(1-\nu)}{\beta}\left[\hat{q}_{t-1}^* + \hat{\imath}_{t-1}^R - \hat{\pi}_{T,t}\right]\right)\end{aligned} \tag{43}$$

$$\hat{q}_t^s = \frac{q}{q+q^*}\hat{q}_t + \frac{q^*}{q+q^*}\hat{q}_t^* \tag{44}$$

$$\hat{f}_t^s = \frac{f^*}{f^*+f+f^g}\hat{f}_t^* + \frac{f}{f^*+f+f^g}\hat{f}_t + \frac{f^g}{f^*+f+f^g}\hat{f}_t^g \tag{45}$$

$$\hat{p}_{N,t} = \hat{C}_{T,t} - \hat{Y}_{N,t} \tag{46}$$

$$\hat{p}_{N,t} = \hat{p}_{N,t-1} + \hat{\pi}_{N,t} - \hat{\pi}_{T,t} \tag{47}$$

$$\hat{\pi}_t = (1-\gamma)\hat{\pi}_{T,t} + \gamma \hat{\pi}_{N,t} \tag{48}$$

$$\Delta \hat{S}_t = \hat{\pi}_{T,t} - \hat{\pi}_{T,t}^* \tag{49}$$

$$\Delta \hat{\varepsilon}_t = \gamma\left(\hat{\pi}_{T,t} - \hat{\pi}_{N,t}\right) \tag{50}$$

$$\hat{C}_t = \gamma \hat{Y}_{N,t} + (1-\gamma)\hat{C}_{T,t}, \tag{51}$$

where $p_{N,t} = \frac{P_{N,t}}{P_{T,t}}$ is the relative price of the non-traded good, $\varepsilon_t = \frac{S_t P_{T,t}^*}{P_t}$ is the real exchange rate, and κ is the standard New Keynesian Phillips Curve coefficient. The parameters $\phi_1 \equiv -\frac{V_q(q)}{qV_{qq}(q)}$ and $\phi_1^* \equiv -\frac{V_q(q^*)}{q^* V_{qq}(q^*)}$ are the elasticities of domestic and foreign liquidity demand with respect to the convenience yield. They determine how strongly convenient-asset demand responds to changes in the marginal convenience yield. The parameter $\phi_2 \equiv \phi_1 \frac{1-\nu}{\nu}$ governs the sensitivity of domestic liquidity demand to the spread between the ordinary bond and the convenient asset. $z \equiv \frac{\phi_2}{\phi_1+\phi_2}$ determines the relative

[6] For any variable X_t, $\hat{x}_t \equiv x_t - x = \log(X_t) - \log(X) \simeq \frac{X_t - X}{X}$, where $x \equiv \log(X)$ and X denotes the steady-state value of X_t.

weight of the interest-rate and liquidity terms in the domestic aggregate-demand. Finally, $\varphi_2^* \equiv \phi_1^* \frac{V_q(q^*)-1}{V_q(q^*)}$ governs the response of foreign liquidity demand to expected return differential.

The linearized equilibrium contains 13 equations in 19 endogenous variables: $\left\{\hat{\pi}_{N,t}, \hat{Y}_{N,t}, \hat{\imath}_t^R, \hat{q}_t, \hat{q}_t^*, \hat{q}_t^s, \hat{p}_{N,t}, \Delta\hat{S}_t, \hat{C}_{T,t}, \hat{\pi}_{T,t}, \hat{\pi}_{T,t}^*, \hat{\pi}_t, \hat{\imath}_t^{R*}, \Delta\hat{\epsilon}_t, \hat{f}_t, \hat{f}_t^*, \hat{f}_t^g, \hat{f}_t^s, \hat{C}_t\right\}$. Policy closes the system in two steps. First, four instruments receive four rules: the government in H sets $\hat{\imath}_t^R$ by the Taylor rule and chooses the paths of $\hat{q}_t^s$ and $\hat{f}_t^g$, and the government in F chooses $\hat{f}_t^s$. Second, foreign monetary policy stabilizes traded inflation, $\hat{\pi}_{T,t}^* = 0$, implemented with $\hat{\imath}_t^{R^*} = 0$, which fixes the two remaining foreign variables. This leaves 17 equations in 17 unknowns, and the system is closed.

4 Calibration

To solve the equilibrium system from subsection 3.1, the model is calibrated at quarterly frequency. The strategy combines standard parameters from the open-economy New Keynesian literature with Switzerland-specific moments, and anchors the steady state to the acute phase of the Global Financial Crisis. All steady-state quantities are expressed as ratios to quarterly nominal GDP to ensure stationarity and comparability across series. Interest and inflation rates are reported at annual rates throughout.

The calibration uses three kinds of inputs. Measured moments come from Swiss data: the policy-rate anchor, the convenience-yield proxy, and the asset positions of residents, nonresidents and the central bank. Literature parameters cover the discount factor, the Phillips-curve slope, the share of non-traded goods, the demand elasticities, and the portfolio cost. One targeted moment remains: the size of the shock, matched to the observed appreciation. Two further choices are a normalization, $Y = 1$, and an assumption, the rest of the world's 95% share of the foreign bond.

The steady state around which the model is linearized is the crisis regime of the acute phase, not a long-run average. A linear model is a local approximation, and the regime this paper studies is a different point from normal times: the convenience yield was ten times its normal level. Section 5.2 repeats the experiment at the normal-times level and in between.

4.1 Steady State Identification

The calibration solves for the steady state recursively, using the observed policy rate i^R as the anchor. The remaining steady-state objects follow from the equilibrium relationships in subsection A.3. The convenience yield ν appears as the spread that separates the ordinary bond rate i from the policy rate.

The policy rate i^R is anchored to the acute GFC episode, September 2008 to June 2009, when the measured convenience yield peaked while the SNB policy rate was still positive. Over this period, the mean policy rate is $i^R = 1.0212\%$ annual. The choice is deliberately conservative. For most of the sample, the Swiss policy rate was at or near zero, so calibrating on a near-zero rate would place the economy at the edge of the zero lower bound from the outset. Fixing i^R at the clearly positive level of the acute phase instead makes the bound harder to reach, so that when a flight-to-safety shock nonetheless drives the policy rate to the bound (Section 5.3), the result is not an artefact of a steady state chosen close to zero.

The convenience yield is the spread between the ordinary bond rate and the policy

rate, the pecuniary return investors forgo for the asset's convenience services,

$$\nu \equiv 1 - \frac{1 + i^R}{1 + i} \approx i - i^R, \tag{52}$$

so that, given i^R and ν, the ordinary bond rate is $i = i^R + \nu$. Inflation follows from the domestic Euler equation (78),

$$\Pi_N = \beta\,(1 + i), \qquad \pi = \Pi_N - 1. \tag{53}$$

With $i^R = 1.0212\%$ annual and $\nu = 300$ basis points, the ordinary bond rate is $i = 4.02\%$ annual, the spread $i - i^R = 3.00\%$ equals ν by construction, and inflation comes out at 2.00% annual, at the upper edge of the SNB's 0 to 2 percent price-stability range. The implied steady-state real policy rate is mildly negative ($i^R - \pi = -0.98\%$), a feature of the model rather than an anomaly: the convenient asset earns a below-inflation return precisely because it yields convenience services, while the real return relevant for saving, $i - \pi = 2.02\%$, is positive and standard. Table 1 collects the steady-state rates.

Table 1: Steady state rates (annual)

	Value (% annual)	Source
Policy rate i^R	1.021	Acute GFC mean (2008-09 to 2009-06)
Convenience yield ν	3.000	GFC peak
Ordinary bond rate i	4.021	Derived from (52)
Inflation π	2.001	Derived from (53)

4.2 The Convenience Yield

The convenience yield ν is the central structural parameter of the model. It plays a dual role: it sets the steady-state spread in (52), and it governs the strength of the convenience channel in aggregate demand through $\phi_2 = \phi_1(1 - \nu)/\nu$ and $z = 1 - \nu$. A higher convenience yield lowers ϕ_2 and so raises the coefficient $z/\phi_2 = \nu/\phi_1$ on the convenient asset in the AD equation (40): the larger the wedge, the more forcefully foreign demand for the convenient asset is transmitted to the domestic economy. The coefficient rises linearly in the wedge, from 0.014 at the normal-times premium to 0.15 at the crisis peak, an order of magnitude, so the same flight-to-safety shock has very different consequences in tranquil and in stressed states.

The convenience yield ν is calibrated to represent the acute crisis state the model is designed to study. Its magnitude is disciplined by Swiss money-market data: the spread between three-month CHF Libor and three-month Swiss Confederation money market

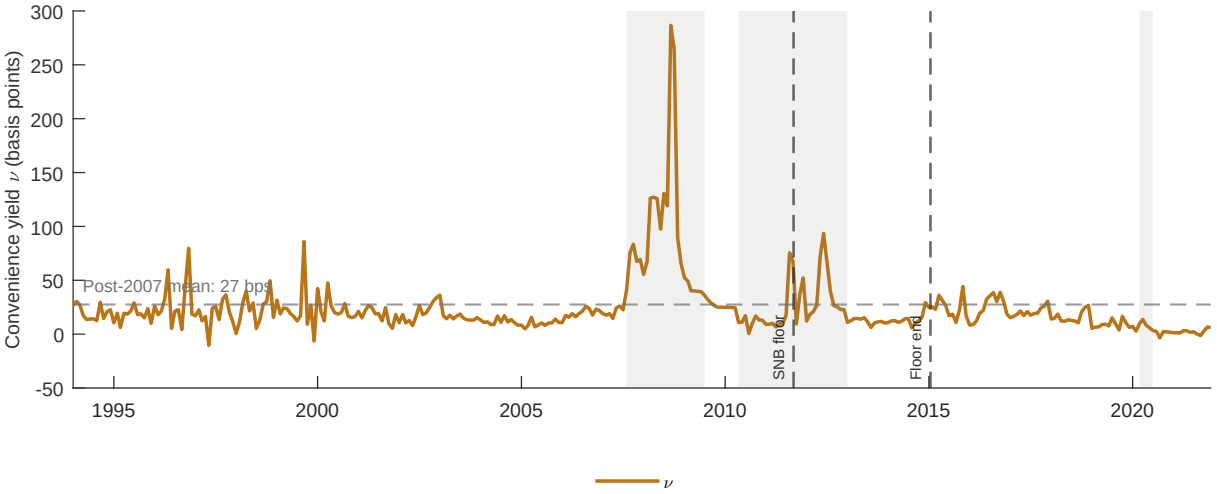


Figure 3: Convenience yield ν estimated from the spread between three-month CHF Libor and Swiss Confederation money market claims, in basis points. Dashed lines indicate the post-2007 and full-sample means. Vertical lines mark the GFC, the introduction and removal of the EUR/CHF floor, and the COVID-19 episode.

debt register claims (GMBF) serves as a high-stress proxy for the convenience yield, computed as in (52).[7] Figure 3 plots the resulting series. The premium is modest in normal times, with a post-2007 mean of 27 basis points, but spikes sharply in risk-off episodes, reaching roughly 287 basis points at the height of the crisis in late 2008. The series is illustrative rather than a precise estimate of the structural parameter: it locates the order of magnitude of the convenience yield and, in particular, the scale it reaches in acute stress.

The proxy is imperfect in both directions. CHF Libor is meant to stand for the rate on claims without convenience services, but it also carries credit, counterparty, term-premium, and market-segmentation components, which push the measured spread above the convenience yield. During the GFC the Libor market was itself thin, so Libor carries its own liquidity discount and the spread nets out part of the very premium it is meant to capture, and the Confederation claims are not the most extreme safe-haven instrument available to global investors. The series is therefore an imperfect high-stress proxy for the convenience yield rather than a structural estimate.

The baseline calibration sets ν to 300 basis points annual, in line with the crisis peak in the data. The quarterly model uses the per-quarter equivalent $\nu/4 = 75$ basis points, so the steady-state rates are exactly consistent with the wedge the quarterly Euler equations see. The interpretation is deliberately conditional: the baseline characterizes a stressed regime[8] rather than the economy in normal times.

Section 5.2 examines how the transmission mechanism changes across convenience yields ranging from normal to extreme-stress levels, $\nu \in \{27, 150, 300, 500\}$ basis points.

[7] I thank Angelo Ranaldo for this suggestion.

[8] The state in which a flight-to-safety shock is most consequential.

Table 2: Asset positions and their data counterparts

	Data	CHF bn	GDP ratio
Resident holdings q	M3	638	4.22
Nonresident holdings q^*	IIP liabilities: debt securities, customer deposits, deposits at the SNB	243	1.61
Supply q^s	$q + q^*$	881	5.82
Central bank position f^g	SNB foreign currency investments	49	0.32
Private net position f	NIIP $- f^g + q^*$	905	5.99

Four-quarter averages over 2007Q3 to 2008Q2, in CHF bn and as ratios to quarterly nominal GDP. Source: SNB data portal (Swiss National Bank, 2025a,b, 2026a,b); ratios are the author's calculation.

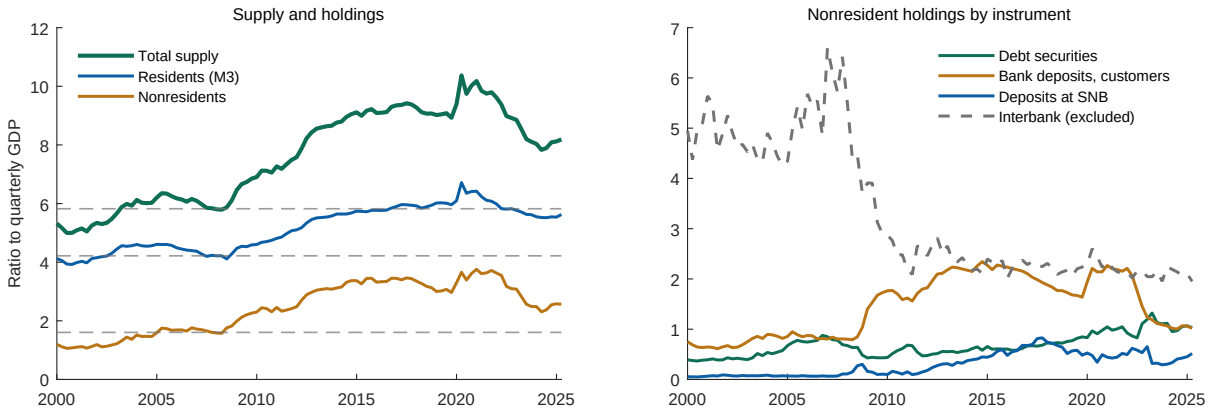
The lowest value is the post-2007 normal-times average and the highest exceeds the GFC peak. As ν rises, the amplification coefficient increases and the domestic response to a given shock strengthens, so that the mechanism is operative precisely in the stressed states the paper studies.

4.3 Portfolio Positions

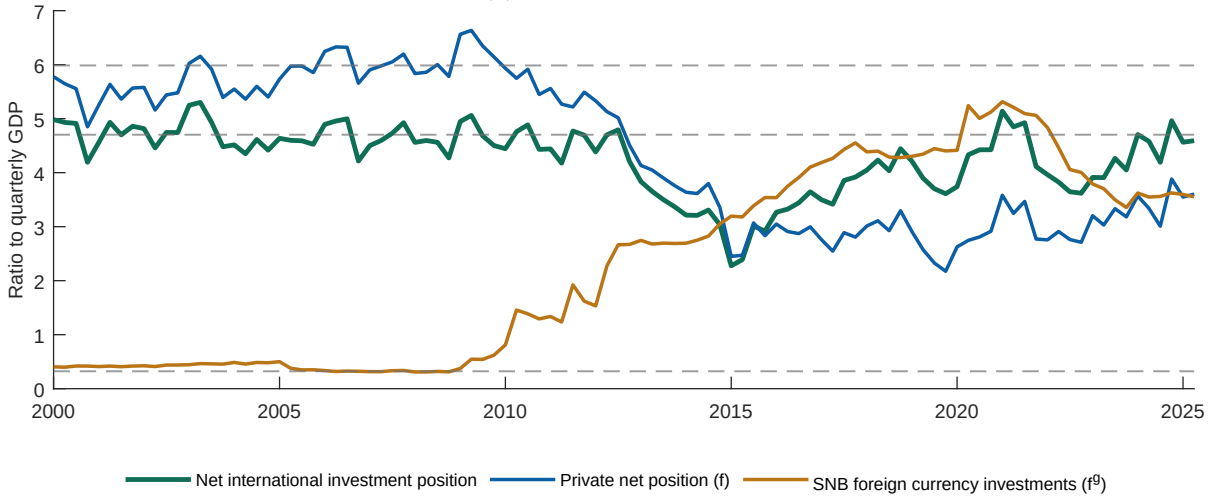
The model needs to pin down four stocks: resident holdings of the convenient asset q, nonresident holdings q^*, the private foreign position f, and the central bank's foreign position f^g. All four are measured from Swiss data. Each series is divided by nominal GDP of the same quarter, and the steady-state value is the average of that ratio over the four quarters before the shock, 2007Q3 to 2008Q2. Table 2 lists the series and Figure 4 plots them.

The convenient asset is a safe nominal claim in Swiss francs, and the data counterparts are the positions with that property. On the resident side q is M3: the currency held by resident non-banks and their franc deposits at banks, time deposits included.[9] On the nonresident side q^* is the sum of three liability positions of the international investment position (IIP): debt securities issued by Swiss residents and held abroad, deposits of

[9] M3 is currency plus bank deposits, time deposits with the money-market paper and securities liabilities banks owe to their customers included. It is broader than the model's asset, and I use it on purpose. In the model the convenient asset is the liability of the consolidated government, in Switzerland the central bank's reserves and currency. Households cannot hold reserves directly. What they hold is the bank deposit, a claim on reserves, convertible on demand and backed by them, so I read the deposit as the household-facing form of the central bank's liability. The franc deposit is also where the safe-haven demand shows up: in the acute phase nonresident customer deposits rose by half, nonresident sight deposits at the SNB doubled, and the inflow ended on the SNB's balance sheet when the banks parked it there. What the measure does not resolve is that the model's supply is fixed by policy, while banks can create deposits.



(a) The convenient asset



(b) Foreign positions

Figure 4: Asset positions, ratios to quarterly GDP. Panel (a), left: supply q^s, resident holdings q (M3) and nonresident holdings q^*, with the 2007Q3 to 2008Q2 averages dashed. Panel (a), right: nonresident holdings by instrument, and the interbank position that is excluded. Panel (b): net international investment position, private net position f and SNB foreign currency investments f^g. Source: SNB data portal (Swiss National Bank, 2025a,b, 2026a,b); q^*, f and the ratios are the author's calculation.

nonresident customers at banks in Switzerland, and sight deposits of nonresidents at the SNB, the central bank's own liability to foreigners. Direct investment, portfolio equity, derivatives and interbank deposits are not included: the first three carry no convenience yield, and interbank deposits are the funding of the banking system rather than a holding of the asset. M3 covers residents only, so q and q^* do not overlap.[10]

The central bank's foreign position f^g is the SNB's foreign currency investments, plotted relative to quarterly GDP in Figure 4b. The private foreign position f is the net position of the private sector in every instrument other than the convenient asset, defined so that $f + f^g - q^*$ equals the country's net international investment position (NIIP). The model has one foreign instrument, so direct investment and equity positions abroad and at home are in it.

[10]The IIP does not split liabilities by currency, so q^* contains a foreign-currency component, mostly bonds issued by Swiss banks.

The acute phase itself, 2008Q3 to 2009Q2, is not used because it contains the inflow the model generates: between the two windows nonresident holdings rose by 20%, from CHF 243 bn to 291 bn, customer deposits by half and deposits at the SNB twofold, while M3 rose by 3% (Figure 4a).

The measured positions give a foreign share of the convenient asset of $q^*/q^s = 28\%$ and a supply of 5.8 quarters of GDP, about 1.5 times annual GDP, the stock relative to which the balance-sheet operations of Section 5.4 are expressed. For the foreign bond market I assume that the rest of the world holds 95%, $f^*_{ss} = 19\,(f_{ss} + f^g_{ss}) = 119.8$. The share enters only as a scale and the results do not depend on it.

4.4 Remaining Parameters

The parameters β, γ, and κ are taken from the literature. Following Rudolf and Zurlinden (2014), I set the discount factor $\beta = 0.995$ and the non-tradable share $\gamma = 0.6$. The slope of the New Keynesian Phillips curve is set to a value commonly used in the literature (Galí, 2015), $\kappa = 0.05$. The steady-state output is normalized to $Y = 1$, so that $Y_T = 1 - \gamma$. The elasticities ϕ_1 and ϕ^*_1 are the home and foreign interest elasticities of demand for convenience services. The foreign elasticity is set to $\phi^*_1 = 0.5$, the reference value of the long-run money-demand literature (Lucas (2000), Benati et al. (2021)). The home elasticity is set to $\phi_1 = 0.05$, the value implied at the acute-window anchor by the semi-log money demand of Ball (2001), and inside the 95 percent confidence set that Benati et al. (2021) estimate for Switzerland on quarterly data, whose log-log sample ends in 2011 precisely because money demand flattens once rates approach zero.[11]

Given ϕ_1 and the convenience yield ν, the structural coefficients in the aggregate demand block follow from their definitions,

$$\phi_2 = \phi_1 \frac{1-\nu}{\nu}, \qquad z = \frac{\phi_2}{\phi_1 + \phi_2} = 1 - \nu, \qquad \varphi^*_2 = \phi^*_1 \frac{\nu - 1}{\nu},$$

where the last identity uses $\nu = V_q(q^*)$ from the foreign first-order condition (82). Note that $z = 1 - \nu$ depends on the convenience yield alone. Tradable consumption C_T is recovered from the steady-state current account (83). The portfolio adjustment cost ψ is set as in Schmitt-Grohé and Uribe (2003). Table 3 collects the full calibration.

[11] Ball (2001) estimates the semi-log specification $\ln m = c - \xi i$ and reports $\xi \approx 0.05$ with the interest rate in percent, equivalently $\xi = 5$ with the rate in decimals. Under this form the local log-log elasticity at rate level i is ξi, identical in either convention: $5 \times 0.0102 = 0.05 \times 1.0212 = 0.051$, which rounds to the calibrated value. Values much below the chosen one would leave the Swiss confidence set. Appendix C.3 reports sensitivity over $\phi_1 \in \{0.04, 0.075, 0.3\}$: the qualitative results, including the binding lower bound, survive across the range, while the calibrated value disciplines the size of the portfolio flows. Results are insensitive to ϕ^*_1 once the shock is sized to the observed appreciation.

Table 3: Calibration summary

Parameter / Variable		Value
Discount factor	β	0.995
NKPC slope	κ	0.050
Non-tradable share	γ	0.600
Convenience yield (annual)	ν	0.030
Structural wedge (quarterly)	$\nu/4$	0.0075
Home money demand elasticity	ϕ_1	0.050
Foreign money demand elasticity	ϕ_1^*	0.500
AD convenience coefficient	ϕ_2	6.617
Forward-looking weight	z	0.993
Foreign convenience coefficient	φ_2^*	-66.167
Portfolio adjustment cost	ψ	7.4×10^{-4}
Nonresident holdings of convenient asset	q^*	1.61
Resident holdings of convenient asset	q	4.22
Private foreign position	f	5.99
Rest-of-world holdings of foreign asset	f^*	119.85
Central bank foreign position	f^g	0.32
Consumption ratio	s_C	1.089
Convenient-asset ratio	s_q	4.01
Private asset ratio	s_f	14.96
Government asset ratio	s_{fg}	0.81

4.5 The Shock

The flight-to-safety shock is an increase in the foreign preference shifter θ_t^*, following an AR(1) process with persistence $\rho_{\theta^*} = 0.95$. The remaining exogenous processes have persistence 0.9. Monetary policy follows the Taylor rule with $\phi_\pi = 1.5$ and $\phi_y = 0.1$, which satisfies the Taylor principle and ensures determinacy under the baseline calibration.

The size of the shock is disciplined in Section 5.1 to match the observed appreciation of the franc over the anchor window, 5.9 percent on the broad nominal effective exchange rate from 2008Q3 to 2009Q2, and is then held fixed across all experiments. The shock, the policy rate, and the wedge are disciplined by the same episode, and the stocks by the position the economy held when it began. The calibration is a Swiss-informed crisis benchmark.

5 Results

5.1 The Baseline Economy

Figure 5 displays the impulse responses to a flight-to-safety shock under the baseline calibration. The shock is an increase in the foreign preference shifter θ_t^*, with its size calibrated to reproduce the trade-weighted appreciation of the franc over the acute phase of the global financial crisis. It generates a peak appreciation of 5.9% in the level of the nominal exchange rate, matching the broad nominal effective exchange rate from 2008Q3 to 2009Q2.[12] As the model is linear, all responses scale proportionally with the shock, so its size governs the magnitudes but not the mechanism. All variables are reported as percent deviations from their steady state. For interest rates and inflation, these are reported in annualized terms as levels, relative to their steady-state values.[13] The figure reports the responses of key variables. The full model, over 120 quarters, is shown in Figure 14 in Appendix C.1.

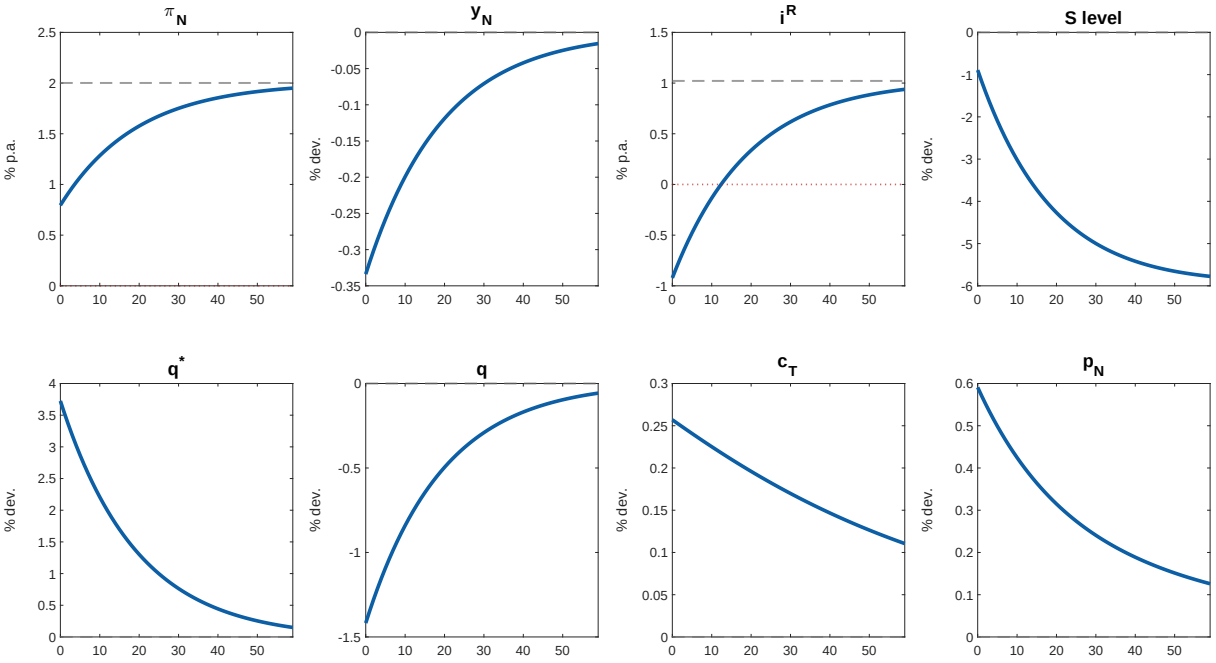


Figure 5: Impulse responses to a flight-to-safety shock under the baseline calibration. All variables are percent deviations from their steady state. Interest rate and inflation are reported in annualized terms relative to their steady-state values, to show when they turn negative.

The figure shows that a single shock generates the joint pattern of the safe-haven narrative: a cumulating appreciation, a disinflation, a policy rate cut, and a contraction in non-traded output. The rest of this subsection traces the mechanism.

[12] The target is conservative by construction: the SNB's first foreign exchange intervention of the crisis era began on 12 March 2009, inside the window's final months, and the sharp depreciation it produced suppresses the measured appreciation.

[13] For rates, a percent deviation is already a change in percentage points.

The shock enters through foreign demand for the asset (42). A higher θ^* raises the value foreign households place on the services of the asset, and their holdings q^* rise by 3.7% on impact. Through market clearing (44), this crowds out domestic households, who reduce their holdings of q by 1.4%. This crowding-out does not reduce household resources: in equilibrium, the same inflow that displaces households from the convenient asset finances higher tradable consumption, so households substitute away from q toward C_T and the foreign bond.

Foreign purchases of the convenient asset are a capital inflow that is absorbed by the current account (43) on three margins. The currency appreciates, households accumulate the foreign bond f, up 0.7%, and tradable consumption C_T rises by 0.26%. The appreciation is the relative-price adjustment that lets the economy absorb the inflow: the extra tradable consumption must be imported, and the currency appreciates to let the imports in. A larger foreign demand for the asset requires the price of home tradables to fall, and by the law of one price (49) a falling tradable price is a nominal appreciation. It amounts to 0.9% on impact, and because tradable inflation stays mildly negative for years afterwards, the level of the exchange rate cumulates to its peak of 5.9%. The appreciation is permanent: the Taylor rule targets inflation rather than the price level, so the episode of below-target inflation is never made up, and the level settles where it peaks, 5.9% stronger (Appendix C.1). The appreciation makes tradables abruptly cheap while sticky non-traded prices fall by less, so the relative price of non-traded goods p_N rises by 0.6%. The rise reflects the collapse in the traded-goods price rather than strong non-traded demand. What stands out is the persistence. Once the foreign return is accounted for, the tradable Euler equation (41) tilts consumption only through the small portfolio-cost term, so the level is set on impact and decays slowly, still up 0.15% after forty quarters, more than half its impact value.

On the domestic side the shock is contractionary. The loss of the convenient asset lowers non-traded demand directly, and the higher relative price lowers it further, both through (40). Non-traded output falls by 0.3% on impact, non-traded inflation falls by 1.2 percentage points, and the central bank cuts the policy rate by 1.9 points. The cut does two jobs. At home, with prices sticky, it passes almost one for one into the real rate that drives non-traded demand and cushions the downturn. Abroad, it lowers the return on the convenient asset, and through (42) the lower return holds part of the foreign demand back. Without the cut, the inflow, the appreciation, the recession, and the disinflation would all be larger. The rate therefore leans against the shock at its point of entry as well as against its domestic consequences. The real effects it leaves depend on the convenience yield, as shown in Section 5.2. The pattern is the Swiss one of the acute phase, a cut to the floor, a contraction, a fall into deflation and an appreciation of this size. I compare

the model responses to Swiss acute phase in Section 5.6.

The foreign demand shock generates a net resource transfer to the home economy. Two opposing forces determine its size. Stronger foreign demand raises the transfer: by supplying more of the convenient asset, the economy obtains additional resources from abroad. The appreciation works in the opposite direction. The claim foreigners already hold commands more home tradable goods once those goods become cheaper, so the outstanding position gains real value and costs more to carry. Under the baseline the demand force dominates, leaving a net inflow that finances higher tradable consumption and the accumulation of foreign bonds. Most of the inflow is initially saved abroad rather than consumed, and it stays positive for a long time, which sustains the elevated path of tradable consumption. Total consumption falls on impact, by 0.1% through (51): the non-traded decline outweighs the tradable gain. The two differ in timing. The recession fades with the shock, the tradable gain persists with the foreign position, so the composite turns positive after five years and is positive in cumulated terms.[14] Appendix C.1 defines this net inflow and evaluates its absorption on impact. In terms of balance-of-payments, the country sells the convenient asset abroad and places the proceeds back into foreign bonds: the financial account recycles nearly all of the net inflow and the trade balance absorbs the rest.[15]

Because the asset provides convenience services, foreign holders accept a return below the shadow rate, so each unit of the outstanding position is carried at $(1-\nu)/\beta$ rather than at $1/\beta$. The difference, ν/β per unit, is the convenience income earned on the stock the rest of the world already holds, the model's counterpart of seigniorage (Appendix C.7, equation (88)).

5.2 The Role of the Convenience Yield

The previous subsection used the baseline steady-state convenience yield. I now vary it to show that the transmission of a flight-to-safety shock depends strongly on the convenience regime. Figure 6 reports the responses to the same shock as in Section 5.1 for $\nu \in \{27, 150, 300, 500\}$ basis points. The shock size and all other parameters are held fixed, so the four economies differ only in the steady-state convenience yield. These values correspond to the calibration scenarios discussed in Section 4.2: the post-2007 normal-times mean, an intermediate stress level, the GFC-peak baseline, and a high-stress case.

The macroeconomic effects increase sharply with the convenience yield. At the normal-times premium of 27 basis points, the shock leaves almost no trace on the domestic economy: non-traded output falls by only 0.03%, non-traded inflation by 9 basis points,

[14] Appendix C.7 reports the composite across experiments.
[15] Appendix C.1 reports the split.

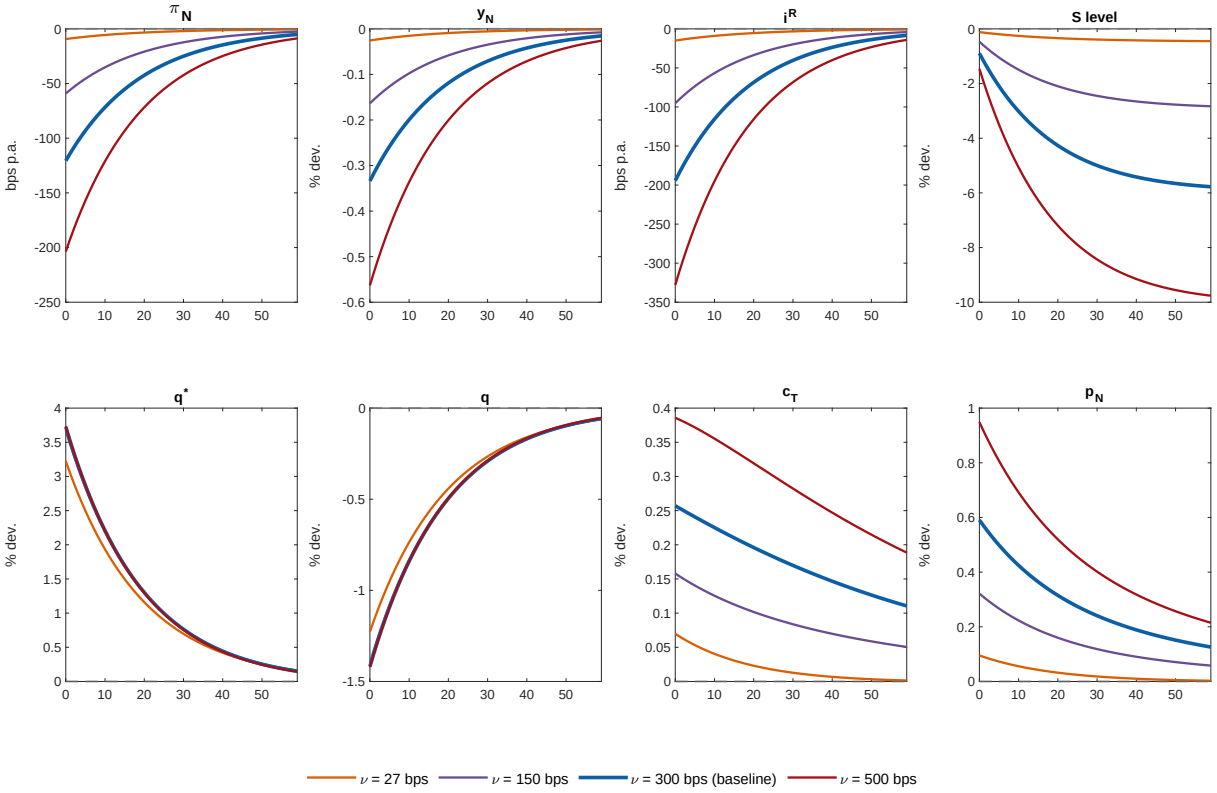


Figure 6: Impulse responses to a flight-to-safety shock for different values of the steady-state convenience yield, $\nu \in \{27, 150, 300, 500\}$ basis points. All variables are percent deviations from each economy's own steady state. Interest rate and inflation are reported in annualized terms. The thick line marks the baseline ($\nu = 300$ basis points). The full set of responses over 120 quarters is reported in Figure 15 in the appendix.

and the policy rate by 15 basis points. The adjustment is absorbed mostly through portfolios and the exchange rate. At the baseline premium of 300 basis points, the same shock generates the crisis pattern described above: a meaningful contraction, disinflation, and a large policy-rate cut. At 500 basis points, the contraction deepens further, non-traded inflation falls by 2.0 percentage points, and the Taylor rule calls for a rate cut of 3.3 percentage points.

The amplification is close to proportional. A higher ν raises the weight of the convenient asset in the non-traded demand block, as discussed in Section 4.2. Moving from 27 to 300 basis points raises the convenience coefficient by a factor of about 11, and the output, inflation and policy-rate responses rise by a factor of about 13. The dynamic responses track the static coefficient: the wedge works through how much domestic demand depends on the convenient asset. The exchange rate cumulates at every premium, since the Taylor rule leaves the price level unanchored, but more so at high premia: the impact jump is a quarter of the eventual appreciation at 27 basis points and a seventh at 300.

The key point is that the gross portfolio reallocation does not scale with its macroeconomic consequences. Foreign holdings of the convenient asset rise by 3.2% on impact

at the lowest premium and by 3.7% at the higher ones.[16] Yet over the same range the contraction in non-traded output and the disinflation increase more than tenfold. This is the dependence on the convenience regime in its purest form: the macroeconomic significance of the flow depends less on its size than on the regime that receives it. In normal times a flight to safety is a portfolio event. In a crisis, the same flight is a macroeconomic one. The same regime dependence appears when the demand elasticity is varied instead of the wedge, holding the foreign side fixed: at the normal-times elasticity the same flight is accommodated by domestic holders at a smaller rise in the wedge, so it generates half the appreciation and half the contraction of the calibrated value, while the portfolio flow behind it is nearly three times larger (Figure 16 in Appendix C.3).

The regime dependence is a property of the linearized model rather than of the calibration.

Proposition 1 (Regime dependence)**.** *Let the supply of the convenient asset be fixed and the policy rate follow a Taylor rule that satisfies the Taylor principle. Let* $\nu_t \equiv 1 - \frac{1+i_t^R}{1+i_t}$ *be the convenience yield, with steady state* ν*. By* (31) *its log deviation is* $\hat{\nu}_t = \hat{C}_{T,t} + \hat{\theta}_t - \hat{q}_t/\phi_1$*, and aggregate demand* (40) *with the relative price* (46) *becomes*

$$\hat{Y}_{N,t} = E_t \hat{Y}_{N,t+1} - \left(\hat{i}_t^R - E_t \hat{\pi}_{N,t+1}\right) - \frac{\nu}{1-\nu}\,\hat{\nu}_t.$$

(i) The domestic block $(\hat{Y}_{N,t}, \hat{\pi}_{N,t}, \hat{i}_t^R)$ *is determined by this equation, the Phillips curve* (39) *and the rule, given the path of* $\hat{\nu}_t$*. These are the equations of the closed-economy New Keynesian model. The flight enters them only through the convenience yield, by the crowding-out* $\hat{q}_t = -\frac{q^*}{q}\hat{q}_t^*$ *and by tradable consumption.*

(ii) The steady-state yield enters the block only as the factor $\nu/(1-\nu)$ *on* $\hat{\nu}_t$*, and for a given path of the inflow and of tradable consumption* $\hat{\nu}_t$ *does not depend on* ν*. The responses of non-traded output, inflation and the policy rate are therefore proportional to* $\nu/(1-\nu)$*: the macroeconomic effect of a unit of inflow is proportional to the steady-state convenience yield to first order, and vanishes as* $\nu \to 0$*.*

The proof is in Appendix B. The proposition says where the open economy enters. The inflow, the exchange rate and the foreign position reach output, inflation and the policy rate through the change in the convenience yield, $\hat{\nu}_t$, and nowhere else. $\hat{\nu}_t$ is the convenience residents lose when foreigners take the asset, plus the relative-price effect of tradable consumption, and ν is its coefficient in demand. The flow and its effect are therefore separate objects. The flow is governed by the foreign demand equation and the current account. The effect is the closed-economy model's response to the change in the

[16] Above 150 basis points the response is flat to the first decimal.

yield, in proportion to ν. On the grid of Figure 6 the inflow is 3.2% to 3.7% of the position at every premium, while the fall in non-traded output on impact rises from 0.03% at 27 basis points to 0.56% at 500, twenty times more. The whole difference is the wedge: at every premium the output response equals roughly 12 times ν times the inflow,[17] and the same holds for inflation and the policy rate with their own constants. The flight is nearly the same size in every regime, its effect is the wedge times the inflow times a constant.[18]

This regime dependence also explains why the lower bound matters precisely in the states in which the shock is most damaging. As ν rises, the policy rate cut required to stabilize the economy grows rapidly. At high premia the Taylor rule prescribes rates far below any plausible lower bound. The next section imposes the bound and shows how the adjustment changes when the interest rate can no longer absorb the shock.

5.3 The Zero Lower Bound

The rate cut prescribed by the Taylor rule in the baseline reaches below zero. In this section the zero lower bound is imposed. The policy rate follows the same Taylor rule as before whenever that rule prescribes a rate at or above the bound, and is held at the bound otherwise,

$$i_t^R = \max\{\, i_t^{R,\text{rule}},\ \underline{i}^R \,\}, \tag{54}$$

where $\underline{i}^R = 0$ is the lower bound.[19] I subject the economy to the same shock as in the baseline, so that the two experiments differ only in whether the policy rate is free to fall. Figure 7 sets the unconstrained economy against the constrained one.

The rule calls for a policy rate cut that the zero lower bound does not allow. The policy rate is held at zero for thirteen quarters, a fall of only 1.0 percentage point from its steady state, while the unconstrained rule would cut by 1.9 points and take the rate to about -0.9%. With the nominal rate above its desired level and expected inflation falling, the short-term real rate is too high relative to the unconstrained path, and the recession is correspondingly deeper. Non-traded output falls by 3.1% on impact against

[17] To within 3% across the grid.

[18] The wedge is the market price of the asset's services, $\nu = \frac{C_T}{1-\gamma} V_q(q)$, the same for domestic and foreign holders. Without it the asset is a perfect substitute for the foreign bond: foreign demand reduces to interest parity, and a shift in foreign preferences for the asset is absorbed by an infinitesimal change in its return. In the limit $\nu \to 0$ the flight has no effect at all, at home or on the current account. Along the way the inflow falls slowly and the domestic effect in proportion to the wedge, which is what the proposition states. The separation of the domestic block relies on log utility with Cobb-Douglas aggregation, under which tradable consumption enters non-traded demand only through the relative price with the weight ν, and on a rule that responds to non-traded rather than headline inflation, so that the appreciation does not reach the policy rate through imported disinflation.

[19] The model is solved in Dynare. The occasionally-binding constraint (54) is handled with the piecewise-linear method of Guerrieri and Iacoviello (2015) (OccBin), which imposes the bound while keeping the rest of the system linear.

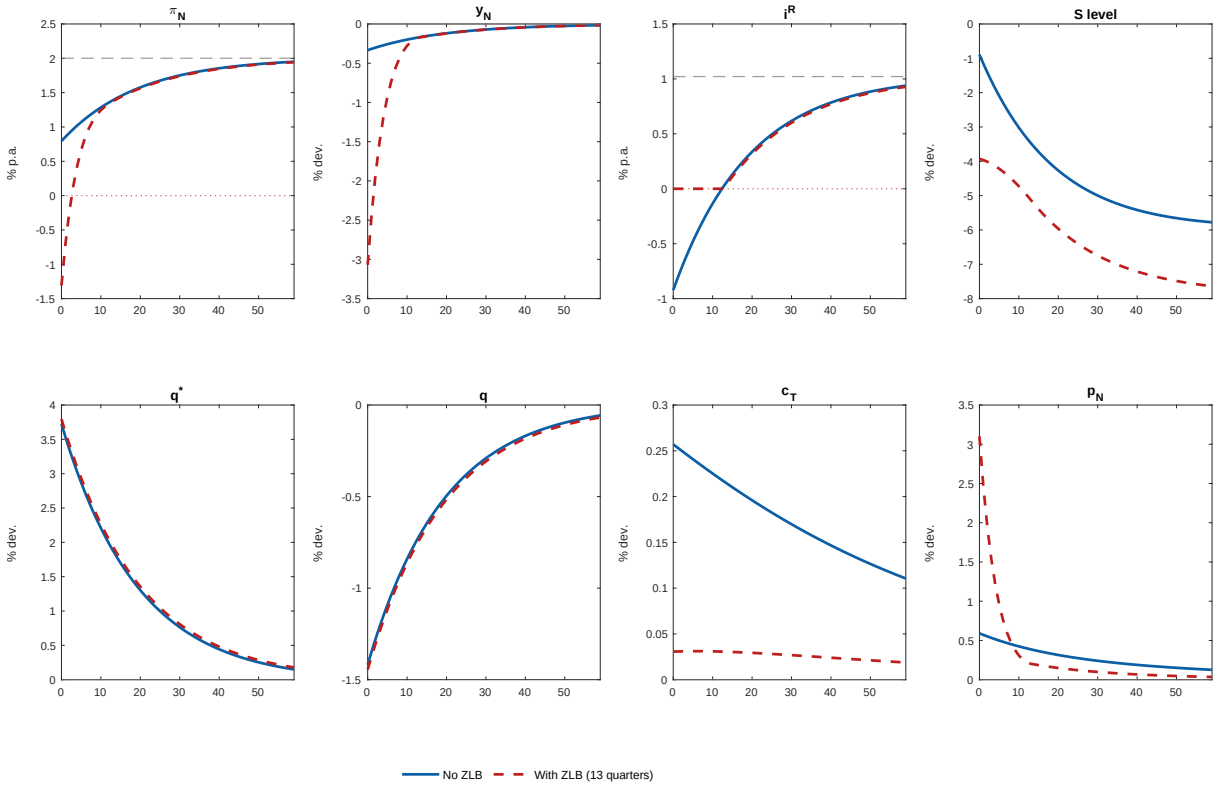


Figure 7: Impulse responses to a flight-to-safety shock, the unconstrained economy (solid) against the zero lower bound (dashed). All variables are percent deviations from their steady state. Interest rate and inflation are reported in annualized terms relative to their steady-state values, to show when they turn negative.

0.3%, and non-traded inflation by 3.3 percentage points against 1.2, which in level terms means outright deflation of about -1.3% against inflation of 0.8% in the unconstrained economy. The constrained economy is in a liquidity trap. The trap has an unusual origin. Where the textbook economy reaches the effective lower bound because domestic demand collapses (Eggertsson and Woodford, 2003), this one is pushed there by foreign demand for its own assets.

The zero lower bound affects prices much more than quantities. Foreign demand is no longer damped by a falling rate through (42), yet it only rises by 3.8% on impact against 3.7%, and the crowding-out of q is 1.4% in both economies. The absorption of the same inflow is what differs. The appreciation is 3.9% on impact against 0.9%, and the relative price of non-traded goods rises by 3.1% against 0.6%. With no policy rate cut to lean against the flight, the current account absorbs it through the exchange rate alone, and in the impact period.

At the bound the net transfer disappears. The two forces that govern it both grow under the constraint, but unequally. New purchases of q^* rise from 3.7% to 3.8% of the steady-state position. The cost of carrying the outstanding position rises with the impact appreciation, from 0.9% to 3.9%, and exceeds the new purchases. The net inflow on impact

is -0.1% of the position against $+2.8\%$ in the unconstrained economy (Appendix C.7). Households sell foreign bonds rather than accumulating them, and tradable consumption rises by 0.03% against 0.26%. The flight brings the same inflow from foreigners, but the appreciation raises the real value already owed to them.

The appreciation and the convenience income act on different margins of the same inherited position. The appreciation raises the number of home goods the position represents, the convenience yield lowers the return paid on each unit. The bound works through the first margin only. The convenience income, earned at ν/β on the position, is higher at the bound than in the unconstrained economy at every date, since the position is larger and the policy-rate path higher (Appendix C.7). The bound does not tax the income. It taxes the position.

The lower bound worsens the domestic economy on every margin: the recession is deeper, disinflation turns into deflation, and the appreciation is more abrupt. At the same time, it eliminates the net resource transfer the economy earns for supplying the domestic convenient asset. Of the two economies the constrained one is the closer to the Swiss acute phase on every domestic variable (Section 5.6). If the central bank cannot use its interest rate, what else can it do? The natural candidate is the balance sheet, since the shock enters through the market for an asset the central bank itself supplies. Section 5.4 addresses this.

5.4 Balance-Sheet Policy

The supply of the convenient asset has so far been held fixed, so that the entire foreign demand shock was absorbed by prices and private portfolios. I now let the central bank use it as an instrument. In place of the exogenous process for $\hat{q}_t^s$, the supply of the convenient asset adjusts each period to implement a target, while the Taylor rule for the policy rate remains in place. The government's foreign asset position stays passive throughout, so the experiment isolates the convenient-asset supply leg of a balance-sheet operation. I consider two targets that correspond to the two channels of the shock: stabilization of the domestic economy, implemented as $\pi_{N,t} = 0$, and stabilization of the exchange rate, implemented as $\Delta S_t = 0$.

Implementation is a liability-issuance and transfer operation, not a foreign exchange intervention. The government issues Q_t^s and the budget constraint rebates the proceeds to households as a transfer. Households recycle the rebate into foreign bonds, so under the domestic target private f rises by 4.4% against 0.7% in the baseline, while the government's foreign position stays fixed. Section 5.5 activates it.

Figure 8 sets the inflation targeting policy against the baseline. Targeting the domestic economy achieves exact stabilization. Non-traded inflation and output remain at steady

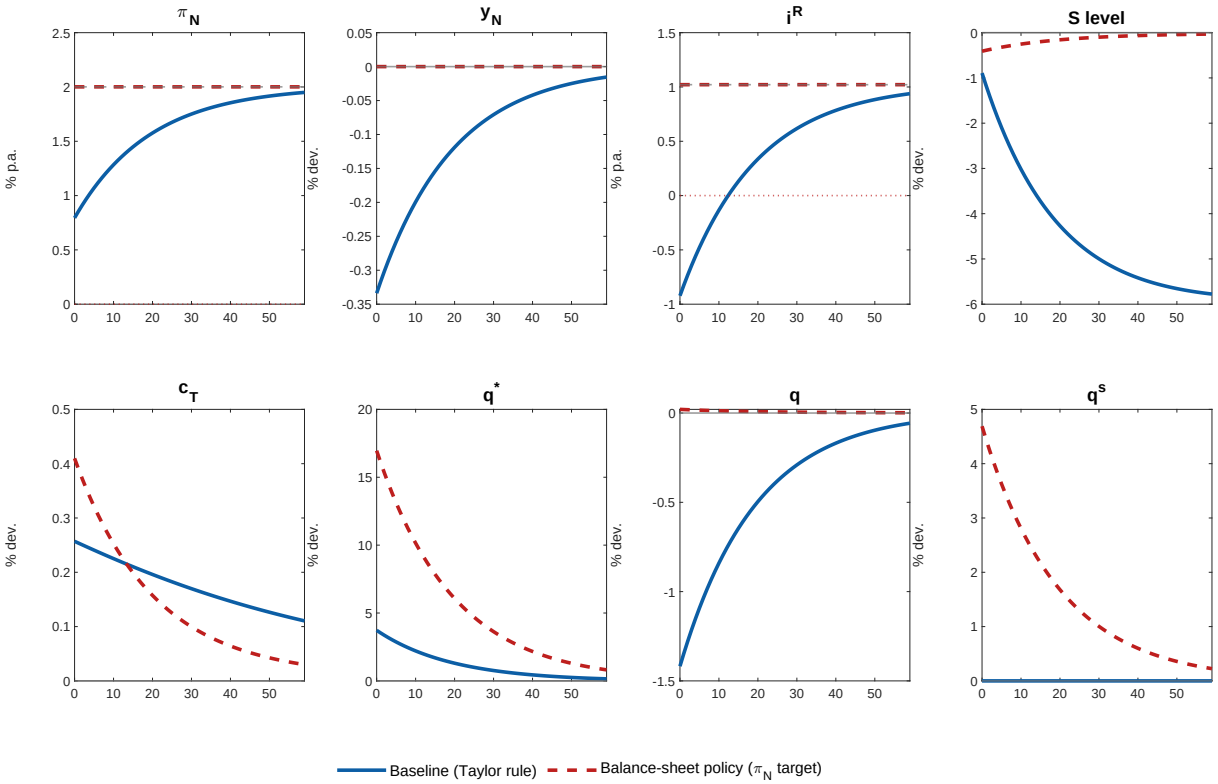


Figure 8: Impulse responses under the balance-sheet policy targeting non-traded inflation (dashed) against the baseline with a fixed supply of the convenient asset (solid). Interest rate and inflation are reported in annualized terms as levels relative to their steady-state values. All other variables are percent deviations from steady state. Targeting non-traded output instead yields identical responses.

state along the entire path, and the policy rate never moves. This is not just a feature of the calibration.

Proposition 2 (Exact stabilization)**.** *Let the policy rate follow the Taylor rule and let the supply of the convenient asset implement $\hat{\pi}_{N,t} = 0$ in every period. The supply that does so is*

$$\hat{q}_t^s = \frac{q}{q+q^*}\,\phi_1 \hat{C}_{T,t} + \frac{q^*}{q+q^*}\,\hat{q}_t^*,$$

and for any path of the flight and any path of the government's foreign position:

(i) $\hat{\nu}_t = 0$ at every date, so $\hat{Y}_{N,t} = \hat{\pi}_{N,t} = \hat{\imath}_t^R = 0$: the supply holds the convenience yield at its steady-state value and closes the channel of Proposition 1 at its source, the policy rate never moves, and the lower bound never binds.

(ii) The flight is absorbed by the current account alone. The exchange-rate level mirrors tradable consumption, $\hat{S}_t = -\hat{C}_{T,t}$, and with $\psi > 0$ both return to steady state.

(iii) The operation meets the whole flight. Foreign demand is $\hat{q}_t^* = \phi_1^* \hat{\theta}_t^* + \varphi_2^* \psi f \hat{f}_t$, the flight term undamped by any rate response, plus a portfolio term.

Targeting $\hat{Y}_{N,t} = 0$ in every period implements the same equilibrium. With $\psi > 0$ the equilibrium exists and is unique.

The proof is in Appendix B. The policy-rate panel shows part (i) directly: the baseline rate dives below zero while the policy rate under inflation targeting stays flat at its steady-state level. With both arguments of the Taylor rule at zero there is no cut to make, so the lower bound, which under a fixed supply of the convenient asset binds for thirteen quarters, never comes into play. The instrument prevents the trap rather than curing it, since the rate cut that would have hit the bound is never needed. The operation behind this is modest and works exactly where the shock enters. The central bank expands the supply of the convenient asset by 4.7% on impact, enough to accommodate a foreign demand that is now larger than in the baseline (q^* rises by 17.0% against 3.7%, since no rate cut damps it through (42)) and to leave domestic holdings at their steady-state level rather than crowded out (q is unchanged against a fall of 1.4%).[20] The over-supply is deliberate and the supply rule of the proposition gives its size. The appreciation still occurs, so the relative price of non-traded goods still rises and weighs on non-traded demand through (40), and the additional domestic holdings, ϕ_1 times tradable consumption, are what offsets it. Targeting non-traded output instead of non-traded inflation delivers responses that are identical, the last part of the proposition. The flight to safety enters non-traded demand through (40) like a demand shock, so output and inflation move together through the Phillips curve (39), and the path of the instrument that stabilizes one stabilizes the other, a divine coincidence in the sense of Blanchard and Galí (2007).[21] The coincidence is exact here because the instrument closes the wedge at its source rather than leaning against its consequences: once the shortage of the convenient asset is met, there is no residual disturbance left for the two objectives to disagree over.

The appreciation, however, does not disappear. The exchange rate still appreciates by 0.4%, now in a single jump on impact, against a cumulated 5.9% in the baseline, as the exchange-rate panel of Figure 8 shows. This is the current account at work: the foreign purchases of the convenient asset are a capital inflow, and by (43) the inflow must be absorbed through some combination of the exchange rate, the trade balance, and the foreign bond position. What remains is small and transitory. The baseline's 5.9% is a permanent shift in the exchange-rate level. Under the target the level mirrors tradable consumption, part (ii), and returns to steady state as consumption decays: a permanent appreciation becomes a transitory one less than a tenth of its size.[22] The

[20] Without the portfolio cost the operation is 5.0% of the stock, about 7% of annual GDP, under either financing of the issuance, see Section 5.5.

[21] The coincidence is a property of the shock, not of the instrument. A flight to safety reaches the domestic block through (40) and nothing else, so for that block it is a pure demand shock, and under a pure demand shock stabilizing inflation stabilizes output. A cost-push term in (39) would break it: output and inflation would then move apart, and one instrument could stabilize one of them only, as in Blanchard and Galí (2007). The results of this section are conditional on the flight being the only disturbance.

[22] Imposing $\pi_{N,t} = 0$ period by period targets the non-traded price level rather than its growth rate, so

economy preserves the net resource transfer that the lower bound would otherwise erode: tradable consumption rises by 0.41% on impact, above the unconstrained baseline's 0.26% and against 0.03% in the trap, with a larger share of it now accumulated in foreign bonds: accommodating the inflow sells more convenience services abroad, so the transfer is not merely preserved but enlarged.

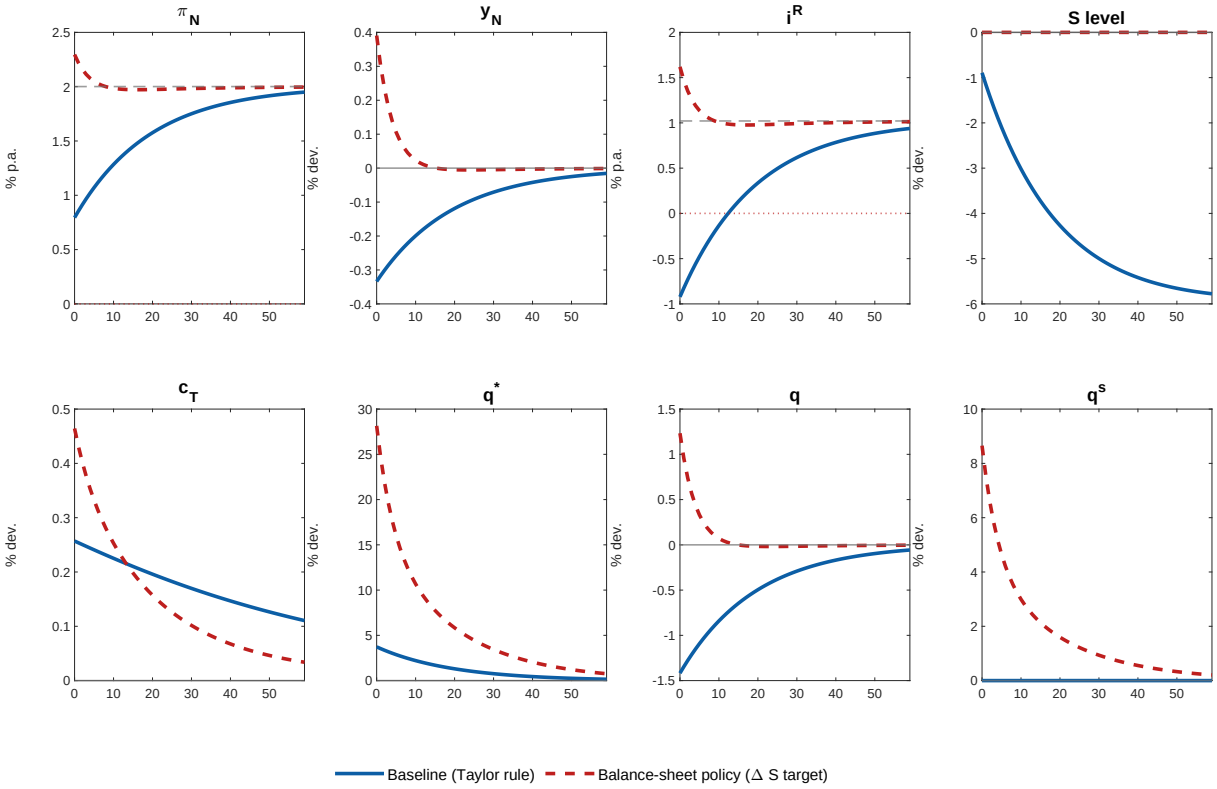


Figure 9: Impulse responses under the balance-sheet policy targeting the exchange rate (dashed) against the baseline (solid). The layout matches Figure 8. The operation required to hold the exchange rate is almost twice the size of the one behind domestic stabilization.

Targeting the exchange rate instead eliminates the appreciation entirely, and Figure 9 shows what doing so costs. Holding $\Delta S_t = 0$ requires the central bank to meet the full foreign demand at an unchanged price: q^* rises by 28.2% against 17.0% under the domestic target and, as the q^s panel shows, the required expansion of the convenient-asset supply peaks at 8.7%, almost twice the operation size behind domestic stabilization. The required expansion also affects the domestic economy. Non-traded output rises by 0.4%, non-traded inflation by 0.3 percentage points, and the two instruments end up pulling against each other, with the balance sheet expanding to hold the currency while the Taylor rule raises the policy rate to 1.6% to lean against the boom the expansion creates. The two margins feed back on each other: the higher policy rate raises the return on the convenient asset and pulls in further foreign demand through (42), so that about a third

no past deflation is left standing and the nominal side is anchored: the exchange rate returns to steady state with the real allocation. Appendix C.5 reports the path.

of the inflow the operation must meet on impact is generated by the rate rule's response rather than by the flight itself. The economy trades an appreciation for a mild boom and a balance sheet nearly twice as large. The percentages refer to the outstanding stock, and the stock is 5.8 quarters of GDP, about 1.5 times annual GDP, so operations of 4.7% and 8.7% of it translate into roughly 7% and 13% of annual GDP.[23]

The two targets are the poles of a family. Let the supply of the convenient asset implement the weighted target $\lambda\pi_{N,t} + (1-\lambda)\Delta S_t = 0$ in every period, with $\lambda \in [0,1]$, so that $\lambda = 1$ reproduces the inflation target and $\lambda = 0$ the exchange-rate target. A single parameter then indexes every stance the instrument can take between the two channels of the shock, and re-solving the model along a grid of λ traces out the attainable set. Figure 10 plots it in outcome space, as the peak appreciation of the exchange-rate level against the impact response of non-traded inflation. The locus characterizes the attainable trade-off between the two targets. Each point is implemented by the same one-parameter rule, so reaching any of them requires nothing beyond the choice of λ. Walking from the domestic pole to the exchange-rate pole trades the residual appreciation of 0.4%, with inflation exactly stabilized, for an impact inflation response of 0.3 percentage points, with the appreciation exactly eliminated. The trade-off in this plane is close to linear, so there is no free lunch between the two objectives. Both are small: the whole menu lies within 0.4% of appreciation and 0.3 points of inflation. What separates the points is the size of the operation behind each one, which the plane does not show. Three properties hold along the entire menu. First, every point lies within a peak appreciation of 0.4%, against 5.9% in the no-policy baseline and 7.9% in the trap, so any stance on the menu removes nine tenths of the appreciation, and its permanent component entirely, before the choice between objectives is posed. Second, the menu never approaches the lower bound. The policy rate is exactly quiescent at $\lambda = 1$ and rises monotonically toward the exchange-rate pole, where it leans against the imported boom at 1.6% on impact, and its level never falls below 0.98% along the locus, so with the second instrument active the bound that dominates the fixed-supply economy is absent from the entire policy space. Third, no point on the menu leaves a permanent appreciation. For any $\lambda > 0$ the weighted target anchors the non-traded price level, which ties the exchange-rate level to the relative price, $S_t = -\lambda\hat{p}_{N,t}$, so the level returns to steady state as the episode fades, and at $\lambda = 0$ it

[23] The floor was a different policy from the exchange-rate target studied here: introduced only after the appreciation to which the shock is calibrated had occurred, it capped a single bilateral rate at the level already reached rather than defending the pre-shock parity, and the trade-weighted franc continued to appreciate while it was in force. The boom is, moreover, a crisis-state prediction: it operates through the same convenience coefficient that gives the shock its bite, and over most of the floor period the premium measured in Section 4.2 had returned toward its normal-times mean, at which Section 5.2 puts the channel an order of magnitude weaker. Finally, the boom is measured relative to an economy otherwise at steady state, so amid the deflationary pressure of those years the same channel appears as deflation avoided rather than inflation observed, which is how the SNB motivated the floor.

never moves.

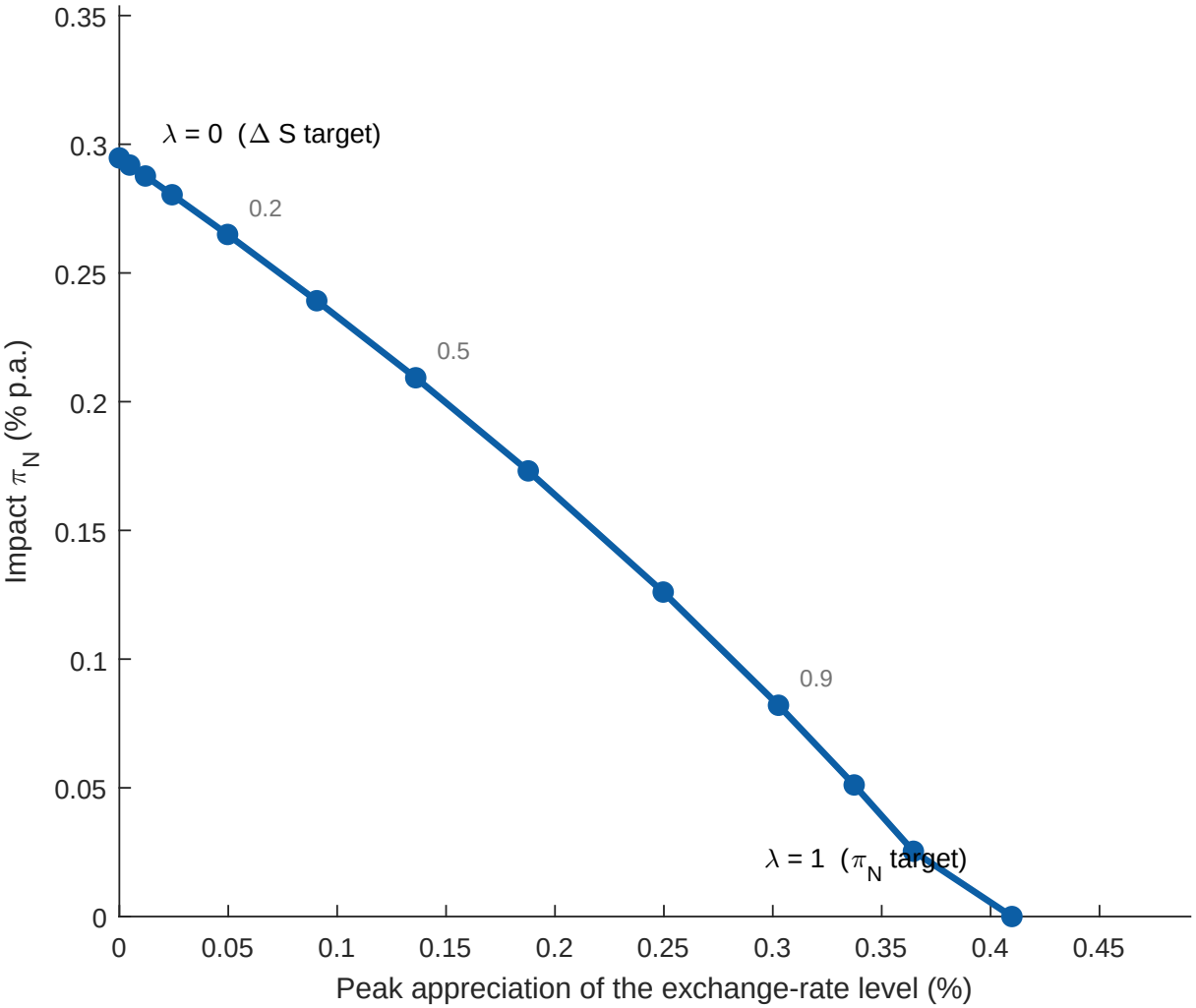


Figure 10: The attainable locus between the two targets. Each point is the outcome under the weighted target $\lambda\pi_{N,t}+(1-\lambda)\Delta S_t = 0$ for one value of $\lambda \in [0,1]$, plotted as the peak appreciation of the exchange-rate level against the impact response of non-traded inflation, with selected values of λ marked along the curve. The pole $\lambda = 1$ reproduces the inflation target of Figure 8, where the appreciation is a single jump on impact, and $\lambda = 0$ the exchange-rate target of Figure 9. The no-policy baseline and the ZLB outcome lie far off the locus, at peak appreciations of 5.9% and 7.9%. Along the locus the policy rate rises from exactly quiescent at $\lambda = 1$ to 1.6% on impact at the exchange-rate pole and its level never falls below 0.98%, so no point on the menu approaches the lower bound.

The shape of the locus carries the main result of the section, and it is a shape in the operation, not in the plane. In outcome space the cost of fighting the residual appreciation is proportionate: the step from $\lambda = 1$ to $\lambda = 0.98$ removes 0.05 points of the 0.4% residual and costs 0.03 points of impact inflation, and the locus continues at that rate to the exchange-rate pole. In the operation the cost is incurred at the first step. The required expansion of the convenient-asset supply behind each point, reported in Table 4, rises from 4.7% of the stock at $\lambda = 1$ to 6.9% at $\lambda = 0.98$, 8.3% at $\lambda = 0.9$ and 8.7% at $\lambda = 0$, and it is monotone in λ along the whole grid. The first step raises the operation by half, 3% of annual GDP, to buy a tenth of the residual. At $\lambda = 0.9$ the operation is

Table 4: Balance-sheet policy: impact responses across regimes. The last three columns implement the weighted target $\lambda\pi_{N,t} + (1-\lambda)\Delta S_t = 0$, with $\lambda = 0$ the exchange-rate target and $\lambda = 1$ the domestic target of the preceding figures. Rates and inflation are annualized, the appreciation is the peak of the exchange-rate level, and the last row is the peak expansion of the convenient-asset supply in percent of the outstanding stock, so 4.7%, 6.9% and 8.7% correspond to roughly 7%, 10% and 13% of annual GDP. Targeting $Y_{N,t} = 0$ instead of $\pi_{N,t} = 0$ yields identical responses. Under the ZLB the appreciation is sharper on impact and cumulates deeper.

			Balance-sheet target		
	Baseline	ZLB	$\lambda = 0$	$\lambda = 0.98$	$\lambda = 1$
Non-traded inflation π_N (% p.a.)	−1.2	−3.3	+0.3	+0.03	0.0
Non-traded output Y_N (%)	−0.3	−3.1	+0.4	+0.1	0.0
Policy rate i^R (% p.a.)	−1.9	−1.0	+0.6	+0.1	0.0
Peak appreciation (%)	5.9	7.9	0.0	0.37	0.41
Tradable consumption C_T (%)	+0.26	+0.03	+0.46	+0.42	+0.41
Convenient-asset supply q^s, peak (%)	0	0	8.7	6.9	4.7

96% of the full defence and a quarter of the residual is gone. Nearly the whole balance sheet that holding the exchange rate requires is needed to lean against it at all. The interior of the menu offers no cheap compromise. The domestic pole is the minimal point of the menu: the smallest operation, no rate move, no inflation, no boom, and the only point at which the rate rule is idle. The locus is a positive object throughout. It characterizes what is attainable conditional on the flight-to-safety shock, in the impact and peak dimensions reported here, for the convenient-asset supply instrument under period-by-period targeting. It does not rank its points.

The menu separates two readings of a balance-sheet expansion. The expansion behind domestic stabilization is an accommodation. It supplies the asset the world demands, and because it meets a demand rather than suppressing a price, it leaves inflation at target. The expansion behind the exchange-rate target defends a price against the inflow, and this part, not the size of the balance sheet, imports the boom. Whether an expansion is inflationary depends on what it accommodates. Table 4 collects the two poles and one interior stance.

5.5 Reserve-Backed Operations

The experiments so far leave the government's foreign position passive: the balance-sheet operations are financed by issuing the convenient asset and rebating the proceeds to households. This subsection activates the other side of the balance sheet. The central bank now backs the issuance with foreign assets, purchasing them one for one with the operation, $\Delta F_t^g = \Delta Q_t^s$, so that reserves rise together with the convenient liability, as on an actual central bank balance sheet. The mandate is unchanged: the supply rule

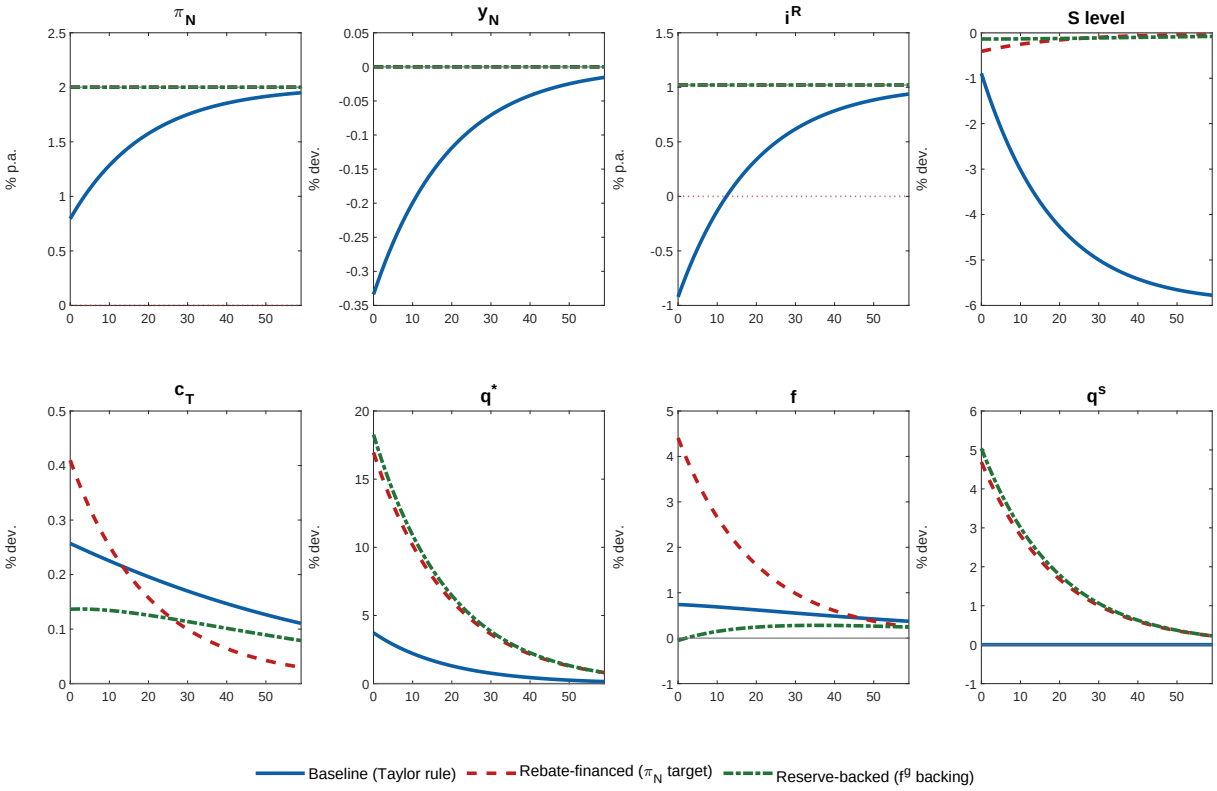


Figure 11: The domestic target with rebate-financed and reserve-backed issuance: no-policy baseline (solid), issuance proceeds rebated to households (dashed), and issuance backed one for one by foreign-asset purchases (dash-dotted). Interest rate and inflation are annual levels, remaining variables percent deviations, 60 quarters. Both cases deliver identical stabilization: the two target lines coincide at zero in the π_N, y_N, and i^R panels. The bottom row shows where they differ.

implements $\pi_{N,t} = 0$ each period with the Taylor rule in place, exactly as in the preceding subsection. The only difference between the two experiments is the financing, and one fiscal assumption must be stated explicitly: the government's foreign position tracks its convenient liability one for one, and the lump-sum transfer balances its budget each period, so the income on the reserves net of the interest on the liability reaches households as it accrues. What the financing changes is who holds the position, not who receives its income.

Table 5 compares the two cases and Figure 11 plots them against the no-policy baseline. Stabilization is identical: non-traded inflation, non-traded output, and the policy rate are at steady state in both columns, so the mandate and the financing are separable. The external accounts are not. Under reserve backing the operation is larger by 7%, from 4.7% to 5.0% of the stock, foreign holdings rise by 18.3% against 17.0%, tradable consumption rises by a third as much, 0.14% against 0.41%, households accumulate no foreign bonds, and government reserves jump by 91% of their stock. The percentage is large because the stock is small, 0.32 quarterly-GDP units in steady state, so the purchase amounts to 0.29 quarters of GDP, about 7% of annual GDP, on impact. The financing changes the size of the operation little and who holds the proceeds entirely.

Table 5: One mandate, two cases: the domestic target under rebate-financed and reserve-backed issuance

	Rebate-financed	Reserve-backed
Convenient-asset supply q^s, impact (%)	+4.7	+5.0
Foreign holdings q^*, impact (%)	+17.0	+18.3
Domestic holdings q, impact (%)	+0.02	+0.01
Tradable consumption C_T, impact (%)	+0.41	+0.14
Total consumption C, impact (%)	+0.16	+0.05
Household foreign bonds f, impact (%)	+4.4	−0.05
Government foreign assets f^g, impact (%)	0	+91
Peak appreciation of S (%)	0.41	0.14
Operation (q^s peak, share of annual GDP)	0.068	0.073

Both columns implement $\pi_{N,t} = 0$ each period with the Taylor rule in place, so non-traded inflation, non-traded output, and the policy rate are at steady state to machine precision in both. In the first column the issuance proceeds are rebated to households, in the second the government purchases foreign assets one for one with the issuance, $\Delta F_t^g = \Delta Q_t^s$. Stabilization is identical, the external accounts are not.

Issuing the convenient asset abroad is borrowing at the policy rate, so the proceeds are financing, not income. The income comes from what the proceeds are invested in. The country borrows from foreigners at the policy rate and reinvests the proceeds in foreign bonds at the foreign rate, and the foreign rate is higher by the wedge, since foreigners accept a return below it in exchange for the asset's services. Borrowing at a low rate to lend at a higher one is a carry trade, and the spread ν earned each period on the stock foreigners hold is the carry. It is the convenience income of Appendix C.7. It accrues on the outstanding stock whoever holds the reinvested proceeds. The financing decides only who that is, households under the rebate and the central bank under reserve backing, and without portfolio frictions this does not matter.

Proposition 3 (Neutrality of the asset leg)**.** *Let $\psi = 0$, and let the rules for the policy rate and for the supply of the convenient asset not condition on $\hat{f}_t$ and $\hat{f}_t^g$ separately. Then the path of the government's foreign position affects no variable other than $\hat{f}_t$: the equilibrium paths of all other variables are the same for every path of $\hat{f}_t^g$. This includes the consolidated position $\hat{F}_t \equiv f\hat{f}_t + f^g\hat{f}_t^g$ and foreign households' bond holdings $\hat{f}_t^*$, which depend on $\hat{F}_t$ alone. In particular:*

(i) An issuance of the convenient asset rebated to households and the same issuance invested in foreign assets deliver the same economy.

(ii) With the supply fixed, a purchase of foreign assets changes nothing but the split of $\hat{F}_t$ between households and the government.

This is the neutrality of Wallace (1981), and it applies to one leg of the operation only. A reserve-backed issuance has two legs: the government issues the convenient asset and

buys foreign bonds with the proceeds. The asset leg is neutral. The government holds the same bond as households, earns the same return on it and passes the income on as a transfer, so households hold the position at one remove and the model is Ricardian with respect to the government's foreign position. The liability leg is not neutral, because the convenient asset is not a perfect substitute for the foreign bond: it carries the wedge, and that is what Section 5.4 works with. Every effect of a reserve-backed operation comes from the asset it issues and none from the assets it buys, and part (ii) says the same thing from the other end: a purchase of foreign assets with the supply fixed does nothing. Appendix C.6 verifies the proposition by setting $\psi = 0$ under either financing. The two cases deliver the same path for every variable. Tradable consumption rises permanently by 0.05% in both, the permanent-income value of the convenience income earned on the position, and the country's foreign claims, 0.29 quarters of GDP on impact, sit with households in one case and with the central bank in the other.

With the calibrated ψ the equivalence breaks, and Table 5 measures by how much. The portfolio cost is a cost of holding foreign bonds that households bear and the central bank does not, so the split of the position, irrelevant a moment ago, now decides who pays it. Rebated, the proceeds push household foreign holdings far above the position they are willing to hold given the cost. Households consume the gain rather than carry the position: tradable consumption rises by 0.41% on impact and decays as the position returns to its desired level, so the gain is consumed within the episode rather than kept as a position whose income is consumed forever. The appreciation is exactly that absorption, since consuming more tradables out of a fixed endowment means importing them and the currency appreciates to let the imports in, so that $\hat{S}_t = -\hat{C}_{T,t}$ holds identically under the target. Invested in reserves, the proceeds never enter household portfolios and no cost is paid on them. Consumption rises to 0.14%, stays there for three years and then declines as households run down the small position they rebuild after the purchase (Appendix C.6).

The size of the operation follows from the same mechanism through the foreign demand curve. Since the exchange rate is the mirror of consumption, the expected exchange-rate change that prices foreign demand in (42) is the expected consumption growth rate, which the tradable Euler equation ties to the household's foreign position: part (iii) of Proposition 2. Under the rebate, consumption is high and falling, so the exchange rate is on a path back toward steady state, an expected depreciation that lowers foreign demand. Under reserve backing consumption is low and flat, so the appreciation is smaller and reverses slowly. With no depreciation to expect, the same flight brings 8% more foreign demand, which the operation must accommodate. The decomposition is exact. At impact the flight alone contributes $\phi_1^* \hat{\theta}^* = 18.25\%$ to foreign demand. The portfolio term subtracts 1.3 points under the rebate, where the franc is expected to fall, and adds

0.02 points under reserve backing, where it is expected to stay put, which reproduces the two columns of Table 5. Without the portfolio cost consumption is flat under both financing cases, the portfolio term vanishes, foreign demand equals the flight term, and the operation is 5.0% of the stock under either financing: the smaller operation behind the rebate-financed target is the counterpart of the consumption gain the rebate sets off. Switzerland ran the reserve-backed branch, and Section 5.6 sets both branches against the acute phase.

5.6 The Acute Phase in the Data

In Table 6 I set the model's branches against the four quarters from 2008Q3 to 2009Q2, the window the shock is fitted to. The comparison is one of orders of magnitude, for two reasons. The model's responses are those to a single shock, the window contains whatever arrived in it. And the SNB began buying foreign currency on 12 March 2009, inside the window, so the observed year is neither the economy with a fixed supply nor the economy that accommodated the inflow. The model's branches should bracket it, not reproduce it.

Table 6: The acute phase, 2008Q3 to 2009Q2, and the model's branches. Orders of magnitude only. The second column is the model's account of the window, the constrained economy of Section 5.3. The last three columns are counterfactuals: the same flight without the bound, and with the domestic target of Section 5.4 under the two financings of Section 5.5. The window contains the first interventions, so the observed year lies between the second column and the last. Model entries are impact responses, except the appreciation rows, which report the level of the exchange rate after four quarters and at its peak, and the rate row, which reports the level at the end of the window.

	Data	Bound	No policy	Domestic target	Reserve backed
S after four quarters (%)	5.9	4.1	1.7	0.4	0.1
S peak (%)	5.9	7.9	5.9	0.4	0.1
i^R end of window (% p.a.)	0.25	0	−0.9	1.0	1.0
y_N (%)	−2 to −3	−3.1	−0.3	0	0
π_N change (pp p.a.)	−4	−3.3	−1.2	0	0
q^* (% of position)	+30	+3.8	+3.7	+17	+18
q^* (% of annual GDP)	+12	+1.5	+1.5	+7	+7
f^g (% of annual GDP)	+6	0	0	0	+7

Data: trade-weighted nominal exchange rate, SNB policy rate target midpoint, real GDP, CPI inflation, IIP liabilities in the three convenient categories of Section 4.3, and SNB foreign currency investments, 2008Q2 to 2009Q2 for the stocks, in percent of the annual GDP of the window. Source: SNB data portal (Swiss National Bank, 2025a,b,c,d, 2026b) and BIS (Bank for International Settlements, 2025); changes are the author's calculation. The model's steady-state rate is the window mean of the SNB target, 1.0%, so the rate row reports levels at the end of the window rather than changes from the eve.

The table says three things. The bound branch reproduces the window on every domestic row to the order of magnitude: a rate at the floor, a contraction of a few percent, a fall into deflation, and an appreciation of the observed size within the window.

The unconstrained economy is closer on the eventual size of the appreciation and further on everything else, which is the lower bound result of Section 5.3 read against one year of data. The stock rows do not match, and with a fixed supply they cannot: the 30% rise in foreign holdings, 12% of annual GDP, needs a supply response, which the fixed-supply branches cannot produce and the target branches can. The SNB began buying foreign currency in March 2009, inside the window, so on the policy rate, output and the central bank's position the observed year lies between the bound branch and the reserve-backed one, as a year in which the balance sheet began to respond should. On the appreciation, inflation and the inflow the data lie beyond every column: the fitted flight is smaller than the window's. The appreciation is fitted and the other rows are read against it, so the table claims no more than that the mechanism is the right size. The target is the eventual peak of the exchange-rate level, which the model reaches after years, matched to the window's four-quarter change. This is conservative: the franc rose a further 27% by 2011Q3. The flight did not end with the window. The franc appreciated by a further 27% by 2011Q3, nonresident holdings and the SNB's position kept growing, and the policy rate reached zero in 2011. That is a flight that did not revert, arriving on positions each wave had enlarged, and it is the subject of Section 5.7.

5.7 Persistence

The previous subsections studied one flight. Between 2008 and 2015 Switzerland absorbed several, and nonresident holdings of the asset rose by 120%. This subsection asks what changes when the flight does not reverse. I run two experiments. In the first the flight is permanent. In the second it lasts a known number of quarters and then stops. Both keep the supply of the convenient asset fixed and the Taylor rule in place, so they extend the economies of Sections 5.1 and 5.3.

A permanent flight is the limit $\rho_{\theta^*} \to 1$. I set $\rho_{\theta^*} = 0.9999$ and keep the shock at 36.5, so θ^* stays at its impact level. Figure 12 reports the responses, and Figure 22 in Appendix C.8 the full model over 120 quarters. Foreign holdings rise by 3.7% and domestic holdings fall by 1.4%, as in the baseline, and they stay there. The flow does not depend on how long the flight lasts. The domestic adjustment does. Domestic households lose 1.4% of their convenient asset forever, so the wedge is permanently higher, by 0.9 percentage points a year, and non-traded demand (40), which responds to the policy rate plus the wedge, needs the real policy rate permanently lower by the same amount. Under a Taylor rule with a coefficient above one the real rate falls only if inflation falls. The economy settles into permanently lower inflation. Non-traded inflation is 1.7 points below target, 0.3% a year, the policy rate 2.6 points below its steady state, at -1.6%, and the exchange rate appreciates at 1.7% a year without end, 18% after ten years. Non-traded

output is unchanged, −0.04%. Tradable consumption is 1.0% higher on impact and 0.8% in the long run, and total consumption 0.4% higher, since the country sells convenience services abroad forever. The unconstrained rule absorbs a permanent flight at no output cost. It does so by holding inflation below target forever.

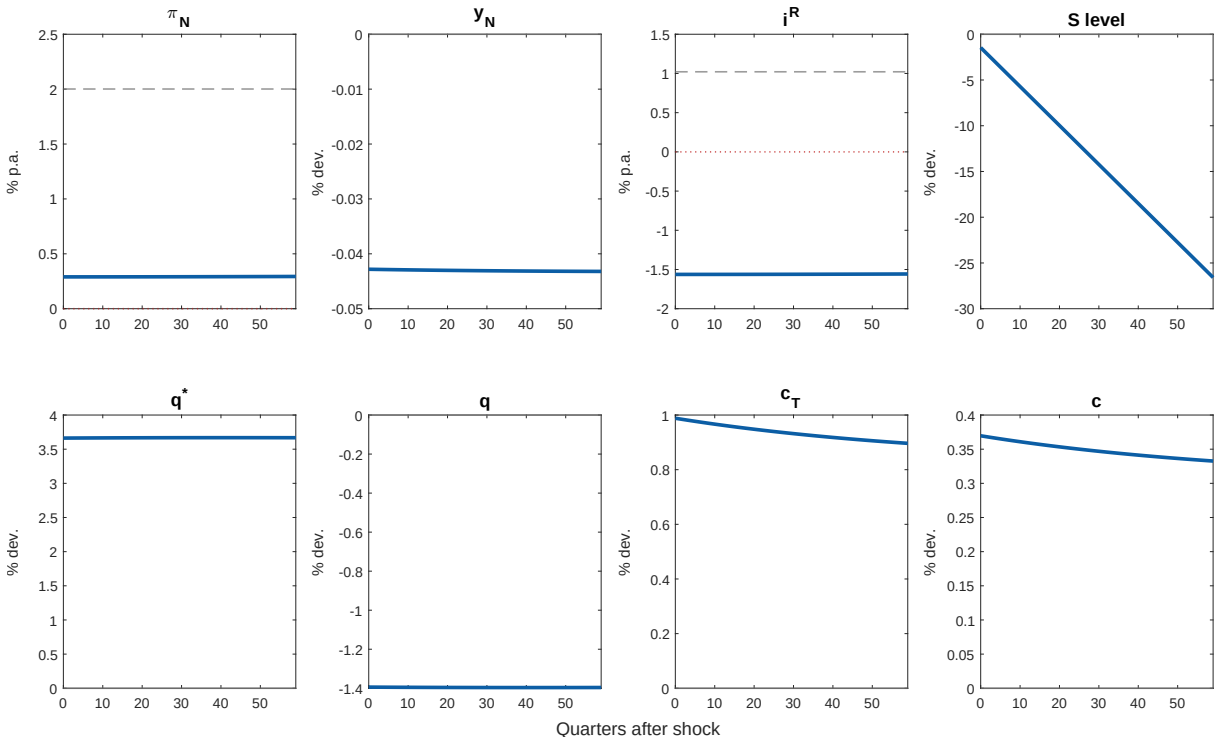


Figure 12: Impulse responses to a permanent flight-to-safety shock, $\rho_{\theta^*} = 0.9999$, in the unconstrained economy with a fixed supply of the convenient asset. All variables are percent deviations from their steady state. Interest rate and inflation are reported in annualized terms relative to their steady-state values. The exchange-rate level is the cumulated change and does not settle.

The bound rules this path out, and the statement is exact.

Proposition 4 (A flight that does not end)**.** *Let the supply of the convenient asset, the government's foreign position, the foreign bond supply and tradable output stay at their steady-state values, let the policy rate follow the Taylor rule with the floor, $\hat{\imath}_t^R = \max\{\phi_\pi \hat{\pi}_{N,t} + \phi_y \hat{Y}_{N,t}, \underline{\imath}\}$ with $\underline{\imath} < 0$, and let the flight be permanent, $\hat{\theta}_t^* = \bar{\theta}^* > 0$ for all t. Under a permanent flight the deviations do not return to zero. A steady state is a constant solution of the linearized system, measured from the original steady state, with a constant rate of depreciation allowed. Write $\hat{r} \equiv \hat{\imath}^R - \hat{\pi}_N$ for the real rate.*

(i) In every steady state $\hat{f} = 0$, $\hat{q} = -\frac{q^*}{q}\hat{q}^*$, and the real rate, foreign holdings and tradable consumption solve

$$(1-\nu)\hat{r} = \frac{\nu}{\phi_1}\hat{q} - \nu\hat{C}_T, \qquad \hat{q}^* = \phi_1^*\bar{\theta}^* - \varphi_2^*\hat{r}, \qquad s_c\hat{C}_T = s_q\left[\left(1 - \tfrac{1-\nu}{\beta}\right)\hat{q}^* - \tfrac{1-\nu}{\beta}\hat{r}\right],$$

three equations that contain neither the rule nor the floor. Their solution is $\hat{r} =$

$-a_r\bar{\theta}^*$, *with* a_r *given in the proof and positive whenever* $\nu s_q < \beta s_c$ *and* $\frac{q^*}{\phi_1 q} + \frac{s_q}{s_c}\left(1 - \frac{1-\nu}{\beta}\right) > 0$. *A permanent flight lowers the steady-state real rate on the convenient asset in proportion to its size, whatever the nominal regime.*

(ii) In every steady state $\hat{Y}_N = \frac{1-\beta}{\kappa}\hat{\pi}_N$. *Let* $\Phi_0 \equiv \phi_\pi + \phi_y \frac{1-\beta}{\kappa}$ *be the long-run response of the nominal rate to inflation under the rule and* $\Phi_1 \equiv \Phi_0 - 1 > 0$ *that of the real rate. With the rule active,* $\hat{\pi}_N = \hat{r}/\Phi_1$ *and* $\hat{\imath}^R = \Phi_0 \hat{r}/\Phi_1$. *At the floor,* $\hat{\imath}^R = \underline{\imath}$ *and* $\hat{\pi}_N = \underline{\imath} - \hat{r}$.

(iii) Let $\bar{\theta}^*_{\max} \equiv \frac{|\underline{\imath}|}{a_r}\frac{\Phi_1}{\Phi_0}$, *the flight at which the nominal rate the rule prescribes in the long run reaches the floor. For* $\bar{\theta}^* < \bar{\theta}^*_{\max}$ *the economy has two steady states, one with the rule active and one at the floor. At* $\bar{\theta}^*_{\max}$ *the two coincide. For* $\bar{\theta}^* > \bar{\theta}^*_{\max}$ *there is none.*

(iv) For $\bar{\theta}^* > \bar{\theta}^*_{\max}$ *the linearized model has no bounded equilibrium at all, deterministic or stochastic.*

(v) a_r *is proportional to* ν *to first order, so, holding the steady-state ratios and the floor fixed,* $\bar{\theta}^*_{\max} \to \infty$ *as* $\nu \to 0$*: in a low-wedge regime no flight is large enough to push the rule to the floor.*

The proof is in Appendix B. In words, the proposition says five things. First, a flight that never ends raises the wedge for good, and since non-traded demand responds to the policy rate plus the wedge, the real policy rate must fall by the same amount to keep non-traded output at its normal level, 0.9 percentage points a year at the fitted flight. Nothing about monetary policy enters this. Second, monetary policy decides only how the lower real rate comes about. Under the rule it comes through lower inflation, twice the fall in the real rate because the rule reacts more than one for one, with the nominal rate falling three times as much. At the floor the nominal rate is zero and inflation is whatever gives the real rate. Third, each way works only up to a size of the flight, and the size is the same for both: the flight at which the rate the rule prescribes in the long run reaches zero. Below it the economy has two steady states, the pair of Benhabib et al. (2001), the intended one and one at the floor, and the flight moves them toward each other. At the threshold they merge. Above it there is none: the rule's steady state asks for a rate below zero, which is where the unconstrained economy of Figure 12 goes, and at the floor inflation would stay so close to target that the rule would leave the floor. At the calibration the threshold is 14.3, two fifths of the fitted flight. Fourth, above the threshold there is no path at all along which the economy stays bounded, constant, cyclical or random. Averaged over time, such a path would have to show a real rate as low as the flight demands and no lower than the rule and the floor allow, and the two

are incompatible. Where the economy goes is a question for the nonlinear model that the paper does not answer. Fifth, none of this happens without the wedge. The fall in the real rate per unit of flight is proportional to ν, so as $\nu \to 0$ no flight is large enough to take the rule to the floor. At the normal-times premium of 27 basis points the threshold is 161, four times the fitted flight. A permanent flight is the case in which the rate cannot do the job, and only in the crisis regime.

Between a flight that fades and one that never ends lies a flight that lasts. I hold θ^* at its impact level for N quarters and set it to zero afterwards. The duration is known, the solution is perfect foresight, and the bound enters as a complementarity condition. Figure 13 shows a flight of twelve quarters, the time between the Lehman flight and the floor, and Figure 23 in Appendix C.8 the full model. The rate is at the floor for seven quarters. Non-traded output falls by 2.1% on impact and non-traded inflation by 2.1 points, to -0.1% in level, the rule would cut to -3.0% and cannot, and the exchange rate appreciates by 2.6% on impact and 2.9% at its peak. Table 11 in the same appendix reports flights of eight, twelve, sixteen and twenty quarters. The inflow is 3.7% to 4.1% of the position in every column, as before. The cost grows faster than the duration. The impact recession is 0.9% at eight quarters, 2.1% at twelve, 5.0% at sixteen and 13% at twenty, and the floor binds for three, seven, eleven and fifteen quarters. The reason is the trap. With the rate at the floor, every quarter ahead in which inflation is expected below target raises the real rate today, and a longer flight puts more of them ahead. Because the exit is anticipated, expected inflation recovers before it and the rate leaves the floor four to five quarters before the flight ends. A flight of twenty quarters carries the same cumulated shock as the baseline, in which θ^* decays at 0.95. Held at full strength instead of fading, the same flight turns the 3.1% recession at the bound of Section 5.3 into one of 13%. Responses of this size are outside what the linearized model is meant for. I report them because they show how fast the fixed-supply economy approaches the limit of Proposition 4. Beyond twenty-four quarters the solver no longer converges at the fitted size, and Appendix C.8 says what that does and does not establish.

The quantity instrument does not meet this limit. Proposition 2 holds for any path of the inflow, so it holds for one that never ends. The supply holds the wedge at its steady-state value, so there is no permanently higher wedge for the real rate to offset. Under the domestic target non-traded inflation, output and the policy rate stay at steady state for a permanent flight as for a transitory one, and the bound is never approached. The operation is permanent. In the long run the portfolio term in part (iii) vanishes and the operation is $\frac{q^*}{q+q^*}\phi_1^*\hat{\theta}^*$, 5.0% of the stock, 7% of annual GDP, held for as long as the flight lasts. A flight that does not end requires a balance sheet that does not shrink.

This is the configuration Switzerland ran. The flight did not end with the acute

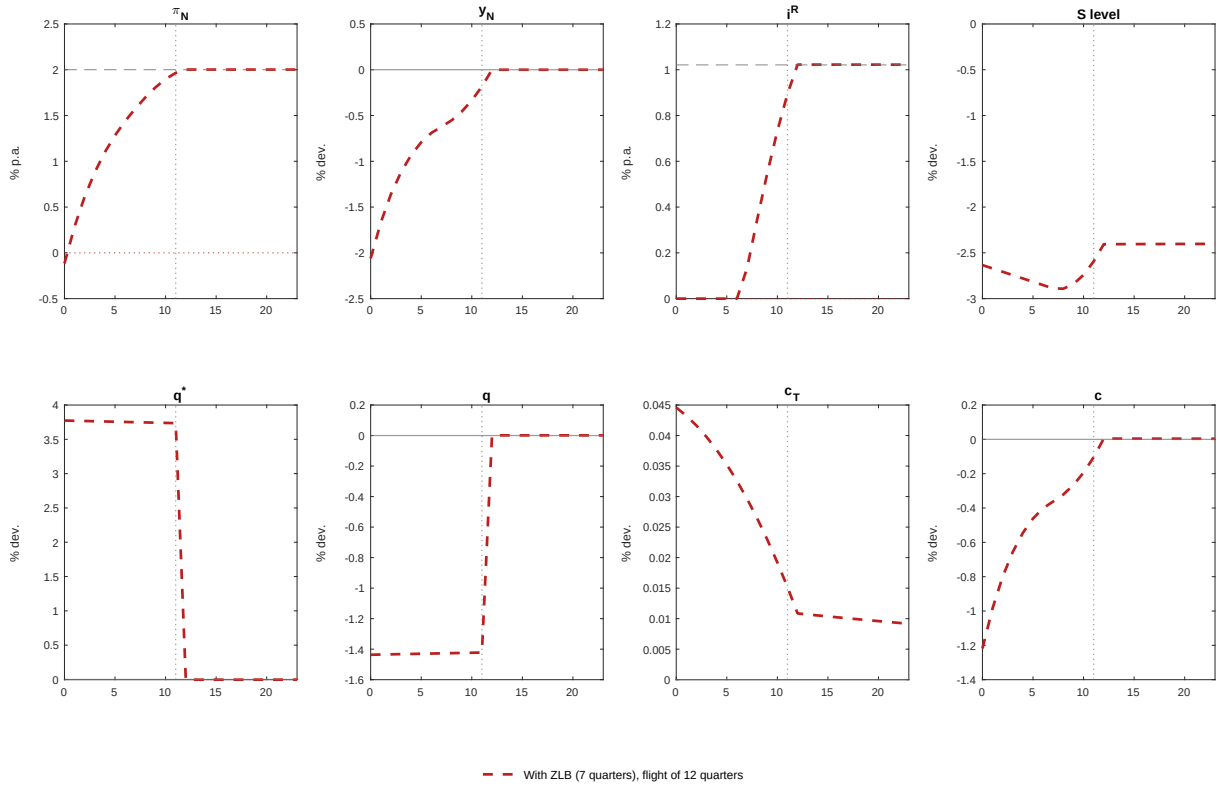


Figure 13: Impulse responses to a flight-to-safety shock that lasts twelve quarters, with the zero lower bound. The foreign preference shifter is held at its impact level for twelve quarters and set to zero afterwards, and agents know the duration. All variables are percent deviations from their steady state. Interest rate and inflation are reported in annualized terms relative to their steady-state values. The policy rate is at the floor for seven quarters and leaves it five quarters before the flight ends.

phase. From 2009Q2 to 2011Q3 the franc appreciated by a further 27%, the policy rate was lowered to zero in August 2011, and the SNB fixed the floor against the euro the following month. Nonresident holdings of the asset rose from 53% of annual GDP in 2009Q2 to 87% in 2015, and the SNB's foreign currency investments from 14% to 67% by the end of 2012. Output did not fall over these years, and inflation fell to zero and briefly below it, not into the deflation the fixed-supply economy predicts for a flight of this length. The model reads the decade as the quantity branch of Section 5.4: a flight that kept arriving, met by a balance sheet that kept growing. The sizes are not comparable, for two reasons. The 7% of GDP is the operation for the flight of 2008, on the stock of 2008. The operation is a share of the stock, 5% of the outstanding convenient asset, and each wave arrived on a stock the previous one had enlarged, nonresident holdings alone from 53% of GDP in 2009 to 87% in 2015, so the same share meant a larger operation each time. And the flights that followed were larger than the one fitted here, and from September 2011 they were met under an exchange-rate target, the branch of Section 5.4 that meets the whole foreign demand and has no fixed size. What the model matches is the direction: a balance sheet that grows with the flight and with the stock, in place of a rate that cannot.

6 Conclusion

This paper studies what a flight to safety does to the economy that receives it, and what its central bank can do about it. In an open-economy New Keynesian model, domestic assets provide convenience services to residents and foreigners, and their holders accept a lower return on them, the convenience yield. A flight works through two channels. It brings income, which households spend on traded goods, and the currency appreciates. It also raises the convenience yield at home: residents hold less of the asset and value what they keep more. Domestic demand responds to the policy rate plus the yield, so the flight works like a monetary tightening the central bank did not choose. Because the yield is the price of a quantity the central bank issues, the supply of the asset is a second instrument.

The flight reaches the domestic economy only through the convenience yield, and its effect is proportional to the level of the yield (Proposition 1). At the normal-times yield of 27 basis points the inflow leaves little trace. At the crisis level of 300 basis points the same flight does thirteen times as much. Fitted to the franc's appreciation in the acute phase of the GFC, the flight takes the policy rate to its lower bound, and the economy falls into a liquidity trap of foreign origin. The rate is pushed to the bound by the world's demand for the economy's assets, not by a collapse of domestic demand. Non-traded output falls nine times as much as without the bound, non-traded inflation falls below zero, and the bound binds for thirteen quarters. The trap also costs the country its income from the asset, because the sharper appreciation raises the real value of what it already owes abroad. The privilege is lost when it would be worth the most. Set against the acute phase, the economy at the bound has the sign of every domestic variable right and the sizes to the order of magnitude, although only the appreciation is fitted.

The balance sheet changes this picture. By supplying the asset the world wants, the central bank holds the convenience yield at its steady-state value. Non-traded output and inflation stay at their steady state and the policy rate never moves, for any path of the flight (Proposition 2). For the flight of 2008 the operation is 7% of annual GDP, close to the 6% the SNB bought over the same four quarters. An appreciation remains, because the income from the asset is still spent, but it is less than a tenth of its peak without policy and reverses. Holding the exchange rate instead takes 13% of annual GDP and leaves a mild boom. A balance-sheet expansion is inflationary only when it supplies more of the convenient asset than the flight takes from residents. What the central bank does with the proceeds matters little. Without portfolio costs, rebating them to households and investing them in foreign assets are equivalent (Proposition 3). With portfolio costs, tradable consumption rises three times as much when the proceeds are rebated.

A flight that does not end is the case interest-rate policy cannot handle alone. Under the Taylor rule a permanent flight holds non-traded inflation 1.7 points below target and the policy rate 2.6 points below its steady state, at an output cost of only 0.04%. With the lower bound, beyond a threshold the economy has no steady state and no bounded path (Proposition 4). At the calibration the threshold is two fifths of the fitted flight, and flights of known length approach it fast. The balance sheet still stabilizes, with a permanent operation. The Swiss decade after 2009 matches this in direction. By 2011Q3 the franc had appreciated by a further 27%. Nonresident holdings rose from 53% of annual GDP in 2009 to 87% in 2015, and the SNB's foreign currency investments from 14% to 67% by the end of 2012. Output did not fall, and inflation fell to zero and briefly below it. The sizes are not comparable to the flight of 2008, because later flights arrived on positions the earlier ones had enlarged, and from 2011 they were met under a minimum exchange rate.

The analysis is positive. It characterizes what monetary policy can do, not what it should do. Its main results are properties of the model, not of the calibration: the channel and its proportionality to the yield, exact stabilization for any path of the flight, the neutrality of the foreign assets bought with the proceeds, and the threshold for a permanent flight. Three questions remain open. Ranking the policies requires a welfare criterion. Beyond the threshold of Proposition 4, where the economy goes is a question for the nonlinear model. And applied to the issuer of the reserve currency, with the rest of the world's policy responding, the mechanism would ask the same questions of the asset the whole world holds. The Swiss case shows the structure of the answer. A run on a safe haven is a demand for something its central bank can produce, and producing it keeps the privilege from turning into a trap.

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A Derivations

This section provides more detailed derivations.

A.1 Derivation of the Expenditure Minimization Problem

Equations (3) and (4) are derived from the following expenditure minimization problem of the domestic household. The household's composite consumption good is defined by the Cobb-Douglas aggregator in equation (1). Given prices $P_{N,t}, P_{T,t}$ and a required level of composite consumption C_t, the household chooses $\{C_{N,t}, C_{T,t}\}$ to minimize nominal expenditure:

$$\begin{aligned} \min_{C_{N,t},\, C_{T,t}} \quad & P_{N,t}C_{N,t} + P_{T,t}C_{T,t} \\ \text{s.t.} \quad & C_{N,t}^{\gamma} C_{T,t}^{1-\gamma} \;=\; C_t. \end{aligned} \tag{55}$$

Let λ_t be the Lagrange multiplier on (1). The Lagrangian is given by

$$\mathcal{L} = P_{N,t}C_{N,t} + P_{T,t}C_{T,t} + \lambda_t \Big(C_t - C_{N,t}^{\gamma} C_{T,t}^{1-\gamma} \Big).$$

The first-order conditions are

$$\frac{\partial \mathcal{L}}{\partial C_{N,t}} : \quad P_{N,t} - \lambda_t \, \gamma \, C_{N,t}^{\gamma-1} C_{T,t}^{1-\gamma} = 0, \tag{56}$$

$$\frac{\partial \mathcal{L}}{\partial C_{T,t}} : \quad P_{T,t} - \lambda_t \, (1-\gamma) \, C_{N,t}^{\gamma} C_{T,t}^{-\gamma} = 0. \tag{57}$$

Dividing (56) by (57) yields the optimal intratemporal allocation:

$$\frac{P_{N,t}}{P_{T,t}} = \frac{\gamma}{1-\gamma} \frac{C_{T,t}}{C_{N,t}} \quad \Longrightarrow \quad \frac{C_{N,t}}{C_{T,t}} = \frac{\gamma}{1-\gamma} \frac{P_{T,t}}{P_{N,t}}. \tag{58}$$

Multiplying (56) by $C_{N,t}$ and (57) by $C_{T,t}$ gives the cost shares:

$$P_{N,t}C_{N,t} = \lambda_t \, \gamma \, C_{N,t}^{\gamma} C_{T,t}^{1-\gamma} = \lambda_t \, \gamma \, C_t, \tag{59}$$

$$P_{T,t}C_{T,t} = \lambda_t \, (1-\gamma) \, C_{N,t}^{\gamma} C_{T,t}^{1-\gamma} = \lambda_t \, (1-\gamma) \, C_t. \tag{60}$$

Adding (59) and (60) gives

$$P_{N,t}C_{N,t} + P_{T,t}C_{T,t} = \lambda_t \, C_t. \tag{61}$$

Define the aggregate price index P_t as the unit expenditure (minimum expenditure per unit of C_t), so that the minimized expenditure is $P_t C_t$. Hence $\lambda_t = P_t$, and the cost-share relationships become

$$P_{N,t} C_{N,t} = \gamma \, P_t C_t, \qquad P_{T,t} C_{T,t} = (1-\gamma) \, P_t C_t. \tag{62}$$

Using (58) in the binding constraint $C_t = C_{N,t}^{\gamma} C_{T,t}^{1-\gamma}$ yields the demands

$$C_{N,t} = \gamma \, \omega \, \frac{P_t}{P_{N,t}} \, C_t, \qquad C_{T,t} = (1-\gamma) \, \omega \, \frac{P_t}{P_{T,t}} \, C_t, \qquad \omega \equiv \gamma^{-\gamma} (1-\gamma)^{-(1-\gamma)}. \tag{63}$$

Plugging these into the binding aggregator implies the Cobb-Douglas price index

$$P_t = \omega \, P_{N,t}^{\gamma} \, P_{T,t}^{1-\gamma}. \tag{64}$$

A.2 Derivation of the New Keynesian Phillips Curve

Final Good Producer Final good is produced by combining a continuum of intermediate goods $i \in [0,1]$ with CES aggregator and elasticity of substitution $\theta > 1$:

$$Y_{N,t} = \left(\int_0^1 Y_{N,t}(i)^{\frac{\theta-1}{\theta}} di \right)^{\frac{\theta}{\theta-1}}. \tag{65}$$

Cost minimization implies demand for each intermediate variety:

$$Y_{N,t}(i) = \left(\frac{P_{N,t}(i)}{P_{N,t}} \right)^{-\theta} Y_{N,t}, \tag{66}$$

with CES aggregate price

$$P_{N,t} = \left(\int_0^1 P_{N,t}(i)^{1-\theta} di \right)^{\frac{1}{1-\theta}}. \tag{67}$$

Intermediate Goods Producers Consider a continuum of intermediate goods firms $i \in [0,1]$, each producing non-traded goods with linear technology:

$$Y_{N,t}(i) = A_t H_t(i),$$

where $H_t(i)$ is labour input and A_t is a common productivity shock. Firms pay a nominal wage W_t. At each period t, only a fraction $1-\alpha$ of firms can reset their price $\tilde{P}_{N,t}$, whereas the remaining fraction α keeps their previous price indexed to steady-state inflation Π.

A firm that can reset its price in t must choose a price that stays potentially for k

periods with probability α^k. The intermediate firm chooses the optimal reset price, $\tilde{P}_{N,t}$, to maximize the expected discounted sum of nominal future profits:

$$\max_{\tilde{P}_{N,t}} E_t \sum_{k=0}^{\infty} \alpha^k M_{t,t+k} \Big[P_{N,t+k}(i) Y_{N,t+k}(i) - W_{t+k} H_{t+k}(i) \Big], \tag{68}$$

where:

- $P_{N,t+k}(i) = \Pi^k \tilde{P}_{N,t}$ at time t+k if firm is stuck with the last reset price;
- $Y_{N,t+k}(i) = \left(\frac{\Pi^k \tilde{P}_{N,t}}{P_{N,t+k}} \right)^{-\theta} Y_{N,t+k}$ is the demand for the firm's good at horizon $t+k$;
- $H_{t+k}(i) = Y_{N,t+k}(i)/A_{t+k}$ is the required labour input;
- $M_{t,t+k}$ is the households' stochastic discount factor from t to $t+k$[24];
- α^k captures the probability that the firm's price remains effective over k periods.

Differentiating with respect to $\tilde{P}_{N,t}$ yields the following FOC:

$$E_t \sum_{k=0}^{\infty} \alpha^k M_{t,t+k} P_{N,t+k}^{\theta} Y_{N,t+k} (\Pi^k)^{1-\theta} \tilde{P}_{N,t} (\theta - 1) - E_t \sum_{k=0}^{\infty} \alpha^k M_{t,t+k} P_{N,t+k}^{\theta} Y_{N,t+k} (\Pi^k)^{-\theta} \theta \frac{W_{t+k}}{A_{t+k}} = 0,$$

which can be solved for $\tilde{P}_{N,t}$ to obtain

$$\tilde{P}_{N,t} = \frac{\theta}{\theta - 1} \frac{E_t \sum_{k=0}^{\infty} \alpha^k M_{t,t+k} P_{N,t+k}^{\theta} Y_{N,t+k} (\Pi^k)^{-\theta} \frac{W_{t+k}}{A_{t+k}}}{E_t \sum_{k=0}^{\infty} \alpha^k M_{t,t+k} P_{N,t+k}^{\theta} Y_{N,t+k} (\Pi^k)^{1-\theta}},$$

whereas $\frac{W_{t+k}}{A_{t+k}}$ are marginal cost of producing one more unit and $\mu \equiv \frac{\theta}{\theta - 1}$ is the steady state markup.

Using $M_{t,t+k} = \beta^k \frac{C_t P_t}{C_{t+k} P_{t+k}}$ and dividing and multiplying with $P_{N,t+k}$[25] yields the optimal reset price chosen at time t:

$$\frac{\tilde{P}_{N,t}}{P_{N,t}} = \mu \frac{E_t \sum_{k=0}^{\infty} (\alpha\beta)^k \frac{1}{A_{t+k}} \frac{W_{t+k} Y_{N,t+k}}{P_{t+k} C_{t+k}} \left(\frac{P_{N,t}}{P_{N,t+k}} \Pi^k \right)^{-\theta}}{E_t \sum_{k=0}^{\infty} (\alpha\beta)^k \frac{P_{N,t+k} Y_{N,t+k}}{P_{t+k} C_{t+k}} \left(\frac{P_{N,t}}{P_{N,t+k}} \Pi^k \right)^{1-\theta}}. \tag{69}$$

The numerator corresponds to discounted expected marginal costs weighted by output, whereas the denominator corresponds to discounted expected revenues weighted by output.

[24] Firms are owned by households.

[25] For example, $P_{N,t+k}^{\theta} (\Pi^k)^{-\theta} = P_{N,t}^{\theta} (\frac{P_{N,t}}{P_{N,t+k}} \Pi^k)^{-\theta}$

Solving for the reset price yields:

$$\left(\frac{\tilde{P}_{N,t}}{P_{N,t}}\right)^{1-\theta} = \frac{1-\alpha\left(\frac{\Pi_{N,t}}{\Pi}\right)^{\theta-1}}{1-\alpha}, \quad \Pi_{N,t} = \frac{P_{N,t}}{P_{N,t-1}}. \tag{70}$$

Aggregate Price Index under Calvo Raising both sides of (67) to the power $1-\theta$ gives the convenient linear form

$$P_{N,t}^{1-\theta} = \int_0^1 P_{N,t}(i)^{1-\theta}\, di.$$

At date t a mass $1-\alpha$ of firms are selected (randomly) and reset their price to the common optimal reset price $\tilde{P}_{N,t}$. The remaining mass α do not reset and keep the prices they last set previously indexed by steady-state inflation: $P_{N,t}(i) = \Pi P_{N,t-1}(i)$. Because of continuum of measure 1 there are two subsets: Resetters, R_t, of measure $1-\alpha$ and Non-Resetters, N_t, of measure α, and the integral above can therefore be solved:

$$\begin{aligned}\int_0^1 P_{N,t}(i)^{1-\theta}\, di &= \int_{R_t} \tilde{P}_{N,t}(i)^{1-\theta}\, di + \int_{N_t} (\Pi P_{N,t-1}(i))^{1-\theta}\, di \\ &= (1-\alpha)\tilde{P}_{N,t}^{1-\theta} + \alpha\Pi^{1-\theta} P_{N,t-1}^{1-\theta}.\end{aligned}$$

The price index is therefore given by

$$P_{N,t}^{1-\theta} = (1-\alpha)\,\tilde{P}_{N,t}^{1-\theta} + \alpha\,\Pi^{1-\theta}\,P_{N,t-1}^{1-\theta}. \tag{71}$$

Aggregate Supply and Recursive Representation Equations (69) and (71) summarize the aggregate supply. They can be combined and simplified for equilibrium. The numerator and denominator sums in (69) can be defined recursively. Firstly, the sum in the numerator can be written as

$$\begin{aligned}K_t^{\text{sum}} &= \mu(\alpha\beta)^0 E_t\left[\frac{1}{A_t}\frac{W_t Y_{N,t}}{P_t C_t}\left(\frac{P_{N,t}}{P_{N,t}}\Pi^0\right)^{-\theta}\right] \qquad (72)\\ &\quad + \mu\sum_{k=1}^{\infty}(\alpha\beta)^k E_t\left[\frac{1}{A_{t+k}}\frac{W_{t+k}Y_{N,t+k}}{P_{t+k}C_{t+k}}\left(\frac{P_{N,t}}{P_{N,t+k}}\Pi^k\right)^{-\theta}\right] \\ &= \mu\frac{1}{A_t}\frac{W_t Y_{N,t}}{P_t C_t} + \mu\sum_{k=1}^{\infty}(\alpha\beta)^k E_t\left[\frac{1}{A_{t+k}}\frac{W_{t+k}Y_{N,t+k}}{P_{t+k}C_{t+k}}\left(\frac{P_{N,t}}{P_{N,t+k}}\Pi^k\right)^{-\theta}\right], \qquad (73)\end{aligned}$$

which can be rewritten recursively as

$$K_t = \mu \frac{Y_{N,t}}{A_t} + \alpha\beta E_t\Big[\left(\frac{\Pi_{N,t+1}}{\Pi}\right)^{\theta} K_{t+1}\Big]. \tag{74}$$

Secondly, the same can be done for the denominator

$$\begin{aligned} D_t^{\text{sum}} &= (\alpha\beta)^0 E_t\left[\frac{P_{N,t}Y_{N,t}}{P_tC_t}\left(\frac{P_{N,t}}{P_{N,t}}\Pi^0\right)^{1-\theta}\right] \\ &\quad + \sum_{k=1}^{\infty}(\alpha\beta)^k E_t\left[\frac{P_{N,t+k}Y_{N,t+k}}{P_{t+k}C_{t+k}}\left(\frac{P_{N,t}}{P_{N,t+k}}\Pi^k\right)^{1-\theta}\right] \\ &= \frac{P_{N,t}Y_{N,t}}{P_tC_t} + \sum_{k=1}^{\infty}(\alpha\beta)^k E_t\left[\frac{P_{N,t+k}Y_{N,t+k}}{P_{t+k}C_{t+k}}\left(\frac{P_{N,t}}{P_{N,t+k}}\Pi^k\right)^{1-\theta}\right] \end{aligned} \tag{75}$$

which can be rewritten as

$$D_t = \gamma + \alpha\beta E_t\Big[\left(\frac{\Pi_{N,t+1}}{\Pi}\right)^{\theta-1} D_{t+1}\Big], \tag{76}$$

where

$$\gamma = \frac{P_{N,t}Y_{N,t}}{P_tC_t}.$$

Equation (69) can now be rewritten as

$$\frac{\tilde{P}_{N,t}}{P_{N,t}} = \frac{K_t}{D_t}.$$

Solving (71) for $\tilde{P}_{N,t}^{1-\theta}$ gives:

$$\tilde{P}_{N,t}^{1-\theta} = \frac{P_{N,t}^{1-\theta} - \alpha\Pi^{1-\theta}P_{N,t-1}^{1-\theta}}{1-\alpha}.$$

Dividing both sides by $P_{N,t}^{1-\theta}$ gives the reset price relative to the aggregate price:

$$\left(\frac{\tilde{P}_{N,t}}{P_{N,t}}\right)^{1-\theta} = \frac{1-\alpha(\Pi_{N,t}/\Pi)^{\theta-1}}{1-\alpha},$$

where $\Pi_{N,t} = P_{N,t}/P_{N,t-1}$. Taking the $1/(1-\theta)$ power gives the actual reset price ratio:

$$\frac{\tilde{P}_{N,t}}{P_{N,t}} = \left[\frac{1-\alpha(\Pi_{N,t}/\Pi)^{\theta-1}}{1-\alpha}\right]^{\frac{1}{1-\theta}}.$$

Finally, combining with the recursive representation (A.2) yields:

$$\left[\frac{1-\alpha(\Pi_{N,t}/\Pi)^{\theta-1}}{1-\alpha}\right]^{\frac{1}{\theta-1}} = \frac{D_t}{K_t}. \tag{77}$$

A.3 Deterministic Steady State

This appendix solves for the deterministic steady state around which the model is linearized. The linearized system of Section 3.1 is the model the quantitative analysis solves, and the steady state enters it only through the coefficients: the interest rates, the convenience yield, and the steady-state ratios that weight the linearized equations. The calibration in Section 4 supplies the anchors, the policy rate, the convenience yield, and the portfolio positions, and the remaining steady-state objects follow recursively from the equilibrium conditions below. Table 3 reports the resulting values.

The deterministic steady state is obtained by imposing constant allocations, prices and policy instruments on the equilibrium conditions above. The Euler equation for domestic illiquid bond (30) implies

$$\Pi_N = \beta(1+i), \tag{78}$$

and the liquidity condition (31) becomes in steady state

$$\frac{C_T}{1-\gamma} V_q(q) = 1 - \frac{(1+i^R)}{(1+i)}, \tag{79}$$

whereas the steady state convenience yield is given by $\nu \equiv \frac{C_T}{1-\gamma} V_q(q)$. It is the wedge between the return on illiquid domestic bond and the policy return on the convenient asset. The Euler equation for the foreign bond in H (32) implies in steady state

$$\Pi_T = \beta(1+i^{R*}), \tag{80}$$

whereas the foreigners' Euler equation for foreign bond (36) gives

$$\Pi_T^* = \beta(1+i^{R*}). \tag{81}$$

Thus, in steady state, the inflation across countries is equal. The foreign demand for the domestic convenient asset (37) satisfies

$$V_q(q^*) = 1 - \beta \frac{(1+i^R)}{\Pi_T^*}. \tag{82}$$

Current account (38) in steady state is

$$Y_T - C_T = f + f^g - \frac{(1+i^{R*})}{\Pi_T^*}(f + f^g) - q^* + \frac{(1+i^R)}{\Pi_T} q^*. \tag{83}$$

Finally, the steady state must also satisfy the relative-price restrictions and asset

market-clearing conditions

$$\begin{aligned}
\frac{P_N C_N}{P_T C_T} &= \frac{\gamma}{1-\gamma}, \\
P_T &= S P_T^*, \\
\Pi_T &= \Pi_T^*, \\
\Pi_N &= \Pi_T, \\
\Pi &= \Pi_N^{\gamma} \Pi_T^{1-\gamma} = \Pi_T = \Pi_N, \\
f^s &= f^* + f + f^g, \\
q^s &= q + q^*.
\end{aligned}$$

B Proofs

Proof of Proposition 1. With the definition of ν_t, condition (31) reads $\nu_t = \frac{C_{T,t}\theta_t}{1-\gamma}V_q(q_t)$. Taking logs and linearizing around the steady state, with $\phi_1 \equiv -\frac{V_q(q)}{qV_{qq}(q)}$, gives $\hat{\nu}_t = \hat{C}_{T,t} + \hat{\theta}_t - \hat{q}_t/\phi_1$.

From the definitions in Section 3.1, $\phi_2 = \phi_1(1-\nu)/\nu$ and $z = \phi_2/(\phi_1+\phi_2)$, so $z = 1-\nu$, $z/\phi_2 = \nu/\phi_1$ and $1 - z = \nu$. Aggregate demand (40) reads

$$\hat{Y}_{N,t} = (1-\nu)E_t\hat{Y}_{N,t+1} - (1-\nu)\big(\hat{\imath}^R_t - E_t\hat{\pi}_{N,t+1}\big) + \frac{\nu}{\phi_1}\hat{q}_t - \nu\big(\hat{p}_{N,t} + \hat{\theta}_t\big).$$

Substituting $\hat{p}_{N,t} = \hat{C}_{T,t} - \hat{Y}_{N,t}$ from (46) and collecting the terms in $\hat{Y}_{N,t}$ on the left,

$$(1-\nu)\hat{Y}_{N,t} = (1-\nu)E_t\hat{Y}_{N,t+1} - (1-\nu)\big(\hat{\imath}^R_t - E_t\hat{\pi}_{N,t+1}\big) - \nu\Big(\hat{C}_{T,t} + \hat{\theta}_t - \frac{\hat{q}_t}{\phi_1}\Big).$$

The last bracket is $\hat{\nu}_t$, and dividing by $1-\nu$ gives the equation in the proposition.

(i) The Phillips curve (39) and the Taylor rule contain no other variable of the model, so $(\hat{Y}_{N,t}, \hat{\pi}_{N,t}, \hat{\imath}^R_t)$ is determined by these three equations given the path of $\hat{\nu}_t$. The flight $\hat{\theta}^*_t$ appears in none of them. With $\hat{\theta}_t = 0$ it reaches $\hat{\nu}_t$ through $\hat{q}_t$, which market clearing (44) with a fixed supply ties to the inflow, $\hat{q}_t = -\frac{q^*}{q}\hat{q}^*_t$, and through $\hat{C}_{T,t}$, which the current account (43) ties to the inflow.

(ii) The coefficients κ, β, ϕ_π and ϕ_y of the three equations do not involve ν, and ν enters only as the factor $\frac{\nu}{1-\nu}$ on $\hat{\nu}_t$. Under the Taylor principle the three equations have a unique bounded solution for every bounded path of $\hat{\nu}_t$. The system is linear, so this solution is $\frac{\nu}{1-\nu}$ times a linear map of the path of $\hat{\nu}_t$ that does not involve ν. With $\hat{\theta}_t = 0$ and the crowding-out of (i), $\hat{\nu}_t = \hat{C}_{T,t} + \frac{q^*}{\phi_1 q}\hat{q}^*_t$, which for given paths of the inflow and of tradable consumption does not involve ν. The responses are therefore proportional to $\frac{\nu}{1-\nu}$, which equals ν to first order, and they vanish as $\nu \to 0$. □

Proof of Proposition 2. With $\hat{\pi}_{N,t} = 0$ for all t, the Phillips curve (39) gives $\kappa\hat{Y}_{N,t} = 0$, so $\hat{Y}_{N,t} = 0$, and the Taylor rule gives $\hat{\imath}^R_t = 0$. Aggregate demand in the form of Proposition 1 then reads $\hat{\nu}_t = 0$, and with $\hat{\theta}_t = 0$ this is $\hat{q}_t = \phi_1\hat{C}_{T,t}$. Market clearing (44) gives the supply rule in the statement. This proves (i) and the form of the rule. None of these steps uses the current account (43), the only equation in which the government's foreign position appears, so they hold for any path of $\hat{f}^g_t$.

(ii) From (46), $\hat{p}_{N,t} = \hat{C}_{T,t}$. From (47) with $\hat{\pi}_{N,t} = 0$, $\hat{\pi}_{T,t} = \hat{p}_{N,t-1} - \hat{p}_{N,t} = \hat{C}_{T,t-1} - \hat{C}_{T,t}$. By (49) with $\hat{\pi}^*_{T,t} = 0$, $\Delta\hat{S}_t = \hat{\pi}_{T,t}$, and summing from the steady state, $\hat{S}_t = \sum_{s\le t}\Delta\hat{S}_s = -\hat{C}_{T,t}$. Convergence is shown below.

(iii) In the tradable Euler equation (41) the return bracket is $E_t\hat{\pi}_{T,t+1} - E_t\Delta\hat{S}_{t+1}$,

which is zero by (49), so $\hat{C}_{T,t} - E_t\hat{C}_{T,t+1} = \psi f \hat{f}_t$. Substituting $\hat{\imath}_t^R = 0$ and $E_t\Delta\hat{S}_{t+1} = \hat{C}_{T,t} - E_t\hat{C}_{T,t+1}$ into (42) gives $\hat{q}_t^* = \phi_1^*\hat{\theta}_t^* + \varphi_2^*\psi f \hat{f}_t$.

Targeting output. Impose $\hat{Y}_{N,t} = 0$ instead. Then (39) becomes $\hat{\pi}_{N,t} = \beta E_t\hat{\pi}_{N,t+1}$, whose only bounded solution is $\hat{\pi}_{N,t} = 0$, and the rest follows.

Existence and uniqueness. Given $\hat{\pi}_{N,t} = \hat{Y}_{N,t} = \hat{\imath}_t^R = 0$, the remaining unknowns are $\hat{C}_{T,t}$, $\hat{f}_t$, $\hat{q}_t^*$, $\hat{q}_t$, $\hat{q}_t^s$ and the inflation rates, determined by the Euler equation, the current account, (ii), (iii) and the supply rule. The dynamic part is the pair $(\hat{C}_{T,t}, \hat{f}_t)$: the Euler equation $\hat{C}_{T,t} = E_t\hat{C}_{T,t+1} + \psi f \hat{f}_t$, and the current account (43) with $\hat{\imath}_t^R = \hat{f}_t^g = \hat{Y}_{T,t} = 0$, $\hat{\pi}_{T,t} = \hat{C}_{T,t-1} - \hat{C}_{T,t}$ from (ii) and $\hat{q}_t^*$ from (iii). Writing $g \equiv \frac{1-\nu}{\beta}$ and $b \equiv \varphi_2^*\psi f < 0$, the current account reads

$$(s_f - s_q b)\hat{f}_t - \Big(\frac{s_f}{\beta} - s_q g b\Big)\hat{f}_{t-1} + (s_c + s_q g)\hat{C}_{T,t} - s_q g\,\hat{C}_{T,t-1} = \text{exogenous terms}.$$

Eliminating $\hat{f}_t = (\hat{C}_{T,t} - E_t\hat{C}_{T,t+1})/(\psi f)$ gives a second-order equation in $\hat{C}_T$ with characteristic polynomial $A_1\lambda^2 - A_0\lambda + A_{-1}$, where $A_1 = s_f - s_q b > 0$, $A_{-1} = \frac{s_f}{\beta} - s_q g b + \psi f s_q g > 0$ and $A_0 = A_1 + A_{-1} + \psi f s_c$. At $\lambda = 1$ the polynomial equals $A_1 - A_0 + A_{-1} = -\psi f s_c$. For $\psi > 0$ it is negative at $\lambda = 1$ with a positive leading coefficient, so exactly one root lies inside the unit circle and one outside: one stable root for the predetermined position and one unstable root for the free consumption, which is the Blanchard-Kahn condition. The equilibrium exists, is unique, and converges: $\hat{C}_{T,t} \to 0$ and by (ii) $\hat{S}_t \to 0$. At $\psi = 0$ the polynomial vanishes at $\lambda = 1$: a unit root, consumption settles at a permanent level and the exchange rate does not return, Appendix C.6.

Implementation. The economy run with the supply rule of the statement and the Taylor rule, in place of the target, has the same equilibrium and no other. The rule and market clearing force $\hat{q}_t = \phi_1\hat{C}_{T,t}$ at every date, so $\hat{\nu}_t = 0$ identically and the domestic block is the closed-economy block without forcing, whose only bounded solution is zero when the Taylor rule satisfies the Taylor principle. With the domestic block at zero the tradable block is the pair just analyzed. □

Proof of Proposition 3. The household's foreign position $\hat{f}_t$ and the government's $\hat{f}_t^g$ enter the equilibrium conditions in three places: the tradable Euler equation (41) through $\psi f \hat{f}_t$, the current account (43) through $s_f\big(\hat{f}_t - \beta^{-1}\hat{f}_{t-1}\big) + s_f^g\big(\hat{f}_t^g - \beta^{-1}\hat{f}_{t-1}^g\big)$, and foreign bond market clearing (45) through $f\hat{f}_t + f^g\hat{f}_t^g$. With $\psi = 0$ the first vanishes. In the current account both positions earn the same gross return β^{-1}, because the government holds the same foreign bond as households and rebates the balance of its budget to them as a lump-sum transfer, and $s_f/s_f^g = f/f^g$ exactly, so the second and third places depend on the pair only through the consolidated position $\hat{F}_t \equiv f\hat{f}_t + f^g\hat{f}_t^g$. Foreign holdings follow from (45), $f^*\hat{f}_t^* = -\hat{F}_t$ with $\hat{f}_t^s = 0$, and depend on $\hat{F}_t$ only. The rules for the policy rate

and the supply do not condition on $\hat{f}_t$ or $\hat{f}^g_t$ separately by assumption, and the supply rule is otherwise arbitrary. The system in all remaining variables together with $\hat{F}_t$ is therefore the same whatever the path of $\hat{f}^g_t$, and its bounded solution is the same. The path of $\hat{f}^g_t$ only adds the equation that splits $\hat{F}_t$: $\hat{f}^g_t = 0$ under the rebate, $f^g \hat{f}^g_t = (q + q^*)\hat{q}^s_t$ under reserve backing, and an exogenous path under a purchase of foreign assets with the supply fixed. □

Proof of Proposition 4. In a steady state all deviations are constant and expectations equal current values. From (47), $\hat{\pi}_N = \hat{\pi}_T$, and from (49) with $\hat{\pi}^*_T = 0$, $\Delta\hat{S} = \hat{\pi}_T = \hat{\pi}_N$. The tradable Euler equation (41) reduces to $\psi f \hat{f} = \Delta\hat{S} - \hat{\pi}_T = 0$, so $\hat{f} = 0$ since $\psi f > 0$. Market clearing (44) with $\hat{q}^s = 0$ gives $\hat{q} = -\frac{q^*}{q}\hat{q}^*$.

(i) Aggregate demand (40) with constant variables, $z = 1-\nu$, $z/\phi_2 = \nu/\phi_1$ and $\hat{\theta} = 0$ reads $\nu\hat{Y}_N = -(1-\nu)\hat{r} + \frac{\nu}{\phi_1}\hat{q} - \nu\hat{p}_N$. Substituting $\hat{p}_N = \hat{C}_T - \hat{Y}_N$ from (46), the terms in $\hat{Y}_N$ cancel, because the coefficient on the relative price in aggregate demand equals the weight ν and non-traded demand has unit elasticity to the relative price under Cobb-Douglas aggregation, and the first equation of (i) follows. Foreign demand (42) with $\Delta\hat{S} - \hat{\imath}^R = -\hat{r}$ gives the second. The current account (43) with $\hat{f} = \hat{f}^g = \hat{Y}_T = 0$ and $\hat{\pi}_T = \hat{\pi}_N$ gives the third. None of the three contains $\hat{\imath}^R$ or $\hat{\pi}_N$ separately, only $\hat{r}$, so they hold in any steady state whatever the policy regime. Substituting the second and third equations into the first and collecting terms, with $B \equiv \frac{\nu}{1-\nu}$ and $\kappa_q \equiv \frac{q^*}{\phi_1 q} + \frac{s_q}{s_c}\left(1 - \frac{1-\nu}{\beta}\right)$,

$$\hat{r}\left[1 - \frac{\nu s_q}{\beta s_c} - \varphi^*_2 B \kappa_q\right] = -\phi^*_1 B \kappa_q \,\bar{\theta}^*, \qquad \text{so} \qquad a_r = \frac{\phi^*_1 B \kappa_q}{1 - \frac{\nu s_q}{\beta s_c} - \varphi^*_2 B \kappa_q}.$$

In the left-hand bracket the first term is the demand equation, the second the consumption effect of the real rate through the current account, the third the feedback of the real rate on foreign demand, and the right-hand side is the crowding-out per unit of flight. With $\varphi^*_2 < 0$, $\kappa_q > 0$ and $\nu s_q < \beta s_c$ both brackets are positive, so $a_r > 0$. The second condition says that the convenience income on the whole stock is below tradable consumption, the first that the crowding-out of domestic holdings per unit of inflow, $\frac{q^*}{\phi_1 q}$, outweighs the small term in the discount rate, which is itself positive whenever $\nu \geq 1 - \beta$.

(ii) The Phillips curve (39) with constant inflation gives $(1-\beta)\hat{\pi}_N = \kappa\hat{Y}_N$. With the rule active, $\hat{r} = (\phi_\pi - 1)\hat{\pi}_N + \phi_y\hat{Y}_N = \Phi_1\hat{\pi}_N$, and $\hat{\imath}^R = \hat{r} + \hat{\pi}_N = \Phi_0\hat{r}/\Phi_1$. At the floor, $\hat{\imath}^R = \underline{\imath}$ and $\hat{\pi}_N = \underline{\imath} - \hat{r}$.

(iii) A steady state with the rule active requires the prescribed rate to lie at or above the floor, $\Phi_0\hat{r}/\Phi_1 \geq \underline{\imath}$. A steady state at the floor requires the rule's prescription there to lie at or below the floor, $\Phi_0(\underline{\imath} - \hat{r}) \leq \underline{\imath}$. Both inequalities are $\hat{r} \geq \underline{\imath}\,\Phi_1/\Phi_0$. The reason is geometric. In the $(\hat{\pi}_N, \hat{\imath}^R)$ plane a steady state lies on the line $\hat{\imath}^R = \hat{\pi}_N + \hat{r}$, of slope one, and on the kinked rule $\hat{\imath}^R = \max\{\Phi_0\hat{\pi}_N, \underline{\imath}\}$, of slope $\Phi_0 > 1$ above the kink and zero

below it. A line of slope one crosses both branches or neither, and passes through the kink when $\hat{r} = \underline{\imath}\Phi_1/\Phi_0$. Since the real block has the unique solution $\hat{r} = -a_r\bar{\theta}^*$ with $a_r > 0$, the inequality holds strictly for $\bar{\theta}^* < \bar{\theta}^*_{\max}$, with equality at $\bar{\theta}^*_{\max}$, where the two steady states coincide at $\hat{\imath}^R = \underline{\imath}$ and $\hat{\pi}_N = \underline{\imath}/\Phi_0$, and fails above it, where the rule's steady state violates the floor and the floor's steady state violates the complementarity.

(iv) Let $\{z_t\}$ be any bounded equilibrium path, deterministic or stochastic, and write $\bar{z}$ for its time average, $\lim_{T\to\infty} \frac{1}{T}\sum_{t=1}^{T} z_t$ (along a subsequence if the limit does not exist, and in expectation in the stochastic case). Every equation of the model except the rule is linear with constant coefficients. Averaging a bounded sequence and its lead or lag gives the same limit, and the law of iterated expectations removes the conditional expectations, so the averages satisfy the steady-state equations exactly: $\bar{\pi}_N = \bar{\pi}_T = \Delta\bar{S}$, $\bar{f} = 0$, $\bar{Y}_N = \frac{1-\beta}{\kappa}\bar{\pi}_N$, and $(\bar{r}, \bar{q}^*, \bar{C}_T)$ solve the real block of (i), so $\bar{\imath}^R - \bar{\pi}_N = -a_r\bar{\theta}^*$. The rule holds at every date, $\hat{\imath}^R_t = \max\{\phi_\pi\hat{\pi}_{N,t} + \phi_y\hat{Y}_{N,t}, \underline{\imath}\}$, and the maximum is convex, so by Jensen's inequality $\bar{\imath}^R \geq \max\{\Phi_0\bar{\pi}_N, \underline{\imath}\}$. Hence $-a_r\bar{\theta}^* = \bar{\imath}^R - \bar{\pi}_N \geq \max\{\Phi_0\bar{\pi}_N, \underline{\imath}\} - \bar{\pi}_N \geq \underline{\imath}\Phi_1/\Phi_0$, the last step because the function $\max\{\Phi_0 x, \underline{\imath}\} - x$ attains its minimum $\underline{\imath}\Phi_1/\Phi_0$ at the kink. For $\bar{\theta}^* > \bar{\theta}^*_{\max}$ this contradicts $-a_r\bar{\theta}^* < \underline{\imath}\Phi_1/\Phi_0$, so no bounded equilibrium exists.

(v) With $\varphi^*_2 = -\phi^*_1(1-\nu)/\nu$, as in Section 3.1, the last term of the bracket is $\phi^*_1\kappa_q$ and does not depend on ν, so

$$a_r = \frac{\nu}{1-\nu}\,\frac{\phi^*_1\kappa_q}{1 - \frac{\nu s_q}{\beta s_c} + \phi^*_1\kappa_q},$$

which with the steady-state ratios fixed is $\nu\,\phi^*_1\kappa_q/(1+\phi^*_1\kappa_q)$ to first order in ν. Hence $a_r \to 0$ and, with the floor fixed, $\bar{\theta}^*_{\max} \to \infty$ as $\nu \to 0$. □

C Results

This appendix reports the full set of impulse responses for each experiment of Section 5, over 120 quarters and for all model variables, together with quantitative detail that the main text summarizes. Throughout, the shock is an increase in the foreign preference shifter $\hat{\theta}_t^*$ of size 36.5, calibrated so that the peak appreciation of the nominal exchange-rate level is 5.9%, the trade-weighted appreciation of the franc over the acute window 2008Q3 to 2009Q2. The shifter follows an AR(1) with persistence $\rho_{\theta^*} = 0.95$. All variables are reported as percent deviations from their steady state. For interest rates and inflation, these are reported in annualized terms as levels, relative to their steady-state values.

C.1 The Baseline Economy

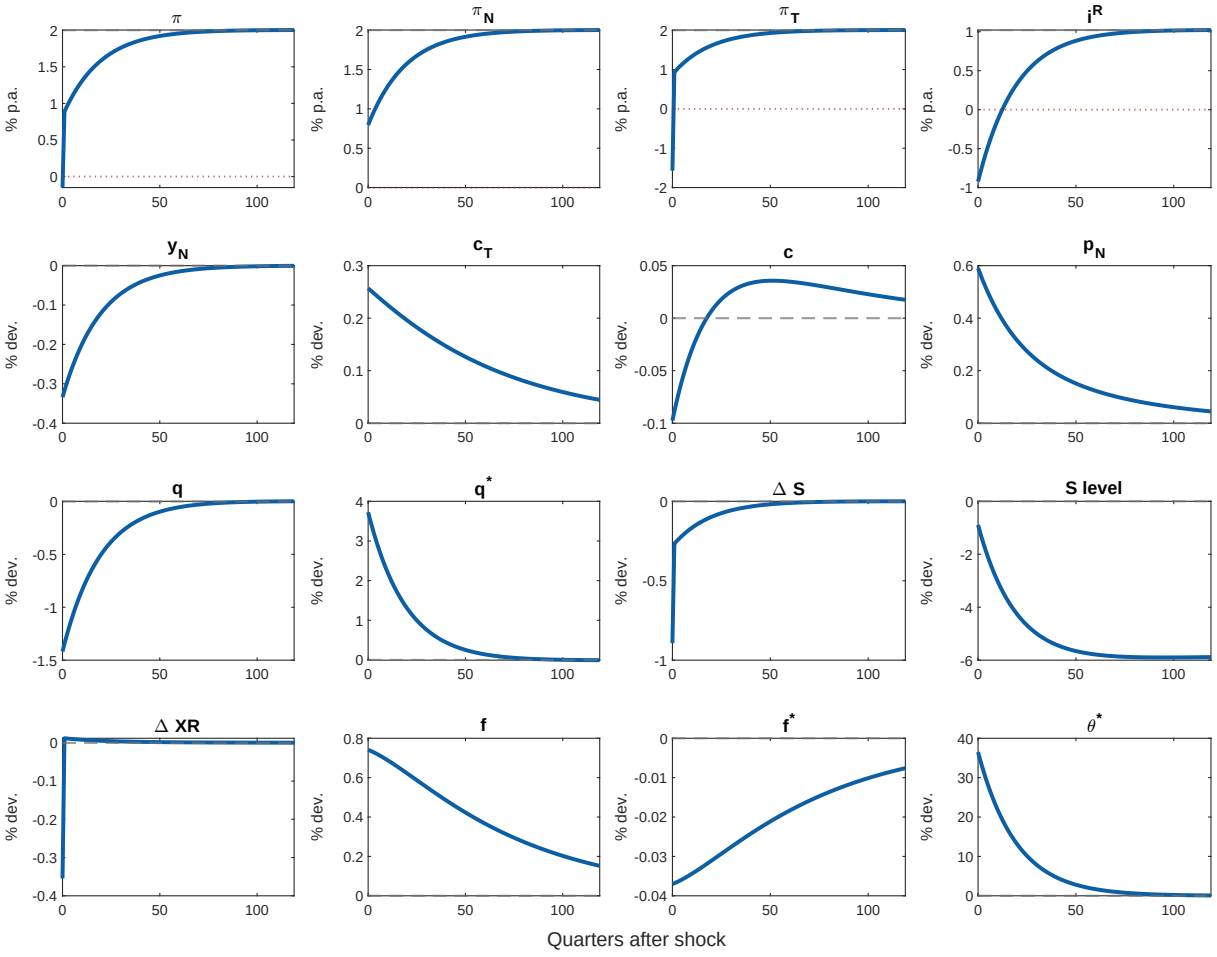


Figure 14: Full set of impulse responses to a flight-to-safety shock under the baseline calibration, over 120 quarters. All variables are percent deviations from their steady state. Interest rate and inflation are reported in annualized terms relative to their steady-state levels.

The model is stationary and all responses return to steady state, but as Figure 14 shows, the return is slow. Tradable consumption and the foreign position are still above

steady state at the end of the figure. The one exception is the level of the nominal exchange rate. Inflation rates return to zero, the price level does not: the episode of tradable deflation leaves the tradable price level permanently lower, the non-traded price level follows it as the relative price returns to zero, and by the law of one price the exchange-rate level is permanently stronger by the same amount. The model has a single steady state in which inflation rates are pinned down and nominal price levels are not.

The rise in foreign demand for the asset is a capital inflow. The current account (43) shows how the inflow is allocated. Using $i^{R*} = 0$, $\pi_T^* = 0$, and the government positions held fixed, the linearized current account can be written as

$$-s_c\,\hat{c}_{T,t} = s_f\Big(\hat{f}_t - \tfrac{1}{\beta}\,\hat{f}_{t-1}\Big) - s_q\,\mathcal{N}_t, \qquad \mathcal{N}_t \equiv \hat{q}_t^* - \frac{1-\nu}{\beta}\Big[\hat{q}_{t-1}^* + \hat{\imath}_{t-1}^R - \hat{\pi}_{T,t}\Big], \tag{84}$$

where $\mathcal{N}_t$ is the period-by-period net resource inflow, allocated between tradable consumption and changes in the foreign-bond position.

Evaluated on impact, the lagged positions are zero and the relation reduces to

$$s_q\,\mathcal{N}_0 = s_f\,\hat{f}_0 + s_c\,\hat{c}_{T,0}, \qquad \mathcal{N}_0 = \hat{q}_0^* + \frac{1-\nu}{\beta}\,\hat{\pi}_{T,0}, \tag{85}$$

where the second term in $\mathcal{N}_0$ is the revaluation of the position foreigners already hold, negative under an appreciation. In numbers, the gross purchase of 3.7 is cut by a carrying cost of 0.9, leaving a net inflow of $\mathcal{N}_0 = 2.8$. These are the two forces of the main text. On the absorption side, the split is governed by the steady-state weights in (85). The bond position rises by three times as much as tradable consumption in percent terms, 0.7 against 0.26, and it is far larger in steady state, $s_f = 15.0$ against $s_c = 1.1$, so 97% of the net inflow is absorbed by purchases of foreign bonds and 3% by higher consumption. The consumption share then grows as the bond position is run down through (41). This is why consumption and the bond position are the persistent panels of the figure while the positions in the convenient asset decay with the impulse. With $\psi = 0$ consumption would be constant at the permanent-income value of the income, so the portfolio cost is what tilts the path toward the present, the mechanism that separates the two financings in Section 5.5. What part of the transfer is convenience-yield revenue and what part a valuation effect is the subject of Appendix C.7.

C.2 The Role of the Convenience Yield

Figure 15 reports the full set of responses over 120 quarters. One implication concerns the net resource transfer. Gross foreign demand saturates at high premia, while the carrying cost of the outstanding foreign position rises with the impact appreciation. The net

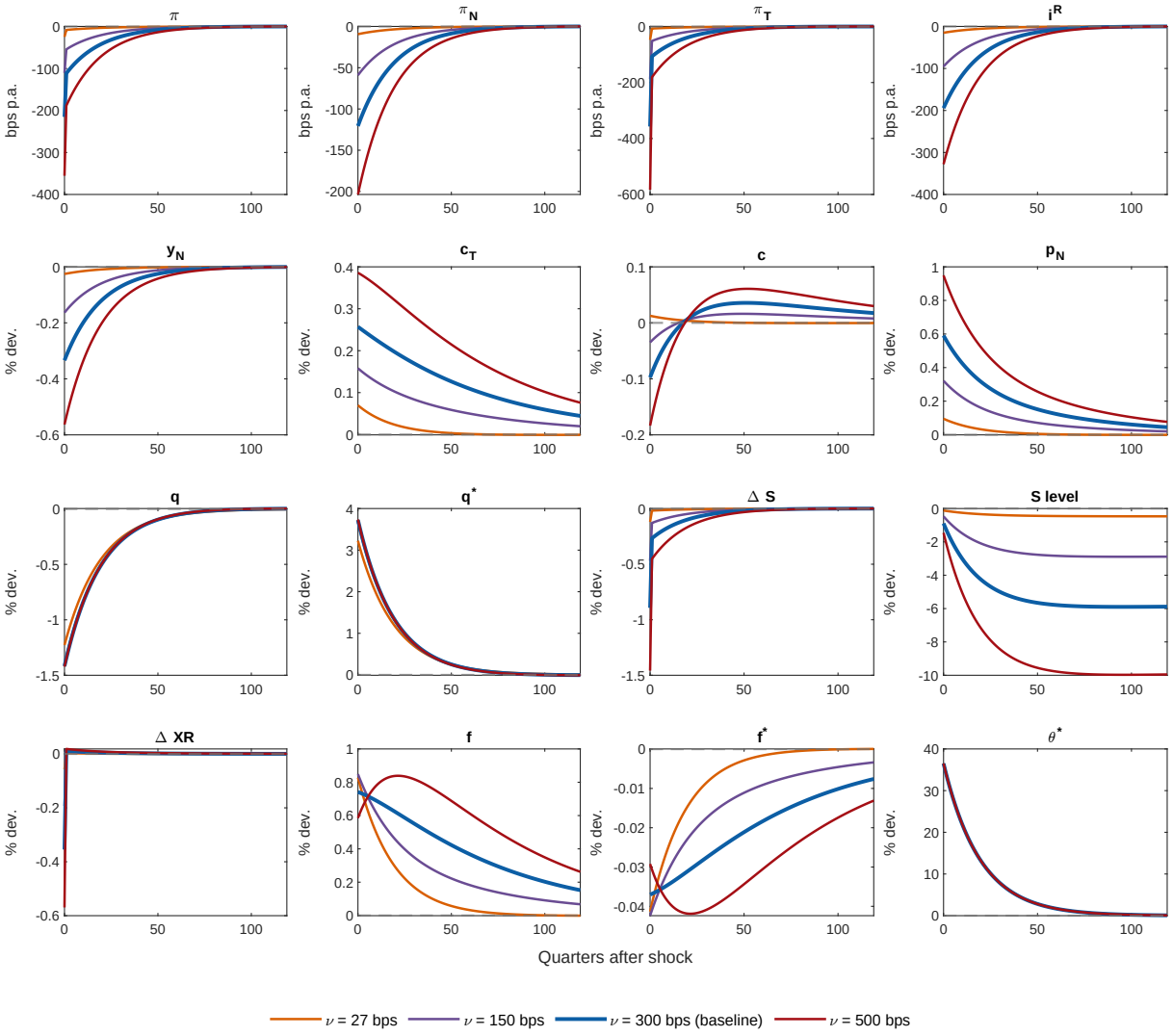


Figure 15: Full set of impulse responses to a flight-to-safety shock for different values of the steady-state convenience yield, over 120 quarters. All variables are percent deviations from their steady state. Interest rate and inflation are reported in annualized terms relative to their steady-state levels. The thick line marks the baseline ($\nu = 300$ basis points).

transfer therefore peaks in the interior of the grid. The impact net inflow of equation (85) rises from 3.1 at 27 basis points to 3.2 at 150 and falls back to 2.8 at the baseline and 2.3 at 500 basis points, where the carrying cost of the deeper impact appreciation absorbs most of the additional financing. It remains positive throughout, so the foreign-bond position rises on impact at every premium, by less at 500 basis points than at the baseline.

Table 7 collects the impact responses. The wedge scales the monetary side of the transmission: appreciation, rate cut, disinflation and the non-traded decline move in proportion to ν, while foreign purchases are 3.2% to 3.7% of the position in every column. The gain in tradable consumption rises with ν, since a larger wedge means more income on the same inflow, but less than proportionally, and the recession rises more. Total consumption therefore falls on impact in every economy with a meaningful wedge, by 0.03% at 150 basis points and 0.18% at 500, and is positive from the start only in normal

times. The timing is the same in the three crisis columns: the recession fades with the shock, the tradable gain persists with the position, so the composite turns positive after five years and its cumulated response is positive and increasing in ν. The two channels of Proposition 1 can be read off the baseline runs: on impact the convenience term $\frac{\nu}{\phi_1}\hat{q}_0$ accounts for 98% of the exogenous push on non-traded demand at 300 basis points and for all of it at 27, where the relative-price term $\nu\hat{p}_{N,0}$ is below 10^{-4}.

Table 7: Sensitivity of the baseline to the steady-state convenience yield ν

	$\nu = 27$ (normal times)	$\nu = 150$	$\nu = 300$ (baseline)	$\nu = 500$
S impact (%)	0.1	0.5	0.9	1.5
S peak (%)	0.5	2.9	5.9	10.0
i^R impact (pp p.a.)	−0.2	−1.0	−1.9	−3.3
π_N impact (pp p.a.)	−0.1	−0.6	−1.2	−2.0
y_N impact (%)	−0.03	−0.16	−0.33	−0.56
q^* impact (%)	+3.2	+3.7	+3.7	+3.7
c_T impact (%)	+0.07	+0.16	+0.26	+0.39
c impact (%)	+0.01	−0.03	−0.10	−0.18
c first positive quarter	1	17	19	20
c cumulative (%-quarters)	+0.2	+1.0	+2.0	+3.1
$\mathcal{N}_0$ (% of position)	3.1	3.2	2.8	2.3

All columns face the common baseline shock, with the convenience yield in basis points per year and all other parameters at their baseline values, so the columns are alternative economies hit by the same flight. c is the composite of (51), its cumulative response the sum of the percent deviations over 120 quarters, and $\mathcal{N}_0$ the impact net inflow of (85).

C.3 Sensitivity to the Demand Elasticity

Figure 16 reports the responses for $\phi_1 \in \{0.04, 0.05, 0.075, 0.30\}$ at the baseline wedge and the common shock of the main text. The foreign coefficient is unchanged across the grid, which isolates the domestic channel: by Proposition 1 the convenience term in aggregate demand carries the coefficient ν/ϕ_1, which falls from 0.15 at the calibrated value to 0.025 at the normal-times one. A higher elasticity lets domestic holders release the asset at a smaller rise in the wedge, so the same flight is accommodated with less pressure on the domestic block. At $\phi_1 = 0.30$ the impact appreciation is 0.6 percent against 0.9 at the baseline, and the long-run level of the exchange rate, pinned by the cumulated deflation of the episode, is 2.9 percent against 5.9. The output contraction and the policy-rate cut are half their baseline size, 0.16 percent and 0.9 points against 0.33 and 1.9, while foreign purchases are three times larger, 10.6 percent of the position against 3.7. In the limit the convenience term $\frac{\nu}{\phi_1}\hat{q}_t$ of Proposition 1 vanishes and only the consumption-scaling term $\nu\hat{p}_{N,t}$ remains, of order $\nu\hat{C}_{T,t}$, so a very elastic domestic demand for the asset makes the flight a portfolio event even at a crisis wedge.

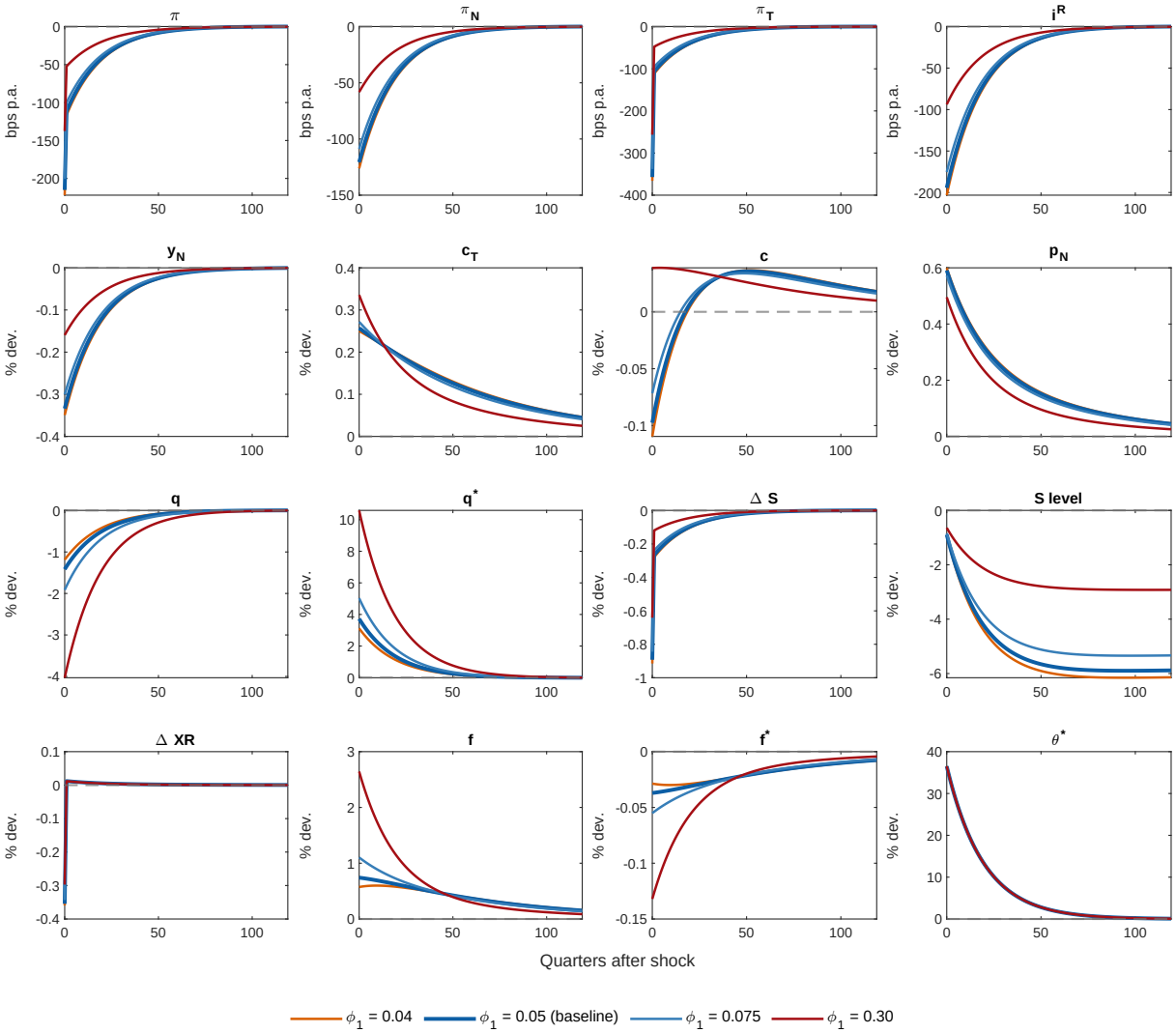


Figure 16: Full set of impulse responses to a flight-to-safety shock for different values of the home demand elasticity, $\phi_1 \in \{0.04, 0.05, 0.075, 0.30\}$, over 120 quarters. The convenience yield is held at its baseline value of 300 basis points and the shock is the common baseline shock, so the four economies face the same flight and differ only in the elasticity. All variables are percent deviations from each economy's own steady state. Interest rate and inflation are reported in annualized terms relative to their steady-state levels. The thick line marks the baseline ($\phi_1 = 0.05$).

The figure holds the shock common across columns. Table 8 holds the appreciation target common instead, rescaling the innovation in each column so that the peak of the exchange-rate level matches the 5.9 percent target. Its columns are alternative calibrations of the same episode rather than different shocks. Both representations are exact given linearity. The values span the sourced range: 0.04 is the floor of the 95 percent confidence set that Benati et al. (2021) estimate for Switzerland on quarterly data, 0.075 is the value implied by the semi-elasticity of Lucas (2000) at the anchor, and 0.30, the value used for normal times, is close to the Swiss point estimate shown in their Figure 5 and lies at the lower end of their cross-country range. The lower bound is reached in every column. Along the common target the domestic block is invariant to the elasticity: the appreciation, the rate cut and the non-traded decline are the same in every column to

the first decimal, because the fit pins the appreciation and the appreciation carries the transmission. What the elasticity disciplines is the portfolio flow behind the observed appreciation, CHF 7 bn at the Swiss floor against 52 bn at the normal-times elasticity, 1% against 9% of annual GDP. At $\phi_1 = 0.30$ the tradable gain behind the larger flow is large enough that total consumption rises on impact, the only column where it does.

Table 8: Sensitivity of the baseline to the home elasticity ϕ_1

	$\phi_1 = 0.04$ (Swiss floor)	$\phi_1 = 0.05$ (baseline)	$\phi_1 = 0.075$ (Lucas)	$\phi_1 = 0.30$ (normal times)
Shock size (stderr)	35.0	36.5	40.3	73.7
S impact (%)	0.9	0.9	0.9	1.3
q^* impact (%)	+3.0	+3.7	+5.6	+21.4
q^* inflow (CHF bn)	7	9	13	52
c_T impact (%)	+0.24	+0.26	+0.30	+0.68
y_N impact (%)	−0.33	−0.33	−0.33	−0.32
c impact (%)	−0.10	−0.10	−0.08	+0.08
Unconstrained cut (pp p.a.)	−1.9	−1.9	−1.9	−1.9

Each column rescales the innovation so that the peak of the exchange-rate level matches the 5.9 percent target, holding all other parameters at their baseline values. The franc inflow applies the impact response to the steady-state stock of CHF 243 bn. The lower bound is reached in every column. The elasticity disciplines the size of the portfolio flow required to generate the observed appreciation, not the transmission to the domestic block.

C.4 The Zero Lower Bound

Figure 17 shows that the zero lower bound binds for thirteen quarters. The deep part of the recession is briefer. Non-traded output recovers from −3.1% on impact to −0.5% by quarter 9, while the bound still binds, because the contractionary impulse fades quickly. The impulse is the shock itself and the appreciation the bound compresses into the impact period. The bound outlasts the recession by five quarters: the rate the rule prescribes stays below zero through quarter 13, so the actual rate lifts off in quarter 14, long after output has normalized. The recovery comes from the impulse fading, not from the bound lifting.

The appreciation follows a different timetable. It is sharper on impact, 3.9% against 0.9%, since no rate cut leans against the flight, and it cumulates deeper. The deflation accumulated during the spell is never undone. Once the relative price completes its adjustment the exchange rate resumes appreciating at the rate of the remaining inflation gap, so the level reaches 7.9% by the end of the horizon, still appreciating slowly, against 5.9% in the unconstrained economy. Both shifts are permanent, by the price-level logic of Appendix C.1. The constrained episode ends with a permanently larger appreciation: the deflation is both sharper and, in cumulated price-level terms, larger.

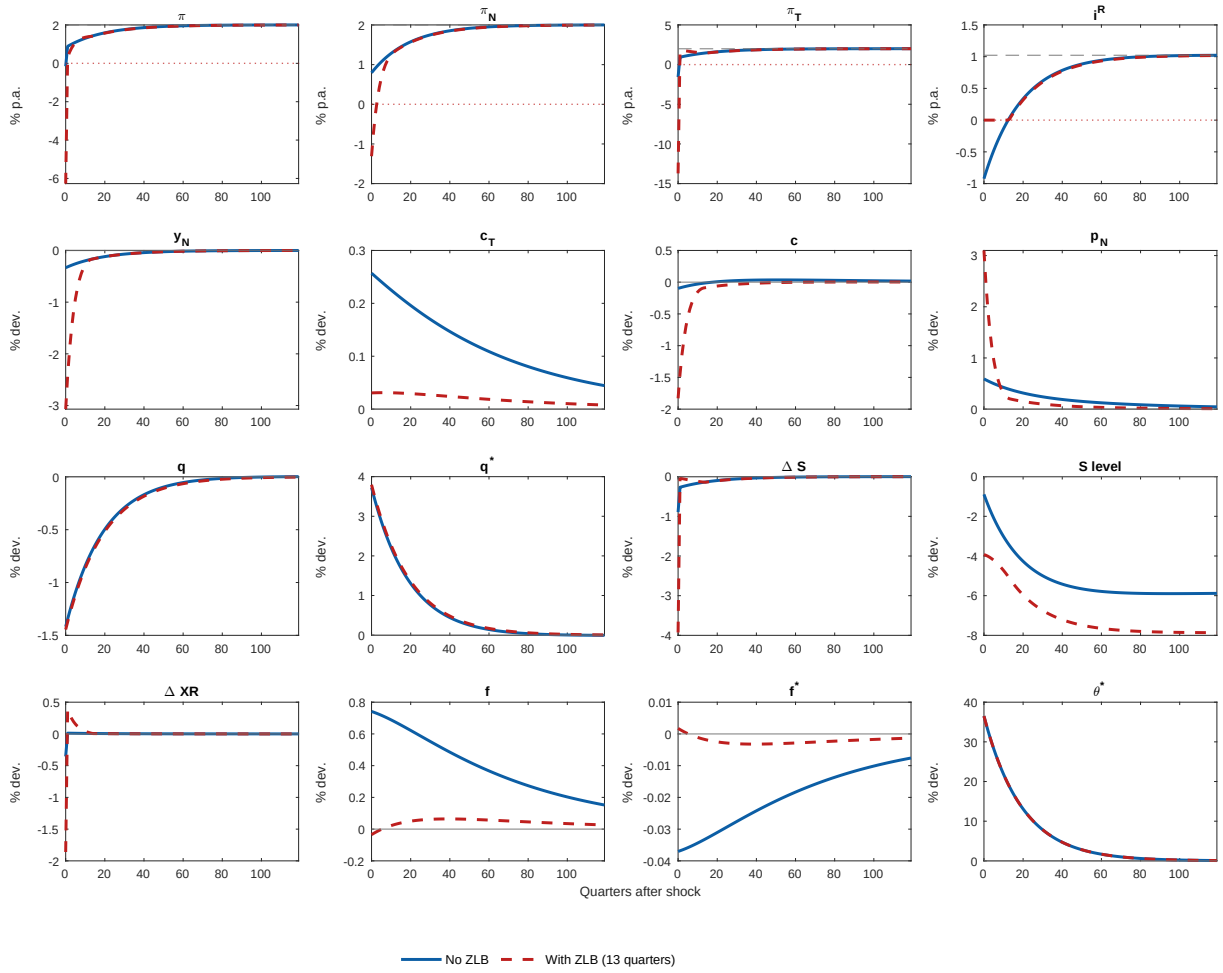


Figure 17: Full set of impulse responses to a flight-to-safety shock, the unconstrained economy (solid) against the zero lower bound (dashed), over 120 quarters. All variables are percent deviations from their steady state. Interest rate and inflation are reported in annualized terms relative to their steady-state levels.

Tradable consumption stays below the unconstrained path at every date. Appendix C.7 shows that this persistent gap reflects the larger valuation burden on the outstanding foreign-held position rather than a reduction in convenience income.

C.5 Balance-Sheet Policy

This Appendix subsection reports the full set of impulse responses under the two balance-sheet targets of Section 5.4 and the grid of the weighted target behind the locus of Figure 10. Figures 18 and 19 show all model variables over 120 quarters for the domestic target and the exchange-rate target against the fixed-supply baseline. The government's foreign asset position f^g stays at zero throughout both experiments.

Two features of the domestic target follow from the model's identities. The oversupply is part (iv) of Proposition 2: domestic holdings must rise by ϕ_1 times tradable consumption, 0.02% on impact, and the 4.7% expansion covers the accommodated foreign

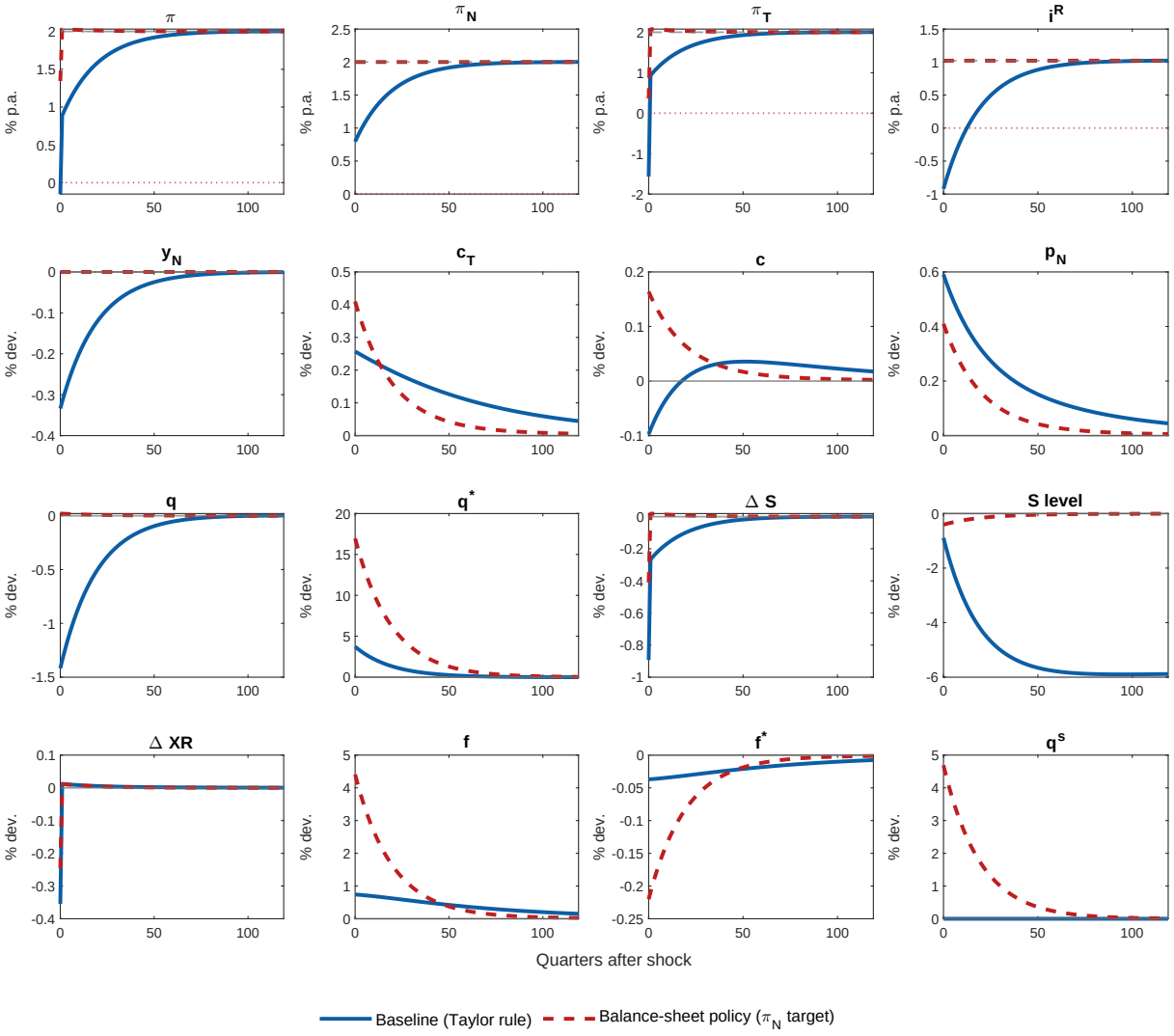


Figure 18: Full set of impulse responses under the balance-sheet policy targeting non-traded inflation (dashed) against the fixed-supply baseline (solid), over 120 quarters. Interest rates and inflation are annualized levels relative to their steady-state values, all other variables percent deviations from steady state.

demand of 17.0% at its weight of 0.28 plus this rise at its weight of 0.72. The concentration of the appreciation in a single impact jump is part (ii). With non-traded inflation at zero the relative price can move only through tradable inflation, and with output at zero the relative price equals tradable consumption, so the entire relative-price adjustment occurs at impact. Thereafter the slow decay of tradable consumption implies a mildly positive tradable inflation whose cumulated sum repays the impact deflation exactly, and the exchange rate returns to steady state over the full horizon. Along the path the level satisfies $\hat{S}_t = -\hat{C}_{T,t}$, so it mirrors consumption pointwise, complete in its return but slow along the way. The peak coincides with the impact jump. The operation behind it is persistent, decaying with a half-life of fifteen quarters.

The allocation of the transfer differs from the baseline. The foreign bond position jumps by 4.4% against 0.7%, while tradable consumption starts above the baseline level

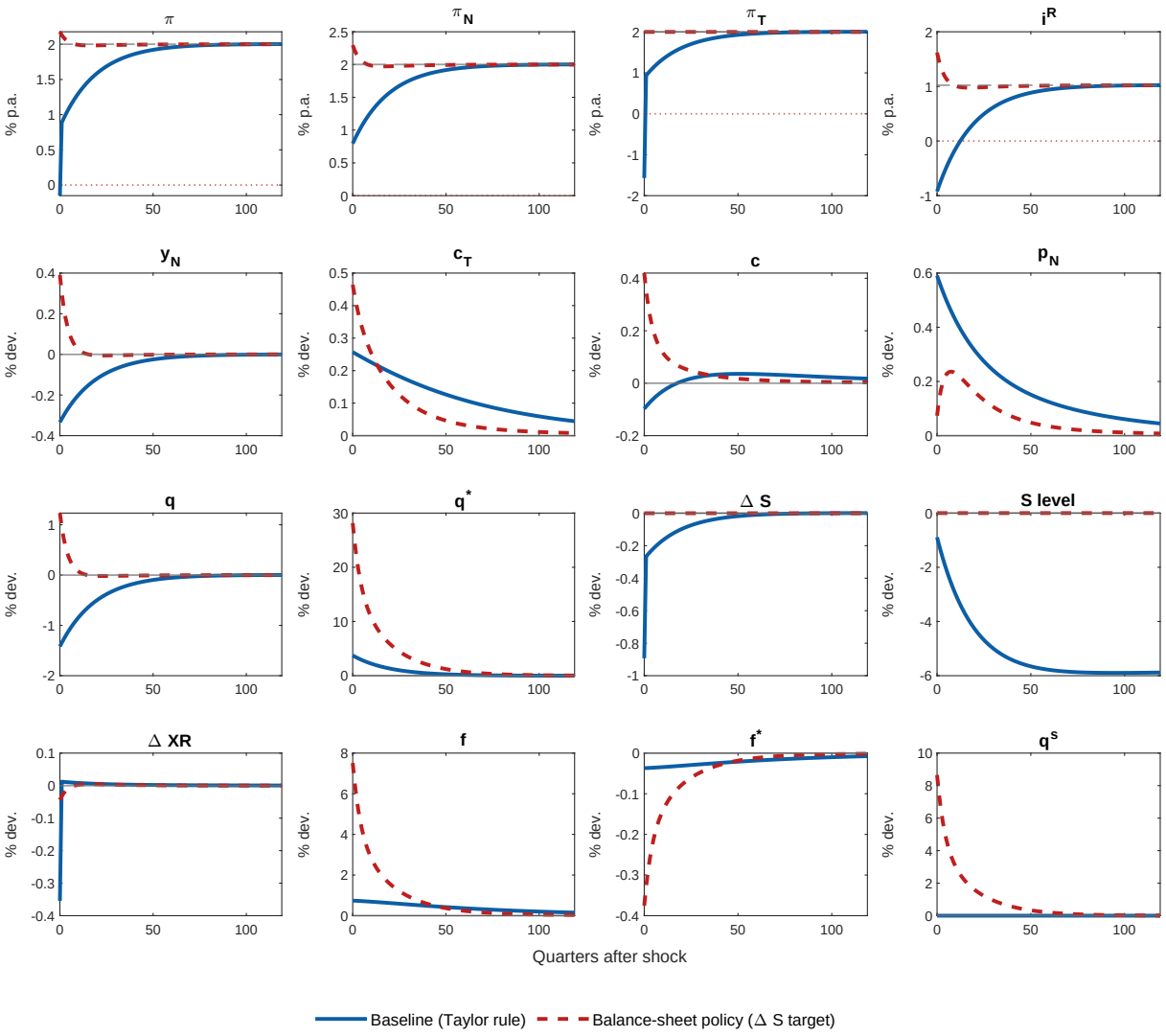


Figure 19: Full set of impulse responses under the balance-sheet policy targeting the exchange rate (dashed) against the fixed-supply baseline (solid), over 120 quarters. The layout matches Figure 18.

and then declines along the Euler equation (41) at the rate $\psi f \hat{f}_t$. The larger net inflow (Appendix C.7) is first accumulated in, and later run down out of, the bond position.

Under the exchange-rate target the operation is larger partly because the target raises the demand it must meet. Holding the exchange rate removes the expected depreciation that damps foreign demand under the domestic target, while the boom raises the policy rate, and both margins raise the return on the asset through (42). Foreign demand is 1.7 times the domestic target's at the same shock, 28.2% against 17.0%, and a third of it is the rate rule's response to the boom. The feedback is the reason the peak operation is 1.85 times larger. The operation is more persistent in its early part than under the domestic target, with a half-life of seven quarters, so that cumulated over the first forty quarters it is 1.2 times the domestic target's, less than the ratio of the peaks.

Table 9 reports the grid of the weighted target $\lambda\pi_{N,t}+(1-\lambda)\Delta S_t = 0$ behind Figure 10. Three facts carry the discussion of the main text. The peak operation is monotone in

λ and front-loaded at the domestic pole: it rises from 4.7% of the stock at $\lambda = 1$ to 6.9% at $\lambda = 0.98$ and 8.3% at $\lambda = 0.9$, where it is already 96% of the full defence, while the residual appreciation has fallen from 0.41% to 0.37% and 0.30%. In outcome space the locus is close to linear, bowed slightly away from the origin. And the policy rate is monotone in λ too, zero at the domestic pole and 0.6 points above steady state on impact at the exchange-rate pole, with a minimum level along every path of 0.98%, so the bound is never approached on the menu.

Table 9: The weighted target $\lambda\pi_{N,t} + (1-\lambda)\Delta S_t = 0$ along the grid. Impact responses of non-traded inflation (pp p.a.), non-traded output (%) and the policy rate (pp p.a.), the peak of the exchange-rate level (%), the peak of the operation (% of the stock), and the minimum level of the policy rate along the path (% p.a., steady state 1.02).

λ	π_N	y_N	S peak	q^s peak	i^R	i^R min
0	0.30	0.39	0.00	8.65	0.60	0.98
0.10	0.28	0.39	0.02	8.65	0.58	0.98
0.20	0.27	0.38	0.05	8.64	0.55	0.98
0.35	0.24	0.37	0.09	8.64	0.51	0.99
0.50	0.21	0.35	0.14	8.63	0.45	0.99
0.65	0.17	0.33	0.19	8.61	0.39	1.00
0.80	0.13	0.29	0.25	8.54	0.30	1.01
0.90	0.08	0.23	0.30	8.30	0.22	1.01
0.95	0.05	0.18	0.34	7.82	0.15	1.02
0.98	0.03	0.11	0.37	6.88	0.08	1.02
1	0.00	0.00	0.41	4.69	0.00	1.02

C.6 The Reserve-Backed Operation

Figure 20 reports the full set of responses. Two panels carry what the main text does not narrate. The f and f^g panels show the composition of the country's foreign claims: under the rebate the position is private, under reserve backing it is public, with households mildly crowded out, and the sum accumulates more in the reserve-backed case, since less of the proceeds is consumed. The composite consumption panel orders the two cases as tradable consumption does, +0.05% against +0.16% on impact, so the financing result is visible in the aggregate and not only in the traded block. Appendix C.7 adds the reserve-backed regime to the impact decomposition of the net transfer, where it shows as the largest net inflow in the table paired with the smallest consumed share.

Figure 21 isolates the role of the portfolio cost by setting $\psi = 0$ in both cases. This removes the only term through which the household's foreign position enters the tradable Euler equation (41): since traded inflation equals the exchange-rate change identically, the return bracket vanishes, and consumption follows $\hat{C}_{T,t} = E_t \hat{C}_{T,t+1}$, a constant path. The two cases then coincide, as Proposition 3 states. Every variable follows the same

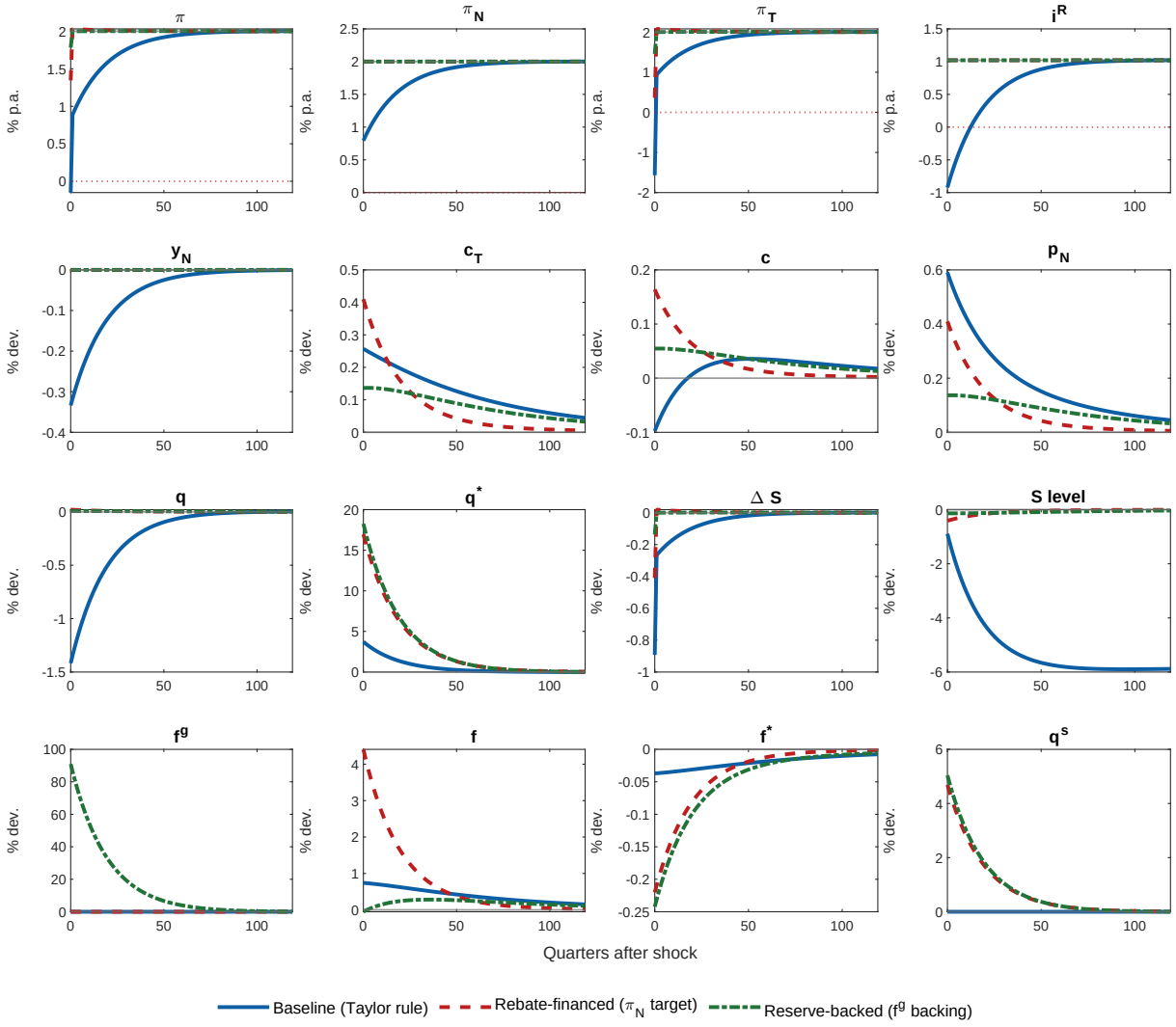


Figure 20: Full set of impulse responses under the domestic target with rebate-financed and reserve-backed issuance, against the no-policy baseline, over 120 quarters. Lines as in Figure 11. Interest rate and inflation panels are annual levels, remaining variables percent deviations. The government foreign position f^g replaces ΔXR in the layout, since it is the object of the experiment. The shock is common to all lines and shown in the baseline overview.

path to 10^{-12}. Consumption rises permanently by 0.045% under both, the permanent-income value of the convenience income earned on the position. Foreign demand at impact is 18.25%, equal to the flight term $\phi_1^*\hat{\theta}^*$ exactly, since flat consumption implies no expected exchange-rate term, and the operation is 5.0% of the stock under both. Only the composition of the country's foreign claims differs, 0.29 quarterly-GDP units held by households under the rebate and by the central bank under reserve backing. The difference between the two cases in Table 5 is therefore generated entirely by the portfolio cost, which pushes households far above their desired foreign position under the rebate and keeps them close to it under reserve backing.

The reserve-backed path at the calibrated ψ shows the same mechanism at a smaller scale. The central bank pays no portfolio cost, but households do, on their own deviation

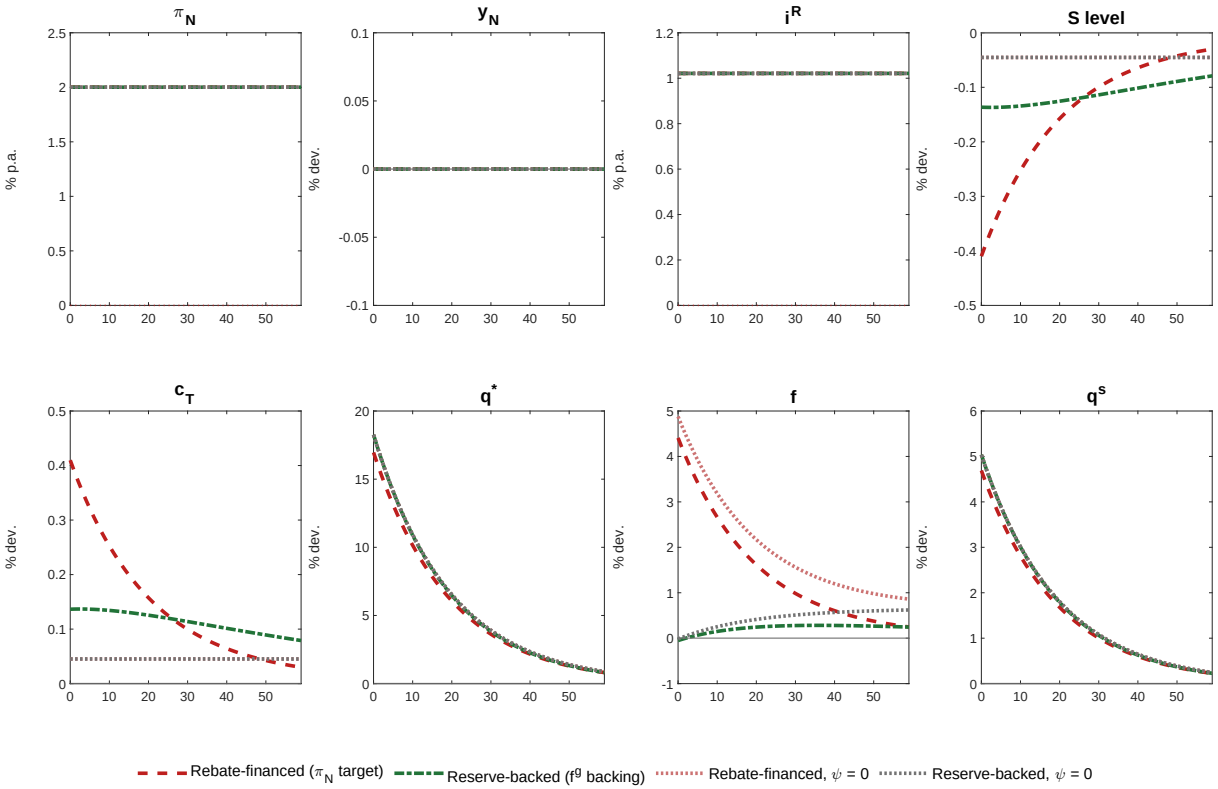


Figure 21: The domestic target under rebate-financed (dashed) and reserve-backed (dash-dotted) issuance at the calibrated portfolio cost, and the same two cases with $\psi = 0$ (dotted), 60 quarters. Lines and units as in Figure 11, with the no-policy baseline omitted for legibility. Without the portfolio cost the rebate-financed and reserve-backed cases coincide: both consumption paths are constant at the same level and the two dotted lines lie on top of each other, except in the f panel, where only the holder of the position differs.

from $\bar{f}$, and the operation moves that deviation. The reserve purchase crowds households out of foreign bonds on impact, $\hat{f}_0 = -0.05\%$, so the Euler equation (41) tilts consumption upward, but only just: consumption is 0.136% on impact and 0.137% in quarter 3. Households then rebuild the position past its desired level, $\hat{f}$ reaching +0.3% around quarter 37, and consumption declines from there, to 0.03% in quarter 120 and still falling. Both return to steady state slowly, since the deviation is small, at most 0.3% against 4.4% under the rebate, and the tilt $\psi f \hat{f}_t$ with it. The identity $\hat{C}_{T,t} - E_t \hat{C}_{T,t+1} = \psi f \hat{f}_t$ holds along the entire path to five decimal places. The flat stretch in Figure 11 is the household's own small portfolio adjustment, induced by the purchase, not a property of the central bank's position. The same identity governs the rebate: there the household holds the proceeds, $\hat{f}_0 = +4.4\%$, the tilt is large and negative throughout, consumption falls from 0.41% toward zero as the position returns to $\bar{f}$, and the windfall is spent within the episode. Without the portfolio cost the tilt is zero in every experiment, consumption is constant, and the foreign position carries a unit root: the baseline at $\psi = 0$ has tradable consumption flat at 0.07% and a permanently higher position, against a path that starts at 0.26% and decays at the calibrated value. Every decaying consumption path in the

paper is the portfolio cost at work, and every flat one its absence.

C.7 The Net Resource Transfer

Section 5.1 describes the net resource transfer the foreign demand shock generates and locates the convenience income in the terms at which the foreign-held position is serviced. This subsection derives both statements from the consolidated intertemporal budget constraint of the home country. The derivation uses only the tradable block of the model, the current account (38), the foreign optimality conditions (37) and (36), the law of one price, and the no-Ponzi conditions, so the constraint holds independently of the nominal rigidity in the non-traded sector.

The current account (38) can be rewritten as

$$q_t^* = \frac{1}{\beta} q_{t-1}^* - \left(\frac{1}{\beta} - \frac{1 + i_{t-1}^R}{\Pi_{T,t}} \right) q_{t-1}^* - (Y_{T,t} - C_{T,t}) + (f_t + f_t^g) - \frac{1 + i_{t-1}^{R*}}{\Pi_{T,t}^*} (f_{t-1} + f_{t-1}^g) + \frac{\psi}{2} (f_t - \bar{f})^2. \tag{86}$$

The term in parentheses on the first line is the shortfall of the realized real return on the foreign-held convenient asset below the shadow rate $1/\beta$, and the foreign Euler equation (37) prices it as the convenience yield $\theta_{t-1}^* V_q(q_{t-1}^*)$. Substituting this condition and the foreign-bond optimality condition (36) into (86), iterating forward, and imposing the no-Ponzi conditions $\lim_{T\to\infty} \beta^{T+1} E_t[q_{t+T+1}^*] = 0$ and $\lim_{T\to\infty} \beta^{T+1} E_t[f_{t+T+1} + f_{t+T+1}^g] = 0$ yields the intertemporal budget constraint (IBC)

$$E_0 \left[\sum_{t=0}^{\infty} \beta^t C_{T,t} \right] = - \frac{1 + i_{-1}^R}{\Pi_{T,0}} q_{-1}^* + \frac{1 + i_{-1}^{R*}}{\Pi_{T,0}^*} (f_{-1} + f_{-1}^g) + E_0 \left[\sum_{t=0}^{\infty} \beta^t \left(Y_{T,t} + \theta_t^* V_q(q_t^*)\, q_t^* - \frac{\psi}{2} (f_t - \bar{f})^2 \right) \right]. \tag{87}$$

Equation (87) separates the transfer into its three components. The term $\theta_t^* V_q(q_t^*)\, q_t^*$ is the convenience-yield revenue, the margin earned each period on the outstanding foreign-held stock. It is the strict analogue of seigniorage: the counterpart, for an asset with a convenience yield, of the revenue from issuing non-interest-bearing money. The first term on the right is the valuation component: an appreciation, a fall in $\Pi_{T,0}$, raises the real value of what the economy owes on the position it has already sold. Everything else is intertemporal trade at the shadow rate, which the no-Ponzi conditions force to net out in present value.

The statement the chapter uses is the flow counterpart of IBC. The net inflow $\mathcal{N}_t$ of

Table 10: The impact decomposition of the net inflow across regimes. Columns follow (88) evaluated at $t = 0$, where $\hat{X}_0 = -\hat{\pi}_{T,0}$ and $\mathcal{N}_0 = \hat{q}^*_0 - \frac{1}{\beta}\hat{X}_0 + \frac{\nu}{\beta}\hat{X}_0$. Units are percent deviations of the foreign position. The last column reports the impact consumption composite from (51), in percent.

	Baseline	ZLB	Domestic target	Reserve backed
New purchases $\hat{q}^*_0$	3.7	3.8	17.0	18.3
Position $\hat{X}_0$	0.9	3.9	0.4	0.1
Income $\frac{\nu}{\beta}\hat{X}_0$	0.007	0.030	0.003	0.001
Net inflow $\mathcal{N}_0$	2.8	−0.1	16.5	18.1
Composite $\hat{c}_0$	−0.10	−1.83	+0.16	+0.05

(84) decomposes identically as

$$\mathcal{N}_t = \underbrace{\frac{\nu}{\beta}\hat{X}_t}_{\text{convenience income}} + \underbrace{\hat{q}^*_t - \frac{1}{\beta}\hat{X}_t}_{\text{net borrowing}}, \qquad \hat{X}_t \equiv \hat{q}^*_{t-1} + \hat{\imath}^R_{t-1} - \hat{\pi}_{T,t}, \tag{88}$$

where $\hat{X}_t$ is the real value of the outstanding position inclusive of interest and revaluation. The first component is the linearized convenience income, the revenue flow of (87). The second is net new issuance less debt service at the shadow rate, and it is where the valuation exposure lives, since the revaluation term $-\hat{\pi}_{T,t}$ enters $\hat{X}_t$ and inflates the service burden when the currency appreciates. Grouping the two $\hat{X}_t$ terms recovers the carrying rate $(1-\nu)/\beta$ of the main text, so the decomposition does nothing more than split that rate into service at the shadow rate and the convenience discount.[26]

Table 10 evaluates the decomposition on impact, where the lagged deviations are zero and $\hat{X}_0 = -\hat{\pi}_{T,0}$, so the comparison across regimes requires no horizon. The product $\frac{1-\nu}{\beta}\hat{X}_0$ is the carrying cost of Appendix C.1, and at the quarterly wedge the rate is 0.9975, which is why the carrying cost and the revalued position coincide at one decimal, 0.9 in the baseline. Under the zero lower bound the sharper impact appreciation enlarges the revalued position more than fourfold. The convenience income rises with it and stays above its unconstrained counterpart at every date, since the margin is earned on a larger base at an uncut rate. The net inflow turns negative, because the same enlarged position is serviced at the full shadow rate and the service exceeds the new purchases. Under the domestic target the dominance runs the other way. The accommodated purchase is the largest among the rebate-financed regimes while the revalued position is below the baseline's, so the net inflow is six times the baseline's, with the convenience income above the baseline's from the first quarter on. Under reserve backing the same mandate earns the largest inflow in the table, more than six times the baseline's, with the smallest

[26] The decomposition is an identity and was verified on the simulated paths of every regime to machine precision.

revaluation, since the nearly flat exchange rate spares the outstanding position. The pairing of its two entries is the financing result of Section 5.5: the largest net inflow in the table next to the smallest consumed share.

The same experiments can be read in total consumption, reported in the last column of Table 10. Non-traded market clearing sets $C_{N,t} = Y_{N,t}$, so the composite follows (51). On impact the baseline delivers -0.1%: the decline in the sheltered sector outweighs the tradable inflow while the rate responds, and the composite recovers only as the recession fades and the position accumulates. At the bound it is -1.8%. Under the domestic target it is $+0.2\%$, the inflow without the recession. One number per experiment summarizes the mechanism: the same flight lowers total consumption on impact when the rate alone responds, far more so when the bound binds, the balance-sheet policy turns the impact into a gain, and reserve backing earns the largest inflow while consuming the least of it.

The constraint also shows why the instrument is the liability of the balance sheet and not its assets. The foreign-income terms in (87) depend on the consolidated position $f_t + f_t^g$ only, which is Proposition 3 read from the budget constraint: with $\psi = 0$ the split of that position between households and the government is irrelevant. The supply of the convenient asset is different in kind. Its traction comes from unsatiated foreign demand, $V_q(q_t^*) > 0$, not from portfolio frictions. A foreign-exchange intervention that pays for foreign assets by issuing the convenient asset, the form the SNB's operations took, bundles the two: an asset leg that is intertemporal trade at the shadow rate, and a liability leg that is the instrument of Section 5.4. The experiments there hold the government's foreign position passive to study the liability leg alone, and Section 5.5 adds the asset leg.

C.8 Persistence

This subsection reports the two experiments of Section 5.7. In the first the foreign preference shifter follows the baseline process with $\rho_{\theta^*} = 0.9999$ in place of 0.95, and the model is solved linearly without the bound. In the second the shifter is held at 36.5 for N quarters and set to zero afterwards, agents know N, and the bound enters as the complementarity condition $\hat{\imath}_t^R \geq \underline{\imath}^R$ in a perfect-foresight solution over 200 quarters, so that the terminal condition binds far from the episode.[27]

Figure 22 shows that the permanent flight moves the economy to a new position within a quarter and keeps it there. The position is the steady state of Proposition 4, and three equations determine it. The Phillips curve (39) with constant inflation gives $\hat{Y}_N = \frac{1-\beta}{\kappa}\hat{\pi}_N$,

[27] The solver is Dynare's `perfect_foresight_solver` with the `lmmcp` option. Holding the shifter flat under the AR(1) requires an innovation of $(1 - \rho_{\theta^*}) \times 36.5$ in quarters 2 to N and of $-\rho_{\theta^*} \times 36.5$ in quarter $N + 1$, all anticipated at $t = 1$.

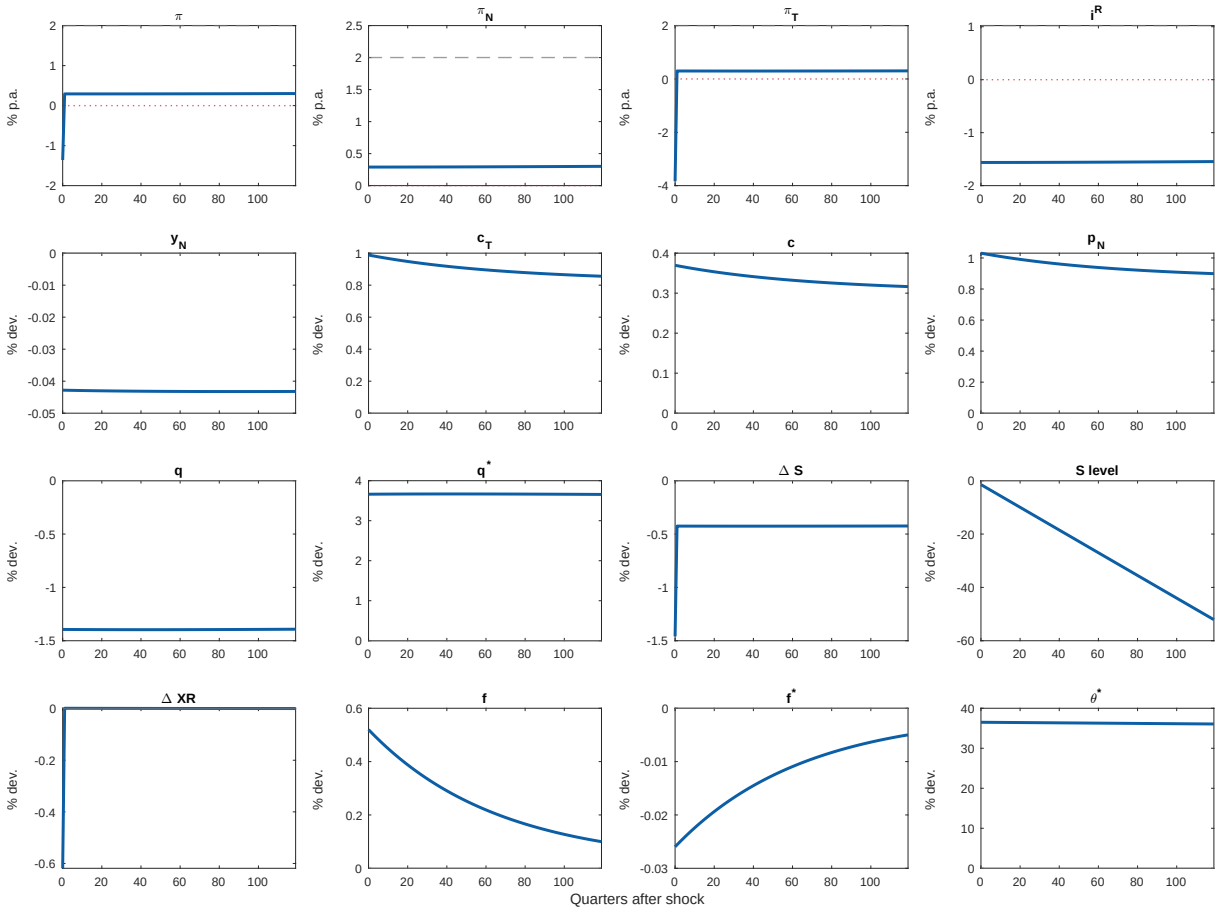


Figure 22: Full set of impulse responses to a permanent flight-to-safety shock, $\rho_{\theta^*} = 0.9999$, in the unconstrained economy with a fixed supply of the convenient asset, over 120 quarters. All variables are percent deviations from their steady state. Interest rate and inflation are reported in annualized terms relative to their steady-state levels. The exchange-rate level is the cumulated change and does not settle.

which at $\beta = 0.995$ and $\kappa = 0.05$ is a tenth of the inflation deviation: a permanent deflation costs 0.04% of output. Aggregate demand (40) with constant variables gives the real rate that keeps non-traded demand at this level, $(1-\nu)(\hat{\imath}^R - \hat{\pi}_N) = \frac{\nu}{\phi_1}\hat{q} - \nu\hat{p}_N - \nu\hat{Y}_N$. Domestic holdings fall by 1.4% through market clearing, so the convenience term is -0.21 per quarter, the relative-price term 0.01, and the real rate must fall by 0.22 per quarter, 0.9 percentage points a year. This real rate does not depend on the nominal regime, part (i) of Proposition 4. The Taylor rule closes the system: $\hat{\imath}^R - \hat{\pi}_N = (\phi_\pi - 1)\hat{\pi}_N + \phi_y\hat{Y}_N$, so a real rate 0.22 below steady state requires inflation 0.43 below it per quarter, 1.7 points a year, and a nominal rate 0.65 below it, 2.6 points a year. The exchange rate appreciates at the rate of tradable deflation, 1.7% a year, since the relative price is constant. Foreign holdings rise by 3.7%, as in the transitory baseline, so the inflow carries no information about the duration of the flight.

Table 11 collects the finite flights. Two regularities hold across the columns. First, the rate leaves the floor four to five quarters before the flight ends, at every duration. Agents know the exit date, expected inflation turns up ahead of it, and the rule prescribes a

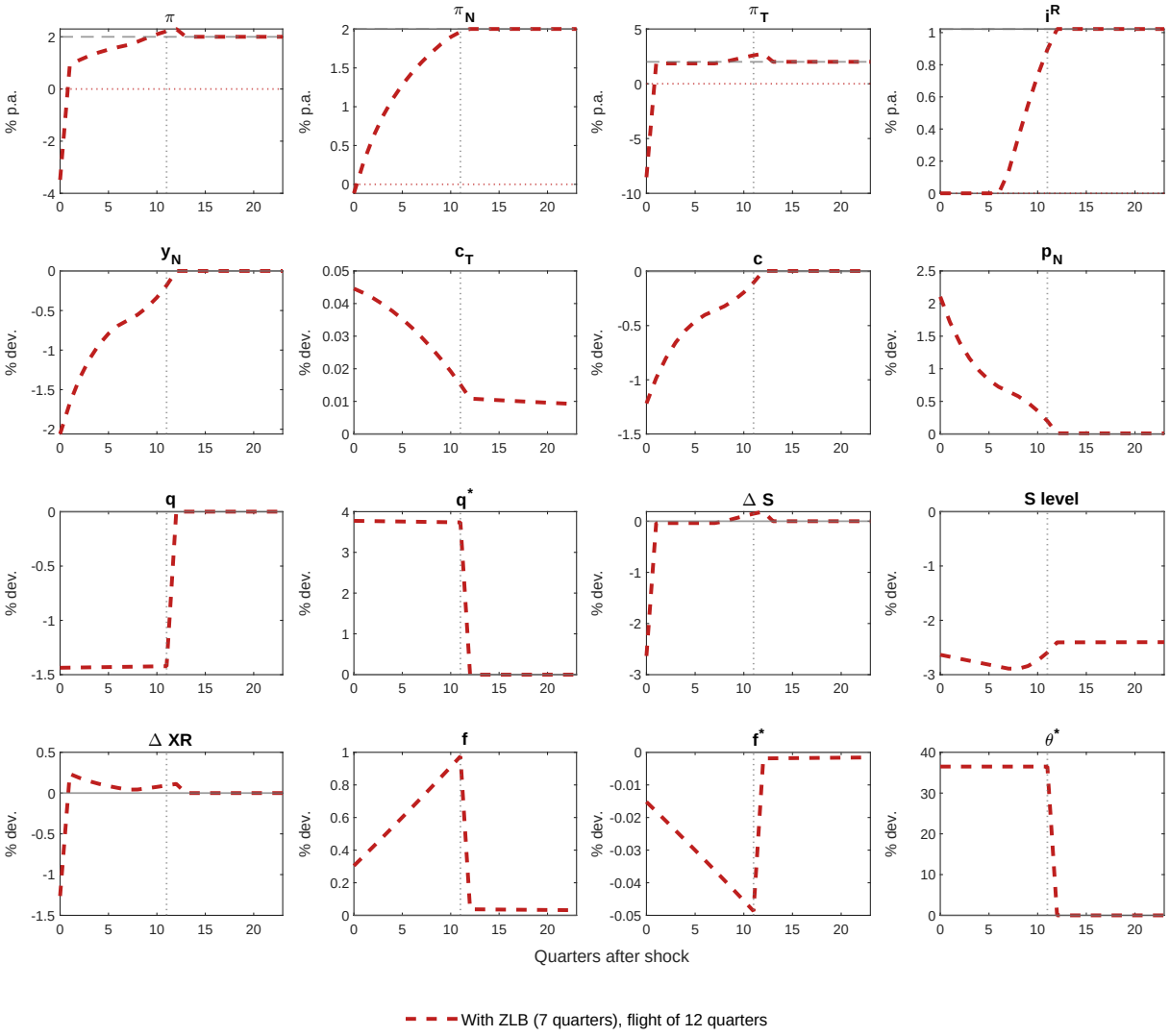


Figure 23: Full set of impulse responses to a flight-to-safety shock that lasts twelve quarters, with the zero lower bound, over 24 quarters. The foreign preference shifter is held at its impact level for twelve quarters and set to zero afterwards, agents know the duration, and the dotted vertical line marks the last quarter of the flight. All variables are percent deviations from their steady state. Interest rate and inflation are reported in annualized terms relative to their steady-state levels.

rate above the floor while the shifter is still at its impact level. The spell at the floor is the duration minus this lead. Second, the impact responses grow with the duration faster than the duration. The impact recession is 0.9% at eight quarters, 2.1% at twelve, 5.0% at sixteen and 13% at twenty, and a flight of twenty-four quarters, outside the table, gives 41% with twenty quarters at the floor. Each four quarters of flight multiply the recession by two and a half. The mechanism is the one of Eggertsson and Woodford (2003). With the rate at the floor, every quarter of expected deflation ahead raises the current real rate, the deeper fall in demand lowers current inflation through (39), and the next quarter inherits a larger expected deflation. The inflow itself hardly moves: foreign holdings rise by 3.7% to 4.1% of the position in every column, so what grows with the duration is the trap, not the flow. The twenty-quarter column is outside the range for

Table 11: A flight-to-safety shock of known duration with the zero lower bound, impact responses and the spell at the floor

	$N = 8$	$N = 12$	$N = 16$	$N = 20$
Quarters at the floor	3	7	11	15
Quarters off the floor before the end	5	5	5	5
i^R change (pp p.a.)	-1.0	-1.0	-1.0	-1.0
$i^{R,\text{rule}}$ change (pp p.a.)	-1.7	-4.0	-9.7	-24.9
y_N impact (%)	-0.9	-2.1	-5.0	-13.1
π_N impact (pp p.a.)	-0.9	-2.1	-5.1	-13.1
S impact (%)	1.2	2.6	6.2	15.6
S peak (%)	1.3	2.9	6.6	17.2
q^* impact (%)	$+3.7$	$+3.8$	$+3.8$	$+4.1$
q impact (%)	-1.4	-1.4	-1.5	-1.5
c_T impact (%)	$+0.08$	$+0.05$	-0.15	-0.8
c impact (%)	-0.5	-1.2	-3.1	-8.2

The foreign preference shifter is held at 36.5 for N quarters and set to zero afterwards, agents know N, and the bound is imposed as a complementarity condition. $i^{R,\text{rule}}$ is the rate the Taylor rule prescribes on impact, i^R the rate at the floor. The peak of S is reached late in the horizon for $N = 20$, where the level is still appreciating slowly. The solver does not converge for $N = 28$ at the fitted size, and for $N = 40$ it converges only for shocks up to 13, a third of the fitted size.

which the linearization is meant. I report it for the rate of growth, which is what places the limit. For $N = 28$ at the fitted size the solver does not converge, and for $N = 40$ it converges only for shocks up to 13, a third of the fitted size. Non-convergence of a Newton-type solver on the complementarity system is not a proof that no equilibrium exists, and I do not claim one. What is established is the permanent case, Proposition 4: below a permanent flight of 14.3, two fifths of the fitted size, the piecewise-linear model has two steady states, at 14.3 one, and above it no steady state and no bounded equilibrium at all. The size at which the forty-quarter flight stops converging, a third of the fitted shock, is close to this threshold.

A flight of twenty quarters at 36.5 carries the same cumulated shock as the baseline AR(1), $36.5/(1-0.95) = 730$. Under the bound the baseline produces a recession of 3.1% and a spell at the floor of thirteen quarters (Appendix C.4), the twenty-quarter flight one of 13% and fifteen quarters. The difference is the time profile. The AR(1) flight is at its strongest when the expected trap is longest and fades as the trap runs, the finite flight holds its full strength until the end.